Leveraged Learning: entropy destroyed per bit received.

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September 9, 2026

In 1948 Claude Shannon introduced the world to information entropy [1]. He quantified the amount of information in a signal. What if we wanted to quantify learning in the same spirit, as the increased understanding of a problem? This work is an attempt to follow that thread. It models the problem as a boolean map, and the understanding as the aggregate ability to predict that map. And learning, as the speed at which that understanding increases, specifically as information is supplied to the learner. Some interesting relationships can be formed between the two.

Allow us to enumerate the seminal tenets of this treatise.

1. Learning is the incremental ability to answer questions correctly that follow a pattern.
2. The number of questions reviewed is not the right measure of a learner's investment. What counts is the information received from asking them. Not all questions are created equal: about some, the learner already knows something a priori, and their answers count for *less*.
3. Conversely, with instincts about the world, one can infer *more* than a single bit from a yes–no question. One answer settles expectations on any number of yet unseen questions.
4. This ability to extrapolate from prior information is codified here as **leverage**.

$$\text{Leverage of Yes/No} = \frac{\text{Expected predictive gain}}{\text{Expected surprisal received}}. \tag{1}$$

The expected surprisal of a yes–no answer is its entropy: at most 1 bit. Its expected predictive gain can exceed 1 bit through deductions about other questions. We model successful learning as this ratio being high.

Summary of results

1. **Setup.** Let there be Q questions q, which can each beget A answers a, for a total of $N = A^Q$ maps. The true map is ψ, and each candidate map ϕ_j has prior belief p_j. Answers $a = \psi(q)$ fed one-by-one eliminate maps. We study the surprisal of each answer, vs the remaining entropy over all Q questions.
2. **The answer matrix.** What the prior belief *predicts* at any point in the learning process is the answer matrix, one column per question, holding the total weight of the maps that would answer a to q. Its column entropies added measure the total remaining predictive uncertainty under separate logarithmic scoring,

$$M_{a,q} = \sum_{j\,:\,\phi_j(q)=a} p_j, \qquad H(M) = \sum_q H\big(M_{\cdot,q}\big).$$

3. **Leverage.** Asking q and hearing a costs the surprisal $s = -\log_2 M_{a,q}$ and, by culling inconsistent maps ϕ_j, settles some of $H(M)$. The two are not equal and the columns of questions never asked move with every update. Let us have seen ℓ questions. The ratio of total reduced uncertainty to surprisal is **leverage**,

$$L_\ell \;=\; \frac{H\big(M^{(0)}\big) - H\big(M^{(\ell)}\big)}{\sum_{k\le\ell} s_k}.$$

   $L = 1$ is what we expect from a uniform prior, one bit destroyed per bit paid. An individual run can exceed this baseline through luck. But ifthe excess is *expected*: that is due deduction from good prior instinct.

4. **The finite average.** Average over the truth $\psi \sim p_j$ and uniformly over the order in which questions arrive. The whole trajectory is then generated by a single sequence, the mean entropy G_ℓ of the answers to ℓ questions,

$$\mathcal{L}_\ell \;=\; \frac{Q\,G_1 - (Q-\ell)\big(G_{\ell+1} - G_\ell\big)}{G_\ell}.$$

   Here $\mathcal{L}_\ell$ divides expected predictive gain by expected surprisal. Anything above 1 in this $\mathcal{L}_\ell$ is deduction. Received surprisal averages to G_ℓ, and what remains is $Q-\ell$ copies of one **increment**. This is exact at every size and for every prior, with no assumption about where the belief came from. The sequence is a combinatoric function of the prior, summing over the $\binom{Q}{\ell}$ sets $\mathcal{S}$ of ℓ questions and the A^ℓ answer patterns y they admit,

$$G_\ell \;=\; -\binom{Q}{\ell}^{-1} \sum_{|\mathcal{S}|=\ell}\sum_{y} P_{\mathcal{S},y}\log_2 P_{\mathcal{S},y}, \qquad P_{\mathcal{S},y} \;=\; \sum_{j\,:\,\phi_j(\mathcal{S})=y} p_j.$$

5. **The thermodynamic limit.** If we send $n \to \infty$ at a fixed asked fraction $t = \ell/Q$. The increments become a profile $\gamma(t)$, the received entropy its integral, and the limit is obtained,

$$L(t) \;=\; \frac{\eta_0 - (1-t)\,\gamma(t)}{g(t)}, \qquad g(t) = \int_0^t \gamma(x)\,dx.$$

   Bounded and nonincreasing is all that is asked of γ.

6. **An Occam prior's macroscopic limit.** Write a boolean map as a polynomial over $\{0,1\}^n$ and group maps by their degree. Allocate prior weight to shells of varying degrees by a CDF $F(x)$, dividing the p_j uniformly over the maps inside it. A map's weights decrease strictly with degree, making this genuinely Occam, and its macroscopic profile is closed form,

$$\gamma(t) \;=\; 1 - F(t), \qquad L_F(t) \;=\; \frac{t + (1-t)F(t)}{\int_0^t \big(1 - F(x)\big)\,dx}.$$

   which allows us to model a simplicity bias that permits an explicit TDL.

Contents

1 A thought experiment

Let us start with a small thought experiment, by introducing *Werner the learner* (Figure 1).

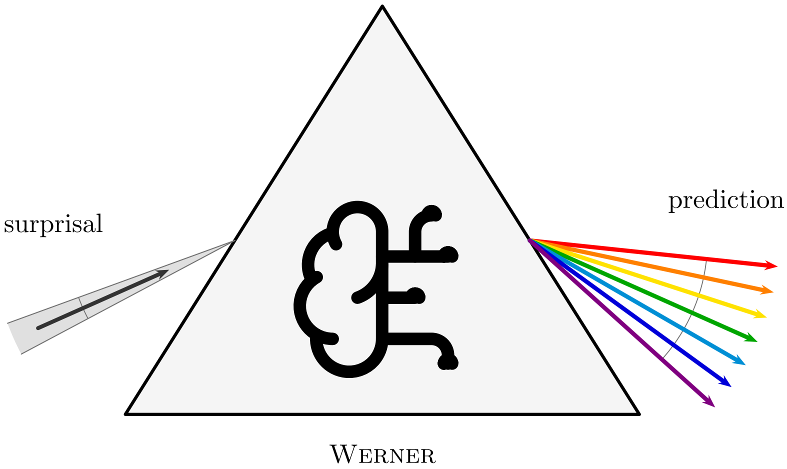


Figure 1: Werner the learner.

In this work, we model learning as the increased ability to give the correct answer to questions, as more of the ground truth is revealed. It is Werner's job to learn a pattern. The most basic shape of a pattern is a **binary map**, and the most basic binary map is a map from one bit to one bit.

The setup of the experiment:

- a **question** $q \in \mathcal{Q} = \{0, 1\}$;
- an **answer** $a \in \mathcal{A} = \{0, 1\}$;
- the **truth** $\psi : \mathcal{Q} \to \mathcal{A}$, the one fixed map nature uses to answer questions. ψ is unknown to Werner. By definition $\psi(q) = a$.

There are exactly four candidate maps ϕ_j that ψ could be (Table 1): each of the two inputs can be sent to either of the two outputs, independently. We index them by reading the truth table $\phi_j(1)\,\phi_j(0)$ as a binary numeral: that number *is* j.

j	$\phi_j(1)$	$\phi_j(0)$	name
0	0	0	constant 0
1	0	1	NOT ($\neg q$)
2	1	0	identity (q)
3	1	1	constant 1

Table 1: The four binary maps from one bit to one bit.

1.1 An unintelligent learner

Now, if Werner is unintelligent, he has no preconceived notion of which maps are more useful, more likely, or somehow more logical than others. In practice, we would have to tell Werner the answer a to every single question q, before he has learned the whole pattern.

We model this with a **uniform prior**: writing $p_j = \Pr(\psi = \phi_j)$ for Werner's belief, every candidate map carries the same weight.

EXAMPLE 1

For the four maps at hand:

$$p_j = \tfrac{1}{4}, \qquad j = 0, 1, 2, 3. \tag{2}$$

What does this belief predict? For each question q separately, we can tabulate the probability of each possible answer a: it is the total belief sum of the maps that would answer a to q,

$$M_{a,q} \;=\; \sum_{j\,:\,\phi_j(q)=a} p_j. \tag{3}$$

Arranged with one row per answer ($a = 0, 1$ from the top) and one column per question ($q = 1, 0$ from the left, descending like the digits of the numeral convention), these numbers form the matrix M. Each column is a probability distribution over answers, so M is column-stochastic. Werner reads it as follows: find the column of the question at hand. It's the marginal distribution over answers, according to his belief.

EXAMPLE 2

Under the uniform prior, two of the four maps answer 0 and two answer 1, at either question:

$$M \;=\; \begin{array}{c|cc} {}_{\mathbf{a}}\diagdown^{\mathbf{q}} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & M_{0,1} & M_{0,0} \\ \mathbf{1} & M_{1,1} & M_{1,0} \end{array} \;=\; \begin{array}{c|cc} {}_{\mathbf{a}}\diagdown^{\mathbf{q}} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0 + p_1 & p_0 + p_2 \\ \mathbf{1} & p_2 + p_3 & p_1 + p_3 \end{array} \;=\; \begin{array}{c|cc} {}_{\mathbf{a}}\diagdown^{\mathbf{q}} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{4}$$

Every column is a fair coin: the unintelligent Werner predicts nothing about any answer before it is revealed.

Werner gets to predict the answer to the question $q = 0$. By his own belief it is a coin toss: $M_{0,0} = M_{1,0} = \frac{1}{2}$. Let us say the answer comes back $a = 1$. The **surprisal** of this answer, to Werner, is 1 bit, because the probability he assigned to it was one half.

In general, when an event that a belief holds with probability P actually occurs, its surprisal is

$$s \;=\; -\log_2 P \quad \text{bits}, \tag{5}$$

here $s = -\log_2 M_{a,q}$ for answer a arriving at question q. Surprisal measures how much the received information deviates from what was already expected: an answer held certain ($P = 1$) carries 0 bits: nothing new [4]. A coin toss carries exactly 1 bit; and the surprisal grows without bound as the answer gets more unexpected, $P \to 0$. It is additive: the surprisal of two independent events occurring is the sum of their surprisals, just as their probabilities multiply.

Moreover, Werner can use a process of elimination (an extreme form of Bayes' rule) to update his prior into a posterior. Semantically: every candidate map that would have answered the question wrongly is now *ruled out*, not merely disfavored. They are inconsistent with what's been learned.

EXAMPLE 3

The truth ψ answered 1 at $q = 0$, and a map with $\phi_j(0) = 0$ cannot be ψ. This kills the constant-0 map ($j = 0$) and the identity ($j = 2$). The two survivors, NOT ($j = 1$) and constant 1 ($j = 3$), both predicted the observed answer, so this round of evidence does not distinguish between them at all (Table 2).

j	$\phi_j(1)$	$\phi_j(0)$	name
~~0~~	~~0~~	~~**0**~~	~~constant 0~~
1	0	1	NOT ($\neg q$)
~~2~~	~~1~~	~~**0**~~	~~identity (q)~~
3	1	1	constant 1

Table 2: The surviving candidate maps after the answer $\psi(0) = 1$.

The surviving ratio of believabilities must be preserved, and the only update that does so is to renormalize the surviving weights to sum to one. (This is Bayes' rule with a 0/1 likelihood: $\Pr(\psi = \phi_j \mid a) \propto \Pr(a \mid \phi_j)\, p_j$, and $P(a \mid \phi_j)$ is 1 for a map that answers a and 0 for a map that does not.)

EXAMPLE 4

The posterior is

$$p' \;=\; \big(0,\ \tfrac{1}{2},\ 0,\ \tfrac{1}{2}\big). \tag{6}$$

Constructing the new answer matrix from p' by the same recipe as before:

$$M' \;=\; \begin{array}{c|cc} {}_{\mathbf{a}}\diagdown^{\mathbf{q}} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p'_0 + p'_1 & p'_0 + p'_2 \\ \mathbf{1} & p'_2 + p'_3 & p'_1 + p'_3 \end{array} \;=\; \begin{array}{c|cc} {}_{\mathbf{a}}\diagdown^{\mathbf{q}} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & 0 \\ \mathbf{1} & \frac{1}{2} & 1 \end{array}. \tag{7}$$

Indeed, Werner knows the answer to $q = 0$: that column has collapsed onto $a = 1$. He is still as agnostic as possible about the other answer: the $q = 1$ column remains a fair coin. He gained 1 bit of surprisal information, and cleaned up his table to the tune of 1 bit: the $q = 0$ column went from a full bit of answer-uncertainty to none, while the $q = 1$ column is unchanged.

After the next question, $q = 1$, he learns $a = 1$, again a coin toss beforehand ($M'_{1,1} = \frac{1}{2}$), so again worth 1 bit of surprisal. Elimination now rules out NOT ($\phi_1(1) = 0$), leaving a single survivor:

$$p'' \;=\; \big(0,\ 0,\ 0,\ 1\big), \qquad M'' \;=\; \begin{array}{c|cc} {}_{\mathbf{a}}\diagdown^{\mathbf{q}} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 0 & 0 \\ \mathbf{1} & 1 & 1 \end{array}. \tag{8}$$

So this is the constant map: $\psi = \phi_3$. Werner is done. Every column is one-hot, and he will be able to answer all questions correctly from now on.

1.2 Intelligent priors

Now, let us assume that Werner knows *something* about the world before he starts.

EXAMPLE 5

Let's restart, with a different ψ. Moreover, let Werner know the map is not constant. His belief then spreads evenly over the two remaining candidates, NOT ($j = 1$) and the identity ($j = 2$):

$$p \;=\; \big(0,\ \tfrac{1}{2},\ \tfrac{1}{2},\ 0\big). \tag{9}$$

Is he any better at predicting $\psi(0)$? Constructing M by the usual recipe:

$$M \;=\; \begin{array}{c|cc} \mathbf{a} \backslash \mathbf{q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p_0 + p_1 & p_0 + p_2 \\ \mathbf{1} & p_2 + p_3 & p_1 + p_3 \end{array} \;=\; \begin{array}{c|cc} \mathbf{a} \backslash \mathbf{q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{10}$$

He is still just as bad as before: every column a fair coin.

So what has all this veteran wisdom about the nature of patterns in the world bought him? His increased ability for *inference*.

EXAMPLE 6

Ask $q = 0$ as the first question, and let the answer again come back $a = 1$. Again a coin toss beforehand, so again worth exactly 1 bit of surprisal. Elimination rules out the identity ($\phi_2(0) = 0$), and the lone survivor is NOT:

$$p' \;=\; \big(0,\ 1,\ 0,\ 0\big), \qquad M' \;=\; \begin{array}{c|cc} \mathbf{a} \backslash \mathbf{q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1 & 0 \\ \mathbf{1} & 0 & 1 \end{array}. \tag{11}$$

This time *both* columns collapse at once. Werner received the same single bit of surprisal as before, but cleaned up 2 bits of table: the answer at $q = 0$ he observed, and the answer at $q = 1$ he *deduced*, without ever asking. Pre-empting the constants means $\phi(1) \neq \phi(0)$, so his prior held the two answers perfectly anticorrelated, and settling one settles the other. He is done after one question instead of two.

We say the information from the first answer was **leveraged**. We will make this more precise in the upcoming chapters.

1.2.1 A one-parameter world

For more structure, let us relax the claim that there are *no* constant maps. Instead, we create a one-parameter world where the prior probability that the map is constant is p, spread evenly within each class (Table 3):

j	$\phi_j(1)$	$\phi_j(0)$	name	p_j
0	0	0	constant 0	$p/2$
1	0	1	NOT ($\neg q$)	$(1-p)/2$
2	1	0	identity (q)	$(1-p)/2$
3	1	1	constant 1	$p/2$

Table 3: The one-parameter prior: constant with probability p.

In vector form, and with the answer matrix constructed by the usual recipe:

$$\big(p_0,\ p_1,\ p_2,\ p_3\big) \ = \ \tfrac{1}{2}\big(p,\ 1-p,\ 1-p,\ p\big), \qquad M \ = \ \begin{array}{c|cc} \mathbf{a} \backslash \mathbf{q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & \frac{1}{2} & \frac{1}{2} \\ \mathbf{1} & \frac{1}{2} & \frac{1}{2} \end{array}\,, \tag{12}$$

fair coins in every column, for every value of p.

How much is there to learn? We measure that by the predictive uncertainty that remains in total. Suppose Werner predicts each answer separately and is scored by logarithmic loss: when $a = \psi(q)$ is revealed, his loss is $-\log_2 M_{a,q}$. Under his own belief, the expected loss on question q is the entropy of its column. Adding the scores over the questions (extensivity) therefore gives the table entropy,

$$H(M) \ := \ \sum_q H\big(M_{\cdot,q}\big), \qquad H\big(M_{\cdot,q}\big) \ := \ -\sum_a M_{a,q}\,\log_2 M_{a,q}. \tag{13}$$

This is an operational choice: $H(M)$ is the aggregate Bayes prediction risk under logarithmic loss [2]. A prediction risk is the expected loss of a stated prediction under a fixed scoring rule, and it is a Bayes risk when the prediction stated is the one that minimizes that expectation. Under logarithmic loss the minimizer is Werner's own column $M_{\cdot,q}$, and the value it attains is that column's entropy, which is why the two coincide here. It scores answers separately, even when they are correlated. One observation can improve many separate predictions at once. Identifying the whole map is a different task, whose uncertainty is its joint entropy. This may never happen. There is a practical argument to measuring the full uncertainty as it stands. In the present example, both columns are fair coins carrying 1 bit each, so the table starts at its maximum: $H(M) = 2$ bits.

Now let us ask a question. Again we could use $q = 0$; by the symmetry of the prior it actually does not matter for this argument. And we are now going to talk about *expected* values, instead of realized ones: averages over the possible maps, denoted $\langle\,\cdot\,\rangle$. We must assume our prior is *a priori* correct: if we had better assumptions, we would have put them in the prior. In other words, we model nature to draw the truth from the very distribution Werner believes. Therefore, the average over ψ simply zips with the weights: $\langle X\rangle = \sum_j p_j\, X(\psi{=}\phi_j)$.

How much surprisal should Werner expect from asking a question q? The answer comes back a exactly when the truth answers a, which under the zipped average happens with probability $\sum_{j:\phi_j(q)=a} p_j = M_{a,q}$ - the very probability Werner assigned to it - and when it does, it costs him $-\log_2 M_{a,q}$ of surprisal. Grouping the average over maps by the answer each map gives:

$$\langle s_q\rangle \ = \ \sum_j p_j\,\big(-\log_2 M_{\phi_j(q),\,q}\big) \ = \ \sum_a \underbrace{\sum_{j\,:\,\phi_j(q)=a} p_j}_{M_{a,q}}\,\big(-\log_2 M_{a,q}\big) \ = \ -\sum_a M_{a,q}\,\big(\log_2 M_{a,q}\big). \tag{14}$$

The right-hand side is exactly the column entropy $H(M_{\cdot,q})$ of equation (13): the expected surprisal of a question is the entropy of its column.

Each answer occurs with exactly the probability Werner assigned to it, so his average surprisal is his own uncertainty about that answer. This identity will do a great deal of work in what follows.

For a two-answer column, this entropy is a function of a single number, the probability x of one of the two answers: the binary entropy function [2]:

$$h_2(x) \ := \ -x\log_2 x \ - \ (1-x)\log_2(1-x), \tag{15}$$

zero at $x \in \{0,1\}$ (a settled answer) and maximal, 1 bit, at the fair coin $x = \frac{1}{2}$. For the first question here the column is a fair coin, so

$$\langle s_1 \rangle \;=\; H\big(M_{\cdot,0}\big) \;=\; h\big(\tfrac{1}{2}\big) \;=\; 1 \text{ bit}, \tag{16}$$

whatever the value of p.

And the other column is now cleaned up. Breaking the update down over the possible maps: with probability $\frac{1}{2}$ the answer is 1 (truth is NOT or constant 1), with probability $\frac{1}{2}$ it is 0 (truth is FALSE or the identity), and

$$M' \;=\; \underbrace{\begin{array}{c|cc} \mathbf{a}\backslash\mathbf{q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & 1-p & 0 \\ \mathbf{1} & p & 1 \end{array}}_{a_1=1,\ \text{prob } 1/2} \quad \text{or} \quad \underbrace{\begin{array}{c|cc} \mathbf{a}\backslash\mathbf{q} & \mathbf{1} & \mathbf{0} \\ \hline \mathbf{0} & p & 1 \\ \mathbf{1} & 1-p & 0 \end{array}}_{a_1=0,\ \text{prob } 1/2} . \tag{17}$$

Either way, the asked column is one-hot and the unasked column is a coin of bias p: what survives on $q = 1$ is precisely the class uncertainty "constant or not". The total entropy of the table is now

$$H(M') \;=\; 0 + h_2(p) \;=\; h_2(p), \tag{18}$$

a reduction of $2 - h_2(p)$ bits, purchased with a single bit of surprisal. Define the **leverage** after one question as the ratio of expected table entropy destroyed to expected surprisal received:

$$\mathcal{L}_1 \;:=\; \frac{2 - h_2(p)}{1} \;=\; 2 - h_2(p) \;\in\; [1,2]. \tag{19}$$

The expected surprisal on the second question is, once more, the entropy of its column,

$$\langle s_2 \rangle \;=\; h_2(p) \;\le\; 1 \text{ bit}, \tag{20}$$

after which the table is empty. Cumulatively over the two questions, $1 + h_2(p)$ expected bits of received information reduced the table by the full 2 bits it started with, so the **cumulative leverage** after two questions is

$$\mathcal{L}_2 \;:=\; \frac{2}{1 + h_2(p)} \;\in\; [1,2]. \tag{21}$$

Our knowledge about the world has allowed us to be more certain after our inference. Set $p = \frac{1}{2}$, and the prior is the uniform one of Section 1.1: $h_2 = 1$, $\mathcal{L}_1 = \mathcal{L}_2 = 1$, and nothing is leveraged at all. As p skews further from $\frac{1}{2}$, the first question unlocks more. The second question carries less and less expected information, and Werner's potential for learning from it goes down with it. The column it could clean up is nearly settled already. In the limit $p = 0$, we recover the non-constant world from the start of Section 1.2. There is no surprisal and no improvement to M left in the second question at all: it contributes nothing to denominator or numerator, and $\mathcal{L}_1 = \mathcal{L}_2$. Figure 2 traces all three curves between these extremes.

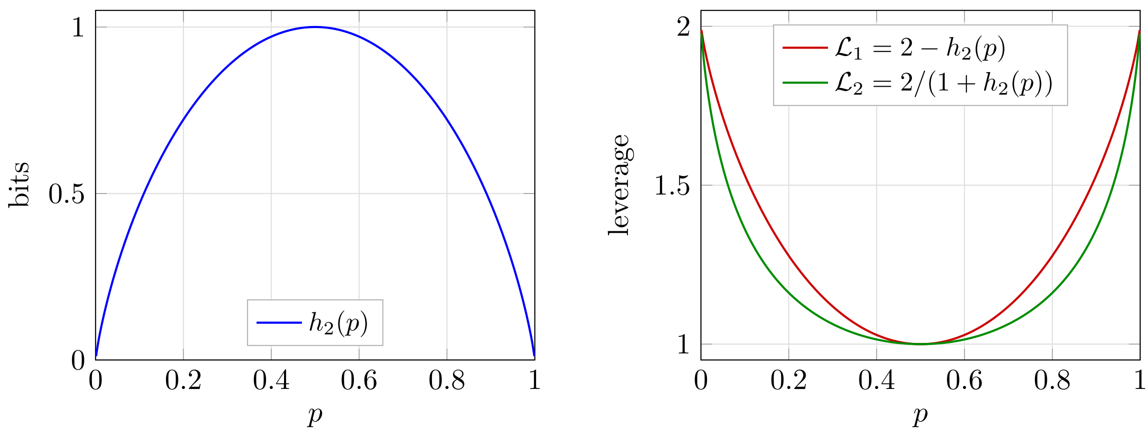


Figure 2: The one-parameter world as a function of the constant-map probability p. Left: the table entropy $h_2(p)$ remaining after one question, in bits. Right: the one-question leverage $\mathcal{L}_1$ and the cumulative leverage $\mathcal{L}_2$ after both questions. At $p = \frac{1}{2}$ (the uniform prior) nothing is leveraged; toward the deterministic-class extremes the first answer doubles its value.

In general, in a larger world, if there are d questions to disagree on between the two maps, the reader can convince themselves that the limit as $p \to 1$ is $\mathcal{L}_1 = d$.

2 The general case

Now let us give the definitions in full generality. Let questions be bit strings of length n, and answers bit strings of length m:

$$q \in \mathcal{Q} = \{0,1\}^n \equiv \mathbb{F}_2^{\otimes n}, \qquad a \in \mathcal{A} = \{0,1\}^m \equiv \mathbb{F}_2^{\otimes m}, \tag{22}$$

where $\mathbb{F}_2$ is the binary field. We write the sizes of those two sets without bars,

$$Q \ := \ |\mathcal{Q}| \ = \ 2^n, \qquad A \ := \ |\mathcal{A}| \ = \ 2^m, \tag{23}$$

so there are Q distinct questions and A possible answers to each. As before, we read bit strings as the numbers they spell, using the two interchangeably: $q \in \{0, \dots, Q-1\}$ and $a \in \{0, \dots, A-1\}$. The truth $\psi : \mathcal{Q} \to \mathcal{A}$ is one of the candidate maps ϕ_j; since each of the Q inputs can be sent to any of the A outputs independently, there are

$$N \ = \ A^Q \ = \ 2^{mQ} \tag{24}$$

of them, and the index j is again the truth table read as a numeral, now with Q digits in base A: $j = \sum_q \phi_j(q)\, A^q$, running from the constant-0 map ($j = 0$) to the constant-($A-1$) map ($j = N-1$). Werner's belief is a distribution over all of them, $\sum_{j=0}^{N-1} p_j = 1$.

EXAMPLE 7

The example we will use to guide the general case is $n = 2$, $m = 1$, which has $N = 2^{1 \cdot 4} = 16$ maps: every boolean function of the two input bits b_1, b_0, with q represented as $b_1 b_0$, or read as an ordinary binary number, $q = 2b_1 + b_0$. Each ϕ_j in Table 4 is shown as its truth table, printed in descending question order $(11, 10, 01, 00)$ so that the answer to question q occupies the digit of weight 2^q. The printed string is exactly j in binary (e.g. AND, which answers 1 only to $q = 11$, reads 1000, so it is $j = 8$). Alongside we conjure an arbitrary prior p_j for illustration.

	$\phi_j(q)$					
j	11	10	01	00	name	p_j
0	0	0	0	0	FALSE (constant 0)	0.394
1	0	0	0	1	$\neg(b_1 \vee b_0)$ (NOR)	0.010
2	0	0	1	0	$\neg b_1 \wedge b_0$	0.010
3	0	0	1	1	$\neg b_1$	0.010
4	0	1	0	0	$b_1 \wedge \neg b_0$	0.131
5	0	1	0	1	$\neg b_0$	0.010
6	0	1	1	0	$b_1 \oplus b_0$ (XOR)	0.010
7	0	1	1	1	$\neg(b_1 \wedge b_0)$ (NAND)	0.010
8	1	0	0	0	$b_1 \wedge b_0$ (AND)	0.213
9	1	0	0	1	$b_1 = b_0$ (XNOR)	0.092
10	1	0	1	0	b_0 (projection)	0.040
11	1	0	1	1	$b_1 \to b_0$ (IF THEN)	0.010
12	1	1	0	0	b_1 (projection)	0.010
13	1	1	0	1	$b_0 \to b_1$ (IF THEN)	0.030
14	1	1	1	0	$b_1 \vee b_0$ (OR)	0.010
15	1	1	1	1	TRUE (constant 1)	0.010

Table 4: The guiding example: all 16 maps of the $n = 2$, $m = 1$ hypothesis space, with a contrived prior ($\sum_j p_j = 1.000$).

2.1 The answer matrix and the update rule

The answer matrix is defined by the same rule as before: the total belief in the maps that would answer a to q:

$$M_{a,q} \;=\; \sum_{j\,:\,\phi_j(q)=a} p_j, \tag{25}$$

now an $A \times Q$ matrix, still with each column a probability distribution over the possible answers: column-stochastic.

It is worth taking a moment to describe the constitution of this matrix. Because of the enumeration of j (the index *is* the truth table) the answer matrix has a natural structure: it is the contraction of the belief vector p with a fixed tensor with $\{0,1\}$ entries E, summed along the map index,

$$M_{a,q} \;=\; \sum_{j=0}^{N-1} E_{a,q,j}\, p_j, \qquad E_{a,q,j} \;:=\; \delta\big(\phi_j(q),\, a\big) \;=\; \delta\Big(\big\lfloor j \cdot 2^{-qm} \big\rfloor \bmod A,\; a\Big). \tag{26}$$

The second form is pure arithmetic on the index: by the numeral convention, $\phi_j(q)$ is nothing but the q-th digit of j in base A. For $m = 1$ this reduces to: bit q of j equals a. So we never need to store E: we read membership of j in the entry $M_{a,q}$ directly off the digits of j. Just as well: E is astronomically large for even moderate m and n. The tensor E is moreover sparse and perfectly regular: each question-slice (q fixed) holds exactly N nonzero entries: every map answers exactly one a, and the maps divide evenly over the A answer rows, N/A apiece. Within a slice the nonzero row follows the odometer pattern of a numeral system: runs of A^q consecutive j share a digit, cycling through all A rows with period A^{q+1}.

The odometer pattern gives every entry of M the same compact closed form: for $m = 1$, bit q of j equals 0 exactly when $j \bmod 2^{q+1} < 2^q$, and equals 1 otherwise.

EXAMPLE 8

Each entry of M collects the eight maps from two to one bit of Table 4 whose truth tables carry the right digit. Filling in the entries - note the period halves column by column. The outer columns are written in their simplified forms, $j \bmod 16 < 8$ being simply $j < 8$ and $j \bmod 2 < 1$ simply "j even".

$$M \;=\; \begin{array}{c|cccc} {}_{\mathbf{a}}\backslash^{\mathbf{q}} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\ \hline \mathbf{0} & \sum_{j<8} p_j & \sum_{j \bmod 8<4} p_j & \sum_{j \bmod 4<2} p_j & \sum_{j\ \text{even}} p_j \\ \mathbf{1} & \sum_{j\geq 8} p_j & \sum_{j \bmod 8\geq 4} p_j & \sum_{j \bmod 4\geq 2} p_j & \sum_{j\ \text{odd}} p_j \end{array}$$
$$=\; \begin{array}{c|cccc} {}_{\mathbf{a}}\backslash^{\mathbf{q}} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\ \hline \mathbf{0} & 0.585 & 0.779 & 0.890 & 0.818 \\ \mathbf{1} & 0.415 & 0.221 & 0.110 & 0.182 \end{array}\,. \tag{27}$$

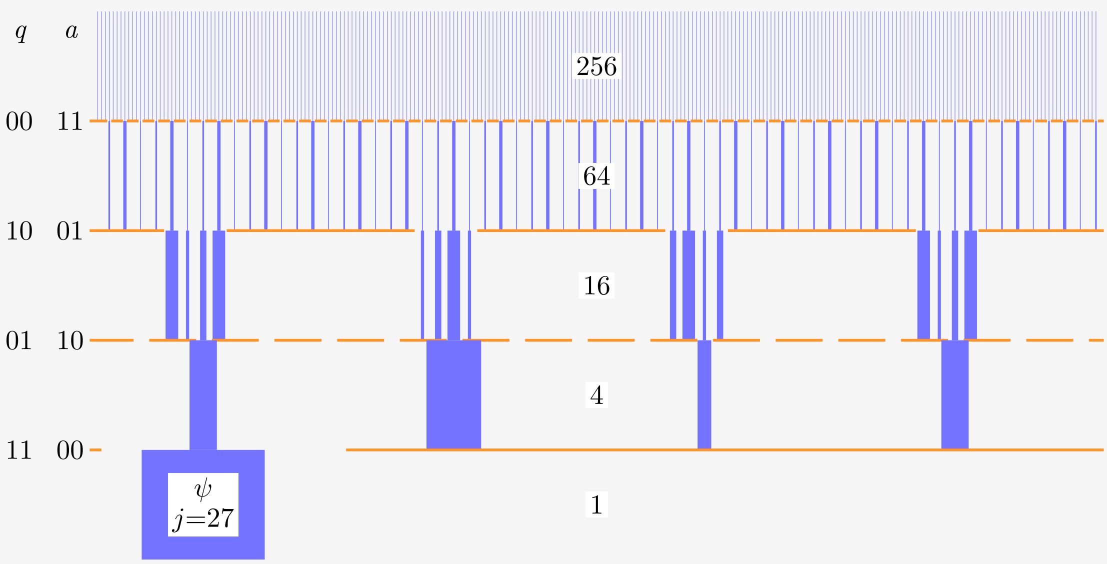
The update rule also carries over verbatim. When question q is asked and the answer $a = \psi(q)$ comes back, the process of elimination reads, in one formula,

$$p'_j \;=\; \frac{\delta\big(\phi_j(q),\, a\big)\; p_j}{M_{a,q}} \;=\; \frac{E_{a,q,j}\; p_j}{M_{a,q}}. \tag{28}$$

The Kronecker delta performs the elimination. Only maps that agree with $\psi(q) = a$ survive and the division renormalizes the survivors, whose total prior mass is exactly $M_{a,q}$ by equation (25). Note that one and the same number prices the answer and funds the update: the event arrives with the probability $M_{a,q}$, costs Werner $-\log_2 M_{a,q}$ bits of surprisal, and its belief mass becomes the normalization. Every update shrinks the support by A^{-1}, radically thinning, especially for large m. The survivors live on a block-wave in j-space. The map determines the offset and frequency. The total set of survivors depends on the set of asked questions, those that fell through the slits, not the order: updating commutes. This is illustrated in the example below.

EXAMPLE 9

The figure below makes the process visual for $n = m = 2$: a schematic prior $\{p_j\}$ over $N = 4^4 = 256$ maps, drawn as vertical lines with thickness representing weight. Each answer stops 3 out of every $A = 4$ surviving lines: maps whose truth tables carry a wrong digit at the asked question. Renormalization inflates the weight of the rest, so the total probability is always unit. Four questions cut $256 \to 64 \to 16 \to 4 \to 1$, and the last line standing is the truth. Note the regularity of the cull: each answer keeps a single residue class of j, so the cadence of the survivors is regular.



The update rule as a cull, for $n = m = 2$. All 256 candidate maps start as vertical lines under a schematic prior. The truth is the bitwise negation $\psi(q) = \neg q$ ($j = 27$, "0123" base 4), and the four questions are asked in the order $00, 10, 01, 11$ ("0, 2, 1, 3" base 4). Each answer terminates the 3/4 of surviving maps whose truth-table digit at the asked question disagrees. Every survivor's thickness, proportional to its weight, is rescaled by the renormalization factor of that update, preserving their ratios. After four questions a single map remains. Interestingly, the first cull's survivors are evenly spaced because it's the "smallest" question. After the three smallest q, even spacing returns.

What an answer does to a learner admits more than one measure. Baldi and Itti score a such a datapoint by the relative entropy between the prior and the posterior it induces, rather than by how unexpected the datum was [4]. Under the paradigm of this work, the elimination of exact maps, the two coincide. Likelihood is either 0 or 1. The posterior is the prior restricted to the survivors, each of them rescaled by the same $1/M_{a,q}$, so the relative entropy of the posterior from the prior is $-\log_2 M_{a,q}$. Surprisal is the measure of information received.

2.2 Entropy bookkeeping

The extensive predictive entropy of the table adds the total logarithmic prediction risks of its columns, exactly as in equation (13):

$$H(M) \;=\; \sum_{q\in\mathcal{Q}} H\big(M_{\cdot,q}\big) \;=\; -\sum_{q\in\mathcal{Q}}\sum_{a\in\mathcal{A}} M_{a,q}\log_2 M_{a,q} \;\;\le\;\; mQ, \tag{29}$$

each column carrying at most m bits (a uniform spread over its A answers). The ceiling mQ is the size of the entire truth table in bits, and it is attained precisely (but not exclusively) by the uniform prior: maximal ignorance prices every answer at full cost.

The belief itself carries an entropy as well, and we give its gap to the table a name:

$$H(p) \;=\; -\sum_{j=0}^{N-1} p_j \log_2 p_j \le mQ, \tag{30}$$

$$C \;:=\; H(M) - H(p) \;\ge\; 0. \tag{31}$$

$H(p)$, Werner's uncertainty about the *identity* of the truth, obeys the same ceiling: a distribution over N maps has entropy at most $\log_2 N = mQ$, attained again by the uniform prior. The information in the prior is not directly useful for prediction, but it is good to track.

Why is $C > 0$? The answers jointly determine the map, so $H(p)$ is the *joint* entropy of all Q answers, while $H(M)$ sums their *marginal* entropies and a joint entropy never exceeds the sum of its marginals. Hence $H(p) \le H(M)$, and the gap C, the **total correlation** of the answers [3], is nonnegative: knowledge stored in correlations *between* answers, invisible to every single column of M.

The same scoring interpretation holds through observation updates. Let $\mathcal{S}$ be the questions already asked and $y = \psi(\mathcal{S})$ the string of answers seen. The current table has $M_{a',q'} = \Pr(\psi(q') = a' \mid \psi(\mathcal{S}) = y)$. Its expected total logarithmic loss is

$$\mathbb{E}\left[\sum_{q'\in\mathcal{Q}} -\log_2 M_{\psi(q'),q'} \;\middle|\; \psi(\mathcal{S}) = y\right] = \sum_{q'\in\mathcal{Q}} H\big(\psi(q') \mid \psi(\mathcal{S}) = y\big) = H(M). \tag{32}$$

Already answered columns contribute zero. Thus $H(M)$ measures the remaining risk of separate predictions, while $H(p)$ measures the remaining uncertainty about the map as a whole. Notably, *in the independent case*, of which the ignorant case is an example,

1. $H(M)$ is the total amount of information we must collect from questions to settle the map.

2. $H(M) = H(p)$, so the entropy equals the risk, without correlations.

This equality pins the scaling. How these quantities respond to an *intelligent* prior is the main question of this work.

Their agreement in the uniform-prior case is a consequence of independence; correlations are what allow one answer to improve several predictions.

2.3 Playing the guiding example

Let us watch these ledgers in motion by playing the guiding example through all four questions. Werner can compute two forecasts for every unasked question, in advance, from his current belief: the expected surprisal $\langle s \rangle$, which by equation (14) is the entropy of that question's column, and the expected drop $\langle \Delta H(M) \rangle$ of the *whole* table if that question is asked next. This is the average over possible a, weighted by their probability (according to M):

$$\langle s \rangle(q) \;=\; \sum_a M_{a,q} \left(-\log_2 M_{a,q}\right) = H(M_{\cdot,q}), \tag{33}$$

$$\langle \Delta H(M) \rangle(q) \;=\; \sum_a M_{a,q} \Big[H(M) - H\big(M^{(q,a)}\big) \Big], \tag{34}$$

where $M^{(q,a)}$ is the table after the update rule (28) processes answer a to question q. We know the entropy of column q will drop to zero, ditching $\langle s \rangle(q)$ entropy from M. But other columns will also respond. The excess of the second over the first is the expected transfer to unasked columns, and it can only be paid out of correlations.

This transfer has a direct expression in mutual information. Work at the current posterior, after the already-seen string $y = \psi(\mathcal{S})$, and consider asking $q \notin \mathcal{S}$ next, with answer $a = \psi(q)$. For any other question q', averaging its entropy drop over the possible a gives

$$\begin{aligned} &H\big(\psi(q') \mid \psi(\mathcal{S}) = y\big) - \sum_a M_{a,q}\, H\big(\psi(q') \mid \psi(\mathcal{S}) = y, \psi(q) = a\big) \\ &\qquad = I\big(\psi(q'); \psi(q) \mid \psi(\mathcal{S}) = y\big). \end{aligned} \tag{35}$$

The asked column loses its whole entropy, so summing gives

$$\langle \Delta H(M) \rangle(q) = H(M_{\cdot,q}) + \sum_{q' \neq q} I\big(\psi(q'); \psi(q) \mid \psi(\mathcal{S}) = y\big). \tag{36}$$

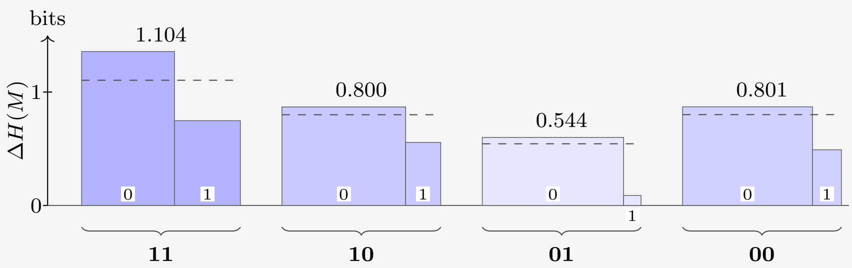
The first term is the expected surprisal received; each term in the sum is the expected improvement on another question. Their sum is the expected withdrawal from the correlation store. The surprisal forecast needs M alone, but the improvement forecast also needs the pairwise joint laws of the asked answer and the other answers. Two priors with the same M can therefore give different forecasts of improvement. We append both forecasts as extra rows under M.

EXAMPLE 10

Werner's belief is the prior of Table 4 as before. He plays greedily: always asking the question with the largest forecast reduction in matrix entropy. The truth is AND ($\psi = \phi_8$).

Question 1. The starting state:

a \ q	**11**	**10**	**01**	**00**
0	0.585	0.779	0.890	0.818
1	0.415	0.221	0.110	0.182
$\langle s \rangle$	0.979	0.762	0.500	0.684
$\langle \Delta H(M) \rangle$	1.104	0.800	0.544	0.801



To see where the forecast entries come from, take $q = 11$: the answer $a = 0$ arrives with probability $M_{0,11} = 0.585$ and would drop $H(M)$ from 2.925 to 1.569; the answer $a = 1$ arrives with probability 0.415 and drops it only to 2.178, so

$$\langle s \rangle(11) = 0.585\,(0.773) + 0.415\,(1.269) = 0.979,$$
$$\langle \Delta H(M) \rangle(11) = 0.585\,(1.357) + 0.415\,(0.748) = 1.104.$$

EXAMPLE 11

Both forecast rows point to $q = 11$: it is the closest thing to a coin on offer ($\langle s \rangle = 0.979$) and moves the whole table most, by a wide margin. Part of $\Delta H(M)$ is just $\langle s \rangle$ itself. Werner asks it, and nature answers $a = 1$ (probability 0.415, surprisal $s_1 = 1.269$). Eight maps die:

j	$\phi_j(11)$	name	p_j	p'_j
~~0~~	~~0~~	~~FALSE (constant 0)~~	~~0.394~~	~~-~~
~~1~~	~~0~~	~~$\neg(b_1 \vee b_0)$ (NOR)~~	~~0.010~~	~~-~~
~~2~~	~~0~~	~~$\neg b_1 \wedge b_0$~~	~~0.010~~	~~-~~
~~3~~	~~0~~	~~$\neg b_1$~~	~~0.010~~	~~-~~
~~4~~	~~0~~	~~$b_1 \wedge \neg b_0$~~	~~0.131~~	~~-~~
~~5~~	~~0~~	~~$\neg b_0$~~	~~0.010~~	~~-~~
~~6~~	~~0~~	~~$b_1 \oplus b_0$ (XOR)~~	~~0.010~~	~~-~~
~~7~~	~~0~~	~~$\neg(b_1 \wedge b_0)$ (NAND)~~	~~0.010~~	~~-~~
8	1	$b_1 \wedge b_0$ (AND)	0.213	0.513
9	1	$b_1 = b_0$ (XNOR)	0.092	0.222
10	1	b_0 (projection)	0.040	0.096
11	1	$b_1 \to b_0$ (IF THEN)	0.010	0.024
12	1	b_1 (projection)	0.010	0.024
13	1	$b_0 \to b_1$ (IF THEN)	0.030	0.072
14	1	$b_1 \vee b_0$ (OR)	0.010	0.024
15	1	TRUE (constant 1)	0.010	0.024

The ledger:

$$H(M)\colon 2.925 \to 2.178, \qquad H(p)\colon 2.707 \to 2.093, \qquad C\colon 0.218 \to 0.085.$$

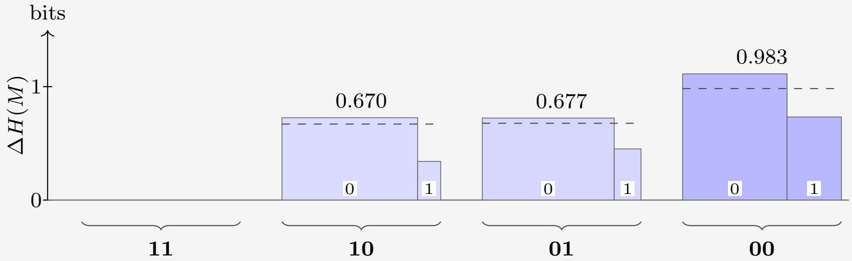
Werner received 1.269 bits but destroyed only 0.748 bits of table. Correlations point the right way only on average, not on every draw. In this case, Werner got the much less informative branch, and paid the premium price for it. The destroyed-per-received ratio of this round is $L_1 = 0.589$: well below 1, anti-leverage.

Question 1 is a reminder that the forecast rows are *expectations*: they hold on average over many repetitions of the experiment, and only if the world really does draw its truth ψ from Werner's prior. On any single run the realized numbers are free to deviate.

EXAMPLE 12

Question 2. The state after the collapse, with fresh forecasts:

a \ q	11	10	01	00
0	0	0.855	0.831	0.658
1	1	0.145	0.169	0.342
$\langle s \rangle$	—	0.596	0.655	0.927
$\langle \Delta H(M) \rangle$	—	0.670	0.677	0.983



Greedy Werner picks $q = 00$. Nature answers $a = 0$, which is the answer Werner favored (probability 0.658), so the surprisal is only $s_2 = 0.604$ bits. Four of the eight die:

j	$\phi_j(00)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.513	0.780
~~9~~	~~1~~	~~$b_1 = b_0$ (XNOR)~~	~~0.222~~	~~-~~
10	0	b_0 (projection)	0.096	0.147
~~11~~	~~1~~	~~$b_1 \to b_0$ (IF THEN)~~	~~0.024~~	~~-~~
12	0	b_1 (projection)	0.024	0.037
~~13~~	~~1~~	~~$b_0 \to b_1$ (IF THEN)~~	~~0.072~~	~~-~~
14	0	$b_1 \vee b_0$ (OR)	0.024	0.037
~~15~~	~~1~~	~~TRUE (constant 1)~~	~~0.024~~	~~-~~

The ledger:

$$H(M)\colon 2.178 \to 1.065, \qquad H(p)\colon 2.093 \to 1.035, \qquad C\colon 0.085 \to 0.030.$$

This time 1.113 bits are destroyed for 0.604 received. Where did the extra 0.509 come from? Partly from the favored answer arriving - the belief dropped 1.058 against only 0.604 paid - and partly from the store: C fell by 0.055. Step ratio 1.842. Cumulatively, however, 1.860 destroyed per 1.873 received gives $L_2 = 0.993$: still a hair below 1. Even a strong second round has not quite repaid the expensive shock of question 1.

Why track C, when its step-by-step changes visibly do not match the amounts deduced? Because on a single realized run the deduced part has *two* sources, and C is the only quantity that separates them. Writing Δ for the realized drops in one step,

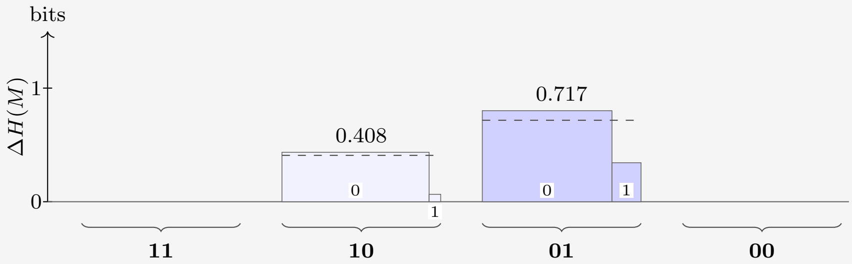
$$\underbrace{\Delta H(M) - s}_{\text{deduced}} = \underbrace{\big[\Delta H(p) - s\big]}_{\text{windfall}} + \underbrace{\big[-\Delta C\big]}_{\text{tower}}, \tag{37}$$

an identity, since $H(M) = H(p) + C$ at every stage. The windfall is luck: the realized surprisal undershooting (or overshooting) the realized drop in belief entropy. Windfall is zero in expectation. The tower term is structure: a genuine withdrawal from the correlation store, and the channel with a global guarantee. Over a complete run its withdrawals sum to exactly the initial C, while the windfalls average away.

EXAMPLE 13

Question 3. Two questions remain:

a \ q	**11**	**10**	**01**	**00**
0	0	0.927	0.817	1
1	1	0.073	0.183	0
$\langle s \rangle$	—	0.378	0.687	—
$\langle \Delta H(M) \rangle$	—	0.408	0.717	—



Greedy Werner picks $q = 01$; nature answers $a = 0$ (probability 0.817, surprisal $s_3 = 0.292$). Two die:

j	$\phi_j(01)$	name	p_j	p'_j
8	0	$b_1 \wedge b_0$ (AND)	0.780	0.955
~~10~~	~~1~~	~~b_0 (projection)~~	~~0.147~~	~~-~~
12	0	b_1 (projection)	0.037	0.045
~~14~~	~~1~~	~~$b_1 \vee b_0$ (OR)~~	~~0.037~~	~~-~~

The ledger:

$$H(M)\colon 1.065 \to 0.264, \qquad H(p)\colon 1.035 \to 0.264, \qquad C\colon 0.030 \to 0.$$

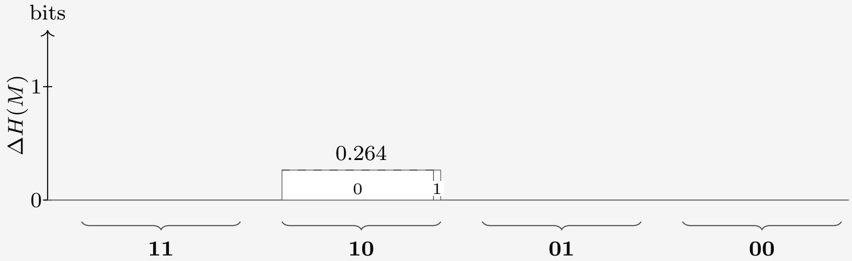
Again a bargain: 0.801 destroyed for a mere 0.292 received, and $L_3 = 1.229$ finally clears unity. And note that C has dropped to zero.

C measures correlations that are not visible from M. If there's only one unknown column, there's nothing to infer. Put another way, usually, the binary maps are an overcomplete basis for M, because their number $N \gg \dim(M) = Q(A-1)$. Expressing M as a linear combination of maps drops knowledge. That's not the case once learning has reduced the number of free parameters to one each.

EXAMPLE 14

Question 4. One question, one heavily loaded coin:

a \ q	**11**	**10**	**01**	**00**
0	0	0.955	1	1
1	1	0.045	0	0
$\langle s \rangle$	—	0.264	—	—
$\langle \Delta H(M) \rangle$	—	0.264	—	—



The two forecasts now coincide: with $C = 0$ there is no transfer left to hope for, and the last question can only be observed, not leveraged. Nature answers $a = 0$ (probability 0.955, surprisal $s_4 = 0.066$):

j	$\phi_j(10)$	name	p_j	p'_j	
8	0	$b_1 \wedge b_0$ (AND)	0.955	1.000	
~~12~~	~~1~~	~~b_1 (projection)~~	~~0.045~~	~~-~~	

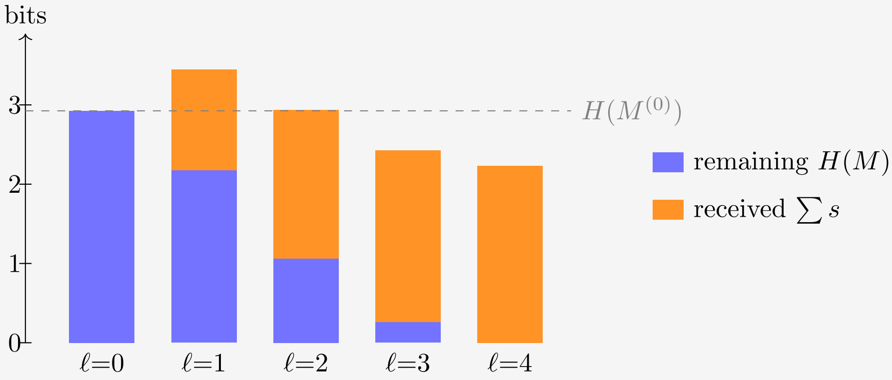
The ledger closes:

$$H(M)\colon 0.264 \to 0, \qquad H(p)\colon 0.264 \to 0, \qquad C\colon 0 \to 0.$$

The truth is AND. The step destroyed 0.264 for 0.066 received (ratio 3.990) - not deduction this time, but luck: the destroyed amount was fixed in advance (the column's full entropy), while the realized surprisal undershot it because the likelier answer arrived. In the split of equation (37): pure windfall, tower 0.

EXAMPLE 15

Cumulatively: 2.925 bits of table destroyed, all of it, for 2.231 bits of surprisal received: $L_4 = 1.311$. Most of that was luck, not intelligence. The figure below stacks the bits after each question: what uncertainty remains, plus what was received. Once it dips below the starting entropy, our prior has done net work.



The realized run through the guiding example: remaining table entropy plus cumulative surprisal received, after each question, against the initial total (dashed). The gap between each stack and the line is the cumulative deduced part. At $\ell = 1$ the stack tops out 0.521 bits *above* the line - the negative deduction of question 1, plainly visible. At $\ell = 2$ the stack still grazes the line from above (0.013 bits); only from $\ell = 3$ on do the stacks fall short of it, as genuine deduction overtakes the early loss.

2.4 Leverage

If the prior is good, meaning it actually models reality (the prevalence of maps), we expect graphs such as the one in Example 15 to drop as questions come in. The ratio of table entropy destroyed per bit of surprisal received deserves a name.

The **leverage**. Run the protocol for ℓ rounds: questions $q_1, \dots, q_\ell$ are asked, the answers $a_k = \psi(q_k)$ are each processed by the update rule (28), producing the tables $M^{(1)}, \dots, M^{(\ell)}$, and round k costs the surprisal $s_k = -\log_2 M^{(k-1)}_{a_k,q_k}$. The cumulative leverage after ℓ rounds is the table entropy destroyed per bit of surprisal received:

$$L_\ell \;:=\; \frac{H\big(M^{(0)}\big) - H\big(M^{(\ell)}\big)}{\sum_{k=1}^{\ell} s_k}\,. \tag{38}$$

A leverage of 1 is the uniform-prior baseline: every bit of table entropy destroyed was paid for at face value, one bit of surprisal each. On an individual run, leverage above 1 can also come from windfall, as the guiding example showed. The aggregate quantity $\mathcal{L}_\ell$ defined in Chapter 3 removes that windfall in expectation: anything above 1 in this $\mathcal{L}_\ell$ is deduction.

2.5 Correlators

This section can be skipped on a first reading: the later chapters do not depend on it.

What kind of operator is the update rule (28)? In map space it is as simple as an operator gets: multiplication by a diagonal 0/1 mask, then renormalization. But there is a change of basis in which the same operation becomes a *convolution* with a two-term kernel, and in which the correlation store C stops being a single number and becomes visible coordinate by coordinate. The main advantage is that the effect of each update is well tabulated beforehand into a small number of targeted parameters.

This section builds that basis for $m = 1$; the final paragraph deals with general m.

First, re-encode each answer as a spin,

$$\sigma_q(\phi_j) \;:=\; (-1)^{\phi_j(q)} \;\in\; \{+1, -1\}, \tag{39}$$

so answer $0 \mapsto +1$ and $1 \mapsto -1$. A pure relabeling, but it buys the one algebraic fact everything below turns on: $\sigma_q^2 = 1$.

The **Walsh–Hadamard transform** is built from a single matrix. For one bit,

$$W \;=\; \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad W_{c,b} \;=\; (-1)^{c\,b}, \quad c, b \in \{0, 1\}. \tag{40}$$

EXAMPLE 16

Acting on a one-bit distribution,

$$W \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \;=\; \begin{pmatrix} p_0 + p_1 \\ p_0 - p_1 \end{pmatrix} \;=\; \begin{pmatrix} 1 \\ \langle \sigma \rangle \end{pmatrix} \;:$$

row $c = 0$ computes the normalization, row $c = 1$ the spin bias. Distribution in, correlators out.

The normalization riding along is a redundancy. The original distribution also carries one variable more than its degrees of freedom.

EXAMPLE 17

Tensoring two copies (the $n = 1$ case) does the same for two bits at once:

$$W \otimes W \;=\; \begin{pmatrix} W & W \\ W & -W \end{pmatrix} \;=\; \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad \begin{array}{l} \leftarrow c = 00\colon\; \text{sums to } 1 \\ \leftarrow c = 01\colon\; \langle \sigma_0 \rangle \\ \leftarrow c = 10\colon\; \langle \sigma_1 \rangle \\ \leftarrow c = 11\colon\; \langle \sigma_1 \sigma_0 \rangle \end{array} \tag{41}$$

The Kronecker product multiplies signs, so for any number l of tensor factors the entry at row c, column j is

$$\left(W^{\otimes l}\right)_{c,j} \;=\; \prod_{i=0}^{l-1} (-1)^{c_i\, j_i} \;=\; (-1)^{\sum_i c_i\, j_i}, \tag{42}$$

with c_i, j_i the i-th bits of the row and column indices; every further tensor factor appends one more bit.

For $m = 1$ a map *is* its truth table of Q bits (the numeral j) so Werner's belief is a vector of length $N = 2^Q$, and one transform computes every correlator at once:

$$\chi \;=\; W^{\otimes Q} p, \qquad \chi_{\mathcal{S}} \;=\; \Big\langle \prod_{q \in \mathcal{S}} \sigma_q \Big\rangle \;=\; \sum_{j=0}^{N-1} (-1)^{\sum_{q \in \mathcal{S}} \phi_j(q)}\, p_j. \tag{43}$$

Each row computes the **correlator** of one subset $\mathcal{S} \subseteq \mathcal{Q}$ of questions. In the subscript we may encode the set as its **mask**: written in the same descending question order as everywhere else, bit q of the subscript records whether $q \in \mathcal{S}$. The row index of the matrix, read as a numeral, is the mask.

EXAMPLE 18

For $n = 2$, the question order is $(11, 10, 01, 00)$ so χ_{0110} is the correlator of $\mathcal{S} = \{10, 01\}$.

the sixteen rows of $W^{\otimes 4}$ probe the sixteen subsets of $Q = \{11, 10, 01, 00\}$. A few entries of the dictionary:

mask	set $\mathcal{S}$	correlator
0000	$\emptyset$	$\chi_{0000} = 1$ (normalization)
0001	$\{00\}$	$\chi_{0001} = \langle \sigma_{00} \rangle$
1000	$\{11\}$	$\chi_{1000} = \langle \sigma_{11} \rangle$
0110	$\{10, 01\}$	$\chi_{0110} = \langle \sigma_{10}\, \sigma_{01} \rangle$
1111	$\{11, 10, 01, 00\}$	$\chi_{1111} = \langle \sigma_{11}\, \sigma_{10}\, \sigma_{01}\, \sigma_{00} \rangle$, the full parity

Sort the N correlators by $|\mathcal{S}|$, the number of questions probed, into **shells**. Shell 0 is the normalization, $\chi_\emptyset = 1$, riding along as a correlator. Shell 1 is M in different clothes:

$$\chi_{\{q\}} \;=\; M_{0,q} - M_{1,q}, \qquad M_{a,q} \;=\; \tfrac{1}{2}\big(1 + (-1)^a\, \chi_{\{q\}}\big), \tag{44}$$

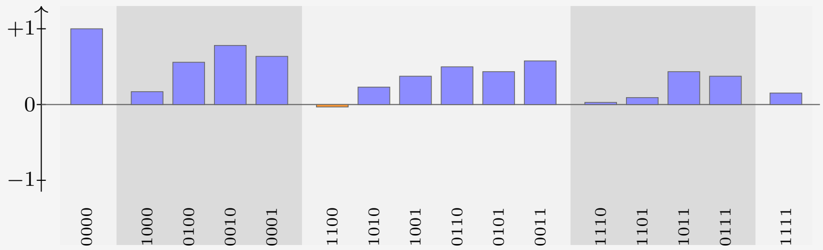
the bias of column q. For $m = 1$, we see that, elegantly, the first shell correlators are the distance from fair coins. A prior invariant under negating every answer simultaneously sends $\chi_{\mathcal{S}} \mapsto (-1)^{|\mathcal{S}|}\chi_{\mathcal{S}}$, so all its odd shells vanish identically, shell 1 among them: such a prior has an agnostic matrix M.

Shell 2 holds the agreement biases of pairs of questions, knowledge M cannot see: two priors can share every column of M and differ here. Higher shells store subtler and subtler parity knowledge. Nothing is lost in the sorting: $W^2 = 2I$, so $(W^{\otimes Q})^2 = NI$. The transform is its own inverse up to a factor N, and knowing every correlator is knowing the belief. The ledger also gives C a signature: a product prior ($C = 0$) factorizes every correlator, $\chi_{\mathcal{S}} = \prod_{q\in\mathcal{S}} \chi_{\{q\}}$, so its upper shells repeat what shell 1 already said; correlated knowledge is precisely *non-factorizing* correlators.

EXAMPLE 19

The prior of Table 4 transforms to the ledger

shell	correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = +0.170$, $\chi_{0100} = +0.558$, $\chi_{0010} = +0.780$, $\chi_{0001} = +0.636$
2	$\chi_{1100} = -0.032$, $\chi_{1010} = +0.230$, $\chi_{1001} = +0.374$,
	$\chi_{0110} = +0.498$, $\chi_{0101} = +0.434$, $\chi_{0011} = +0.576$
3	$\chi_{1110} = +0.028$, $\chi_{1101} = +0.092$, $\chi_{1011} = +0.434$, $\chi_{0111} = +0.374$
4	$\chi_{1111} = +0.152$



Shell 1 reproduces the columns of equation (27): $(1+0.170)/2 = 0.585 = M_{0,11}$, and likewise 0.779, 0.890, 0.818. The negative entry χ_{1100} says questions 11 and 10 disagree slightly more often than they agree; every other pair leans toward agreement. This is a reflection of the strong coherent 0-preference encoded in the prior. A prior leaning toward 1 would give a negative shell 1.

Now look at the *rows* of $W^{\otimes Q}$ as functions of j. They are **block waves**. The observation mask is built of the same material:

$$\delta\big(\phi_j(q), a\big) \;=\; \frac{1}{2}\Big(1 + (-1)^{a+\phi_j(q)}\Big), \tag{45}$$

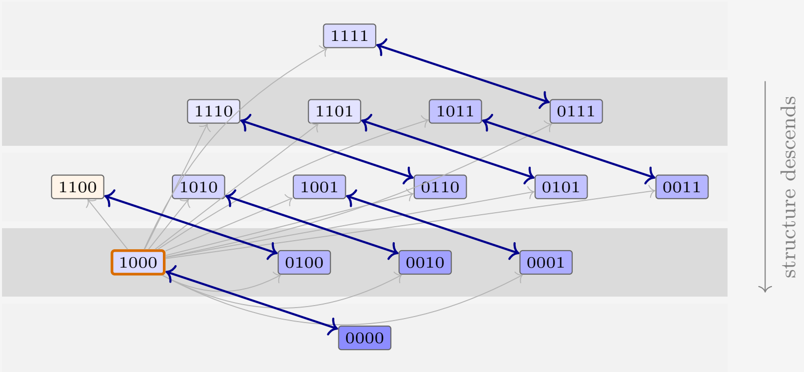
a 0/1 block wave: one constant plus one Hadamard row. And multiplication and convolution swap under this transform exactly as under Fourier: multiplying two functions of j pointwise convolves their ledgers. And the observation mask's ledger has only two nonzero entries - at $\emptyset$ and at $\{q\}$ - so this convolution is no great sum: it combines each correlator with exactly one other.

The update rule, carried out entirely in the ledger, is therefore that two-term combination, followed by renormalization by the new shell 0. In index notation, learning $a = \psi(q)$ sends every set $\mathcal{S}$ to

$$\chi'_{\mathcal{S}} \;=\; \frac{\chi_{\mathcal{S}} + (-1)^a\,\chi_{\mathcal{S}\triangle\{q\}}}{1 + (-1)^a\,\chi_{\{q\}}}, \tag{46}$$

where $\mathcal{S}\triangle\{q\}$ is $\mathcal{S}$ with the membership of q toggled: q is added if absent, removed if present (on masks, bit q is flipped). The numerator is the monomial algebra of $\sigma_q^2 = 1$: multiplying $\prod_{r\in\mathcal{S}}\sigma_r$ by σ_q toggles q's membership of the product, so every correlator is averaged with its **partner**: the correlator differing by q, signed by the answer. The denominator is the $\mathcal{S} = \emptyset$ instance of the numerator, and by equation (44) it equals $2M_{a,q}$: the same number that priced the answer in equation (28) renormalizes the ledger.

EXAMPLE 20



The update rule in the ledger, drawn for the guiding example's first question ($q = 11$). Every set is paired with the partner whose membership of 11 is toggled (arrows); each pair is averaged with the answer's sign and renormalized, per equation (46). The pair $\emptyset \leftrightarrow \{11\}$ at the orange-bordered asked set fixes the normalization and saturates the asked bias; each blue double arrow averages a pair into both of its members, and the net flow of structure is one shell down, toward M. The thin gray arrows from the asked q to every other pill are a reminder that this sets the renormalization. The fill encodes each prior value to scale: darker is larger, orange is negative.

Read formula (46) by cases:

- $\mathcal{S} = \emptyset$: partner $\{q\}$, giving $(1 + (-1)^a \chi_{\{q\}})/(1 + (-1)^a \chi_{\{q\}}) = 1$. Normalization is preserved.

- $\mathcal{S} = \{q\}$, the asked question: partner $\emptyset$, giving $(\chi_{\{q\}} + (-1)^a)/(1 + (-1)^a \chi_{\{q\}}) = (-1)^a$. The bias saturates to certainty - the one-hot collapse of column q, derived rather than decreed.

- $\mathcal{S} = \{q'\}$, an unasked question: the partner is the pair $\{q', q\}$, and one line of algebra puts the shift in terms of the *connected* correlator:

$$\chi'_{\{q'\}} - \chi_{\{q'\}} = \frac{(-1)^a \big(\chi_{\{q',q\}} - \chi_{\{q'\}}\, \chi_{\{q\}}\big)}{2M_{a,q}} = \frac{(-1)^a \operatorname{Cov}\big(\sigma_{q'}, \sigma_q\big)}{2M_{a,q}}. \tag{47}$$

  Unasked columns move in proportion to their covariance with the asked question: knowledge stored one shell up descends into the predictions. A product prior has zero covariance everywhere, so its unasked columns are frozen - the "$C = 0$ means no transfer" fact, now a one-line consequence.

- Higher shells: the same cascade. Each update marries shell-k sets containing q to shell-$(k{-}1)$ sets without it; whatever structure was stored jointly with q moves one shell down, toward M.

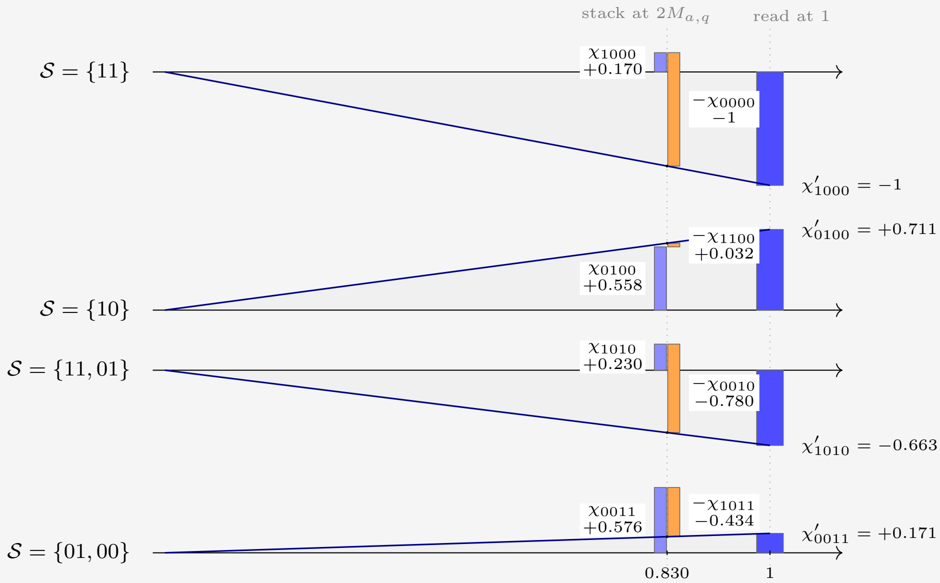
EXAMPLE 21

Question 1 of the walkthrough asked $q = 11$ and received $a = 1$: sign $(-1)^a = -1$, toggled set $\{11\}$, mask 1000. Equation (46) maps the ledger of the previous box to

shell	updated correlators
0	$\chi'_{0000} = 1$
1	$\chi'_{1000} = -\mathbf{1}$, $\chi'_{0100} = +0.711$, $\chi'_{0010} = +0.663$, $\chi'_{0001} = +0.316$
2	$\chi'_{1100} = -0.711$, $\chi'_{1010} = -0.663$, $\chi'_{1001} = -0.316$, $\chi'_{0110} = +0.566$, $\chi'_{0101} = +0.412$, $\chi'_{0011} = +0.171$
3	$\chi'_{1110} = -0.566$, $\chi'_{1101} = -0.412$, $\chi'_{1011} = -0.171$, $\chi'_{0111} = +0.267$
4	$\chi'_{1111} = -0.267$

The asked bias saturates: $\chi'_{1000} = (0.170 - 1)/(1 - 0.170) = -1$. The other singletons move by equation (47): for $q' = 10$ the covariance is $-0.032 - (0.558)(0.170) = -0.127$, so the shift is $(-1)(-0.127)/(2 \times 0.415) = +0.153$, landing on $+0.711$ — and indeed $(1 + 0.711)/2 = 0.855$, the column $M_{0,10}$ of the Question 2 box. Note also that every mask containing the asked bit is now just $(-1)^a$ times its partner ($\chi'_{1010} = -\chi'_{0010}$, etc.): with σ_{11} pinned, no correlator involving question 11 carries independent knowledge anymore.

EXAMPLE 22



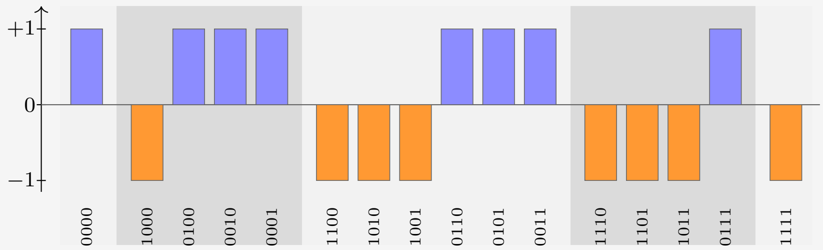
The update rule of equation (46), drawn for the guiding example's first update ($q = 11$, $a = 1$, so $(-1)^a = -1$ and toggled bit 1000). One row per set $\mathcal{S}$. At $x = 2M_{a,q} = 1 + (-1)^a \chi_{1000} = 0.830$ stands the numerator as a waterfall: the bar of $\chi_{\mathcal{S}}$ (blue), and beside it the partner's signed contribution $(-1)^a \chi_{\mathcal{S} \triangle \{q\}}$ (orange), hanging from the level where the first bar ends. The two subscripts differ by the flipped bit. The ray from the origin through the waterfall's endpoint rescales it, by similar triangles, to $x = 1$, where its height is the updated correlator $\chi'_{\mathcal{S}}$.

As Werner moves through the questions, more correlators settle to ± 1. The 1-shell models the belief of the bias of each question. The 2-shell answers questions such as "if $\psi(10) = 0$, then $\psi(01) = 1$". And the higher orders are even more conditional, measuring belief about such statements, in the case of certain answers, etc. As we learn more, the conditions become satisfied or not, and the information becomes less nested.

EXAMPLE 23

The remaining questions of the run (00, 01, 10, all answered 0, so with sign +1) each saturate their own bit the same way, and since XOR toggles commute, the order does not matter - the commutativity of updating, seen again. After all four, the ledger is pure signs, $\chi_{\mathcal{S}} = (-1)^{|\mathcal{S}\cap\{11\}|}$, which is -1 on every set containing 11, $+1$ on the rest: the transform of the point mass on $\psi = \phi_8$:

shell	final correlators
0	$\chi_{0000} = 1$
1	$\chi_{1000} = -1,\ \ \chi_{0100} = +1,\ \ \chi_{0010} = +1,\ \ \chi_{0001} = +1$
2	$\chi_{1100} = -1,\ \ \chi_{1010} = -1,\ \ \chi_{1001} = -1,$ $\chi_{0110} = +1,\ \ \chi_{0101} = +1,\ \ \chi_{0011} = +1$
3	$\chi_{1110} = -1,\ \ \chi_{1101} = -1,\ \ \chi_{1011} = -1,\ \ \chi_{0111} = +1$
4	$\chi_{1111} = -1$



2.5.1 General m

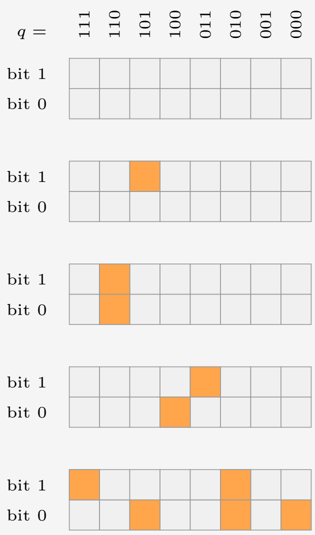
It turns out that taking $m > 1$ is more general but not much more instructive. With multiple output bits it is as if we have m a priori independent maps per question; they just happen always to be asked and answered simultaneously, in unison. Therefore we do not have to think about what one of them teaches us about another. It would mean we have more than 2 options every question, and therefore the factor by which every update thins the hypotheses is not 2 but A. The guiding example helps because the maps are represented as binary numbers, which are more familiar.

For completeness, the main formulas. The truth table is now mQ bits: one **site** (q, i) per bit i of each answer, each carrying a spin $\sigma_{q,i}(\phi_j) := (-1)^{\phi_j(q)_i}$, where $\phi_j(q)_i$ is the i-th bit of the answer. Correlators are indexed by sets $\mathcal{S}$ of sites: masks of mQ bits under the same numeral convention, m of them per question:

$$\chi_{\mathcal{S}} = \Big\langle \prod_{(q,i)\in\mathcal{S}} \sigma_{q,i} \Big\rangle = \sum_{j=0}^{N-1} (-1)^{\sum_{(q,i)\in\mathcal{S}} \phi_j(q)_i}\, p_j. \tag{48}$$

EXAMPLE 24

For $n = 3$, $m = 2$ the sites form a 2×8 grid: one column per question, one row per bit of the answer. A set $\mathcal{S}$ is a pattern of hot cells in this grid, and $\chi_{\mathcal{S}}$ multiplies the corresponding spins:



$\chi_\emptyset = 1$: the normalization

$\langle \sigma_{101,1} \rangle$: the bias of one answer bit

$\langle \sigma_{110,1}\, \sigma_{110,0} \rangle$: the parity of one answer — with the two biases, the block of $M_{\cdot,110}$

$\langle \sigma_{100,0}\, \sigma_{011,1} \rangle$: agreement across two questions

$\langle \sigma_{111,1}\, \sigma_{101,0}\, \sigma_{010,1}\, \sigma_{010,0}\, \sigma_{000,0} \rangle$: a five-site parity, deep in the shells

Any of the $2^{16} = 65536$ hot/cold patterns is a correlator, and sorting by the number of hot cells gives the shells. The mask of $\mathcal{S}$ is the grid read as a 16-bit numeral.

Write $\mathcal{B}_q$ for the block of q's m sites, and $a \cdot \mathcal{T} := \sum_{i \in \mathcal{T}} a_i$ for the parity of the answer a on a subset $\mathcal{T} \subseteq \mathcal{B}_q$. The answer matrix is read off the correlators supported on one block. There are A of them per column:

$$M_{a,q} \;=\; \frac{1}{A} \sum_{\mathcal{T} \subseteq \mathcal{B}_q} (-1)^{a \cdot \mathcal{T}}\, \chi_{\mathcal{T}}. \tag{49}$$

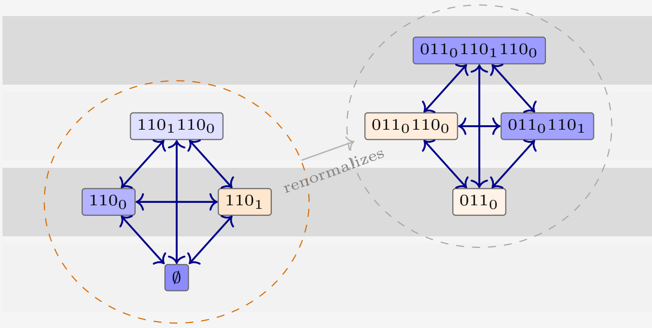
The observation mask at (q, a) is a function of the m bits of one block, so its ledger has A nonzero entries, and hearing $a = \psi(q)$ combines every correlator with its A partners — one per toggle $\mathcal{T}$ within the asked block:

$$\chi'_{\mathcal{S}} \;=\; \frac{\sum_{\mathcal{T} \subseteq \mathcal{B}_q} (-1)^{a \cdot \mathcal{T}}\, \chi_{\mathcal{S} \triangle \mathcal{T}}}{A\, M_{a,q}}, \tag{50}$$

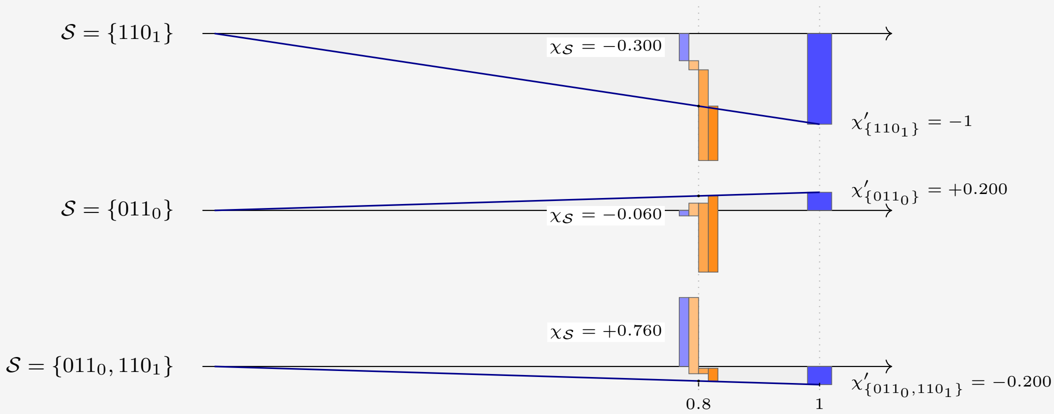
the denominator once more the $\mathcal{S} = \emptyset$ instance of the numerator. For $m = 1$ the three formulas collapse to equations (43), (44) and (46).

EXAMPLE 25

To draw the update, take a sparse prior at $n = 3$, $m = 2$: four maps, $\phi(q) = q \bmod 4$ (weight 0.45), constant 00 (0.35), "11 iff $q \geq 4$" (0.12), constant 11 (0.08). Werner asks $q = 110$ and hears $a = 11$: toggle signs $(+, -, -, +)$ on $(\emptyset, 110_0, 110_1, \text{both})$, where q_i denotes site (q, i), and denominator $AM_{11,110} = 0.8$. The pairs of the $m = 1$ figure become *quartets*:



For $m = 2$, toggling any subset of the asked block's two sites ties four correlators into a clique that averages jointly, per equation (50). The on-block quartet (orange hull) contains the normalization and the entire block of $M_{\cdot,110}$; its signed sum is the denominator that renormalizes every other quartet. Fills to scale for the sparse prior, as before.



The $m = 2$ update, three sets from the sparse prior. The waterfall now has four segments: $\chi_\mathcal{S}$ (blue) and its three signed quartet partners, in toggle order 110_0, 110_1, both (light to dark orange). The ray from the origin rescales the endpoint from $x = AM_{a,q} = 0.8$ to $x = 1$. The asked site saturates to -1; the foreign site 011_0 flips sign, $-0.060 \to +0.200$; and $\{011_0, 110_1\}$ lands on -0.200, the renormalized value $-\chi'_{\{011_0\}}$.

3 Expectations

While the games of the previous chapters may be entertaining, they don't readily divulge any universal laws. When we average (integrate) out the question order, and the specific map drawn, many more robust quantities appear. This is a stepping stone to our terminus: the thermodynamic limit, where we expect questions to flow in and learning to happen on a continuum, where such ensemble-average statements are physically instructive and often quite descriptive of nature.

Let $\langle \ldots \rangle$ denote expectation over both

- the truth $\psi \sim \{p_j\}$,
- uniformly over the question order $\{q_k\}$.

We take the ratio of the expected entropy cleared from the table M to the expected surprisal received:

$$\mathcal{L}_\ell \;:=\; \frac{H\big(M^{(0)}\big) - \big\langle H\big(M^{(\ell)}\big)\big\rangle}{\big\langle \sum_{k=1}^{\ell} s_k \big\rangle}\,. \tag{51}$$

The calligraphic $\mathcal{L}_\ell$ denotes this ratio of expectations, and not the expectation of the individual-run ratio L_ℓ. The two are different quantities: averaging the ratio lets a lucky run with a vanishing denominator dominate, while averaging numerator and denominator separately does not. We already used this aggregate quantity in Subsection 1.2.1. There, the problem was actually symmetric in $q_0 \leftrightarrow q_1$. That is not the case in general.

As stated above, in order to progress, we will be interested in such averages. Both are natural. The question order is uniform, without replacement: Without control of the questions, Werner has no reason to expect one to arrive before another. And the truth is drawn from the prior itself: Werner scores his future against his own beliefs, which is the best he can do.

The bookkeeping, especially for questions asked, simplifies at once. Updating commutes, so after ℓ rounds the state depends only on the asked *set* $\mathcal{S}$, a uniform random subset of size ℓ. Let us say the answers received form a vector

$$y = \{y_k = \psi(q_k)\}, \quad \forall q_k \in \mathcal{S}. \tag{52}$$

Each surprisal is paid at the posterior odds of its own round,

$$s_k = -\log_2 M^{(k-1)}_{y_k, q_k}, \tag{53}$$

and the posterior column is, by construction of the update rule, exactly the conditional probability of the next answer given everything seen so far:

$$M^{(k-1)}_{y_k, q_k} = \Pr(\psi(q_k) = y_k \mid \psi(q_1) = y_1, \ldots, \psi(q_{k-1}) = y_{k-1}). \tag{54}$$

Take $P_{\mathcal{S},y}$ to be the mass of the maps that answer the pattern y on a set $\mathcal{S}$. The sums of the surprisals display the chain rule of probability

$$\sum_{k=1}^{\ell} s_k = -\log_2 \left[\prod_{k=1}^{\ell} P_{\mathcal{S}_1,y}\,\frac{P_{\mathcal{S}_2,y}}{P_{\mathcal{S}_1,y}}\,\frac{P_{\mathcal{S}_3,y}}{P_{\mathcal{S}_2,y}} \cdots \frac{P_{\mathcal{S}_\ell,y}}{P_{\mathcal{S}_{\ell-1},y}}\right], \tag{55}$$

where we use the shorthand that $\mathcal{S}_k$ contains the first k questions, and is answered by the first k elements of pattern y. Each conditioning cancels against the next term's outcome, and $\mathcal{S}_\ell = \mathcal{S}$. Thus we arrive at the expression with no path dependence:

$$\sum_{k=1}^{\ell} s_k \;=\; -\log_2 P_{\mathcal{S},y}, \qquad P_{\mathcal{S},y} \;=\; \Pr\big(\psi(\mathcal{S}) = y\big) \;=\; \sum_{j\,:\,\phi_j(q_k) = y_k\ \forall q_k \in \mathcal{S}} p_j. \tag{56}$$

Averaging (56) over ψ gives the entropy of the block distribution $P_{\mathcal{S},\cdot}$: the distribution over such sets. Everything is generated by one sequence, the **mean block entropy** at size k:

$$G_k \;:=\; \binom{Q}{k}^{-1} \sum_{|\mathcal{S}|=k} H(\psi(\mathcal{S})) \;=\; -\binom{Q}{k}^{-1} \sum_{|\mathcal{S}|=k} \sum_{y} P_{\mathcal{S},y} \log_2 P_{\mathcal{S},y}, \tag{57}$$

the inner sum running over the A^k answer patterns on the block; nothing here is specific to $m = 1$. The endpoints are

$$G_0 \;=\; 0, \qquad G_Q \;=\; H(p). \tag{58}$$

G_1 is by definition the mean column entropy of the first question, so the prior table entropy is

$$H\big(M^{(0)}\big) \;=\; Q\, G_1. \tag{59}$$

EXAMPLE 26

What a block entropy is, on the guiding space ($n = 2$, $m = 1$). Take the block $\mathcal{S} = \{11, 10\}$, two questions. An answer pattern is $y = (y_{11}, y_{10})$, with four options, and by the numeral convention the maps answering each pattern are four consecutive rows of Table 4: the residue classes are read off the two leading truth-table digits.

y	maps j	names	$P_{\mathcal{S},y}$
$(0,0)$	0–3	FALSE, NOR, $\neg b_1 \wedge b_0$, $\neg b_1$	0.424
$(0,1)$	4–7	$b_1 \wedge \neg b_0$, $\neg b_0$, XOR, NAND	0.161
$(1,0)$	8–11	AND, XNOR, b_0, $b_1 \to b_0$	0.355
$(1,1)$	12–15	b_1, $b_0 \to b_1$, OR, TRUE	0.060

The four masses sum to one, and the block entropy is the Shannon entropy of this four-outcome distribution, $H(\psi(11, 10)) = -\sum_y P_{\mathcal{S},y} \log_2 P_{\mathcal{S},y} = 1.723$ bits: Werner's joint uncertainty about the two answers before either is asked. It is one of the six size-two terms averaged into G_2 in the next example.

Two exact laws follow. The first is equation (56) averaged:

$$\Big\langle \sum_{k=1}^{\ell} s_k \Big\rangle \;=\; G_\ell. \tag{60}$$

The second law describes the total column entropy of $M^{(\ell)}$, which is the answer matrix after $\ell = |\mathcal{S}|$ questions have been settled. Fix $\mathcal{S}$ and average the posterior table over the answers. Each unasked q' column's entropy becomes a conditional entropy,

$$H(\psi(q') \mid \psi(\mathcal{S})) = H(\psi(\mathcal{S} \cup \{q'\})) - H(\psi(\mathcal{S})), \tag{61}$$

and the already answered columns contribute zero. Sum the $Q - \ell$ unasked columns, then average over which ℓ questions were asked:

$$\begin{aligned} \big\langle H\big(M^{(\ell)}\big) \big\rangle \;&=\; \binom{Q}{\ell}^{-1} \sum_{|\mathcal{S}|=\ell} \sum_{q' \notin \mathcal{S}} \Big[H\big(\psi(\mathcal{S} \cup \{q'\})\big) - H\big(\psi(\mathcal{S})\big) \Big] \\ &=\; (\ell{+}1) \binom{Q}{\ell}^{-1} \binom{Q}{\ell+1} G_{\ell+1} \;-\; (Q{-}\ell)\, G_\ell. \end{aligned} \tag{62}$$

The first sum collects that way because every set of size $\ell+1$ arises as $\mathcal{S}\cup\{q'\}$ in $\ell+1$ ways, one for each q'. The second because $H(\psi(\mathcal{S}))$ is simply repeated for each of the $Q-\ell$ unasked q', and we invoke the definition of G_k. Substituting $(\ell{+}1)\binom{Q}{\ell+1} = (Q{-}\ell)\binom{Q}{\ell}$ cancels the prefactor and leaves

$$\big\langle H\big(M^{(\ell)}\big)\big\rangle \;=\; (Q-\ell)\,\big(G_{\ell+1}-G_\ell\big). \tag{63}$$

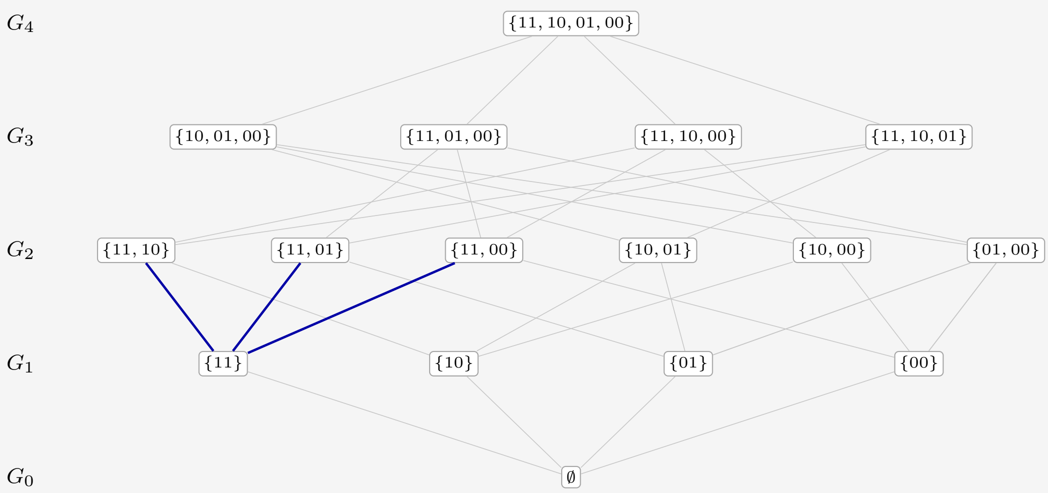
The whole expected learning curve is the discrete derivative of one monotone sequence.

EXAMPLE 27

This example illustrates equation (63): the total column entropy of the table M after ℓ questions, read off a lattice. Take the guiding space, $\mathcal{Q}=\{11,10,01,00\}$ with $Q=4$ and $m=1$, and abbreviate $\psi(11,10)$ for $\psi(\{11,10\})$. Definition (57) averages $H(\psi(\mathcal{S}))$ over the sets of each size, so the five block entropies collect $1,4,6,4,1$ terms apiece:

$$\begin{aligned}
G_0 &= 0,\\
G_1 &= \tfrac14\big[\,H(\psi(11))+H(\psi(10))+H(\psi(01))+H(\psi(00))\,\big],\\
G_2 &= \tfrac16\big[\,H(\psi(11,10))+H(\psi(11,01))+H(\psi(11,00))\\
&\qquad +H(\psi(10,01))+H(\psi(10,00))+H(\psi(01,00))\,\big],\\
G_3 &= \tfrac14\big[\,H(\psi(10,01,00))+H(\psi(11,01,00))\\
&\qquad +H(\psi(11,10,00))+H(\psi(11,10,01))\,\big],\\
G_4 &= H(\psi(11,10,01,00)) \;=\; H(p).
\end{aligned}$$

Those sixteen sets are the nodes of a lattice, one rank per value of ℓ, and each G_ℓ is the average of the entropies on rank ℓ:



Every edge up is the addition of one question, and it weighs the conditional entropy of the answer it adds: the edge from $\mathcal{S}$ up to $\mathcal{S}\cup\{q'\}$ represents $H(\psi(q')\mid\psi(\mathcal{S}))$. Fix a single node. Its upward edges run to the questions still unasked, and their weights are exactly the column entropies of the posterior table. The remaining table entropy of an asked set is the total marginal weight of the up-edges leaving its node. There are $Q-\ell$ of them, drawn in blue above for $\mathcal{S}=\{11\}$; summed and averaged over the rank, they are exactly equation (63)'s $\big\langle H\big(M^{(\ell)}\big)\big\rangle$, a component of the leverage.

Putting together (59), (60), and (63), equation (51) becomes

$$\mathcal{L}_\ell \;=\; \frac{Q\,G_1 - (Q-\ell)\big(G_{\ell+1} - G_\ell\big)}{G_\ell}\,, \qquad 1 \le \ell \le Q, \tag{64}$$

the vanishing factor $Q-\ell$ retiring the undefined increment at $\ell = Q$; at $\ell = 0$ nothing has been asked or paid and the ratio is an empty $0/0$. Some observations.

- Expectation kills the windfall. Each step has $\langle \Delta H(p) - s\rangle = 0$, so the expected deduced part is a pure tower withdrawal: $H(M^{(0)}) - \langle H(M^{(\ell)})\rangle - G_\ell = C_0 - \langle C_\ell\rangle$, with $\langle C_\ell\rangle = \langle H(M^{(\ell)})\rangle - \big(H(p) - G_\ell\big)$. It is nonnegative: anti-leverage is only ever bad luck.
- Deduction saturates one round early: substituting $G_Q = H(p)$ at $\ell = Q-1$ already gives $\langle C_{Q-1}\rangle = 0$. The store is empty before the last question, which can only be observed, never leveraged.
- $\mathcal{L}_\ell \ge 1$ always, but it need not be monotone. The endpoint is forced: $\mathcal{L}_Q = H(M^{(0)})/H(p) = 1 + C_0/H(p)$, the aggregate completion ratio.

There is a broader baseline than uniformity: any product prior over the answers gives $\mathcal{L}_\ell = 1$, whenever the denominator is positive. Its columns may be biased, but observing one cannot improve another, so expected predictive gain equals expected surprisal.

More generally, the finite aggregate leverage satisfies the sharp bounds

$$1 \le \mathcal{L}_\ell \le Q, \qquad G_\ell > 0. \tag{65}$$

For a fixed asked set $\mathcal{S}$, averaging over its answer string y gives the total predictive gain

$$H(M^{(0)}) - \mathbb{E}_y H(M^{(\mathcal{S},y)}) = \sum_{q\in\mathcal{Q}} I\big(\psi(q);\psi(\mathcal{S})\big).$$

Each term is at most $H(\psi(\mathcal{S}))$, proving the upper bound. The terms with $q \in \mathcal{S}$ sum to $\sum_{q\in\mathcal{S}} H(\psi(q)) \ge H(\psi(\mathcal{S}))$, and all other terms are nonnegative, proving the lower bound. Averaging over the uniformly chosen sets and dividing by G_ℓ preserves both inequalities. Independence attains the lower bound; a single nondegenerate random answer copied to every question attains the upper bound. For the second, let ν be a discrete probability distribution on the answer alphabet $\mathcal{A}$, with $\nu(a)$ the weight it puts on the answer a, and let the prior draw one $a \sim \nu$ and return that same a to every question,

$$p_j \;=\; \begin{cases} \nu(a), & \phi_j \equiv a \;\text{ for some } a \in \mathcal{A},\\ 0, & \text{otherwise,}\end{cases} \tag{66}$$

for any ν that is not a point mass. The support of ν fixes the support of p: the maps of positive weight are exactly the constant maps $\phi \equiv a$ with $\nu(a) > 0$, one apiece, and every non-constant map has $p_j = 0$. Every nonempty block then carries $H(\psi(\mathcal{S})) = H(\nu)$, so $G_\ell = H(\nu)$ for all $\ell \ge 1$, the increment vanishes, and $\mathcal{L}_\ell = Q$ at every ℓ. Werner knows the map is constant, so a single observation collapses all the uncertainty at once, as in Section 1.2. It is the clique block law of Appendix B taken at full size, $r = Q$.

Recall (49) from the discussion on correlators, Section 2.5. The joint law of the answers to any set $\mathcal{S}$ of questions is an inverse transform over the correlators with support contained inside $\mathcal{S}$'s blocks. The mean block entropy $G_{|\mathcal{S}|}$ does not depend on higher correlators (involving more questions) than are in $\mathcal{S}$. Since the expected leverage up to ℓ questions is assembled from G_ℓ and $G_{\ell+1}$ alone, it depends only on correlators supported on at most

$\ell+1$ questions. Two priors agreeing in correlators to that order are indistinguishable through round ℓ. Deeper shells are invisible.

There is also a monotonicity property hiding in G. Entropy is *submodular* [2]: a fresh answer is worth at most in a larger context $\mathcal{S}$ what it was worth in a smaller one $\mathcal{V}$,

$$H(\psi(q) \mid \psi(\mathcal{S})) \leq H(\psi(q) \mid \psi(\mathcal{V})); \quad \forall \mathcal{V} \subseteq \mathcal{S}, \tag{67}$$

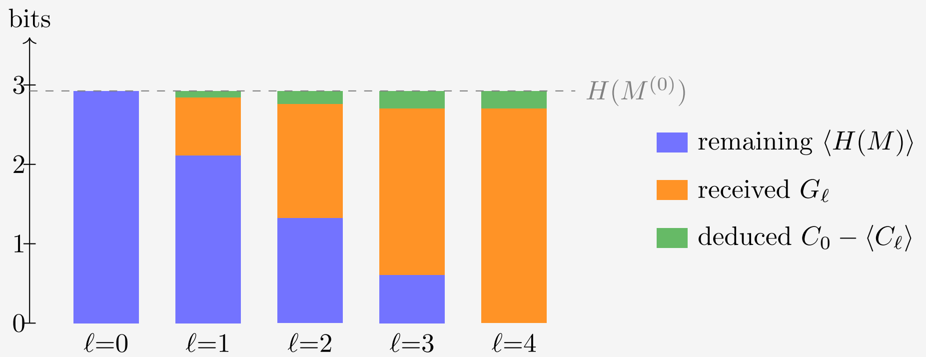
because conditioning on more answers can only explain away part of its novelty, never add to it. Coupling a random ℓ-set to a random $(\ell+1)$-set and averaging, the increments $G_\ell - G_{\ell-1}$, which are the expected surprisals of the ℓ-th answer, decrease with ℓ: answers get cheaper as evidence accumulates. This only holds on uniformly random question order.

EXAMPLE 28

For the guiding prior of Table 4, recall the block-entropy sequence is $G_k = \left\langle \sum_{i=1}^{k} s_i \right\rangle$, and the laws unfold it into the expected run:

ℓ	received G_ℓ	remaining $\langle H(M^{(\ell)}) \rangle$	store $\langle C_\ell \rangle$	leverage $\mathcal{L}_\ell$
1	0.731	2.113	0.137	1.110
2	1.436	1.328	0.056	1.113
3	2.100	0.608	0	1.104
4	2.707	0	0	1.081

The aggregate leverage humps at $\ell = 2$ and lands on $2.925/2.707 = 1.081$. The realized run of Chapter 2 scattered around this gentle expectation: greedy questioning and a lucky truth delivered 1.311. The figure below stacks the expected bits.



Conservation is now exact at every ℓ: remaining plus received plus deduced reaches $H(M^{(0)})$, so the cake always tops out on the dashed line, and no overshoot is possible. The top layer is the expected withdrawal from the correlation store, $C_0 - \langle C_\ell \rangle$; it saturates at $C_0 = 0.218$ already at $\ell = 3$.

3.1 Choosing the questions

The uniform distribution on question order was a symmetry choice, not a strategy. Werner can do better along two independent axes: how far he looks ahead, and whether he commits to an order beforehand or pivots contingent on the answers so far. A **Greedy** Werner asks, at every step, the single question with the largest expected drop of $H(M)$: one step of lookahead. **ℓ-optimal** maximizes the expected knowledge after exactly ℓ questions, looking the whole horizon ahead. All four combinations are straightforward with the machinery at hand. 1-optimal coincides with Greedy at the start.

	order fixed beforehand	**contingent on answers**
greedy	Add planned questions to the docket, each maximizing the immediate expected drop of $H(M)$, averaged over the still-unseen answers. All of this before any answer is seen, so the plan cannot react to luck.	Recompute the forecasts after every answer and ask the current argmax. Question order is a branching tree, each branch contingent on last answer.
ℓ-optimal	Updating commutes, so a fixed plan is just a set $\lvert\mathcal{S}\rvert = \ell$: search the $\binom{Q}{\ell}$ subsets for the smallest expected remainder $\sum_{q' \notin \mathcal{S}} H(\psi(q') \mid \psi(\mathcal{S}))$. The best non-adaptive knowledge; sees correlations that the one-step view misses.	Backward induction over posteriors. The Bayes-optimal policy: recompute optimal further set for remainder at ℓ, after every answer.

Table 5: The four strategies open to Werner: one step of lookahead or ℓ, with the question order fixed in advance or chosen contingent on the answers already heard.

Expected knowledge is non-decreasing left to right and top to bottom. The dynamic program in the last cell is exact but grows quickly: its states are the reachable posteriors, one per pair of asked set and answer pattern, $(1 + A)^Q$ in all. That is 81 for the guiding example and astronomical beyond. Its policy is also horizon-dependent, since the best first question for $\ell = k$ need not open the best plan for $\ell = k + 1$. For the guiding prior of Table 4, contingent greedy outperforms the uniform order, and already attains the contingent ℓ-optimum at every horizon. That is a quirk of this prior, not a law: a prior that stores its knowledge in deep shells can defeat Greedy in either column: a question worth little now may unlock a cascade later.

Both ℓ-optimal protocols produced the same trajectory in the examples tested above, but this is not a law. Adaptive questioning can strictly improve on every fixed set of the same size, as Subsection 3.1.1 demonstrates.

3.1.1 When the first answer chooses the next question

The distinction is that an answer can tell Werner which question will be useful next.

EXAMPLE 29

Take $n = 2$, $m = 1$, and put equal prior weight $1/4$ on the four maps below, with zero weight on the others. The columns retain the descending question order used throughout.

j	$\phi_j(11)$	$\phi_j(10)$	$\phi_j(01)$	$\phi_j(00)$	p_j
0	0	0	0	0	1/4
2	0	0	1	0	1/4
1	0	0	0	1	1/4
5	0	1	0	1	1/4

Ask $q = 00$ first. If $a = \psi(00) = 0$, the survivors are maps 0 and 2, distinguished by asking $q' = 01$. If $a = 1$, they are maps 1 and 5, distinguished by asking $q' = 10$. Thus two adaptive questions identify the map completely and leave $H(M) = 0$.

No fixed pair does so. The pairs $\{00, 01\}$ and $\{00, 10\}$ each leave one fair unknown answer with probability $1/2$, hence an expected remainder of $1/2$ bit. The pair $\{01, 10\}$ leaves maps 0 and 1 indistinguishable when both answers are zero, again an event of probability $1/2$ with one fair unknown answer. These are the three best fixed pairs. Pairs containing the constant question 11 leave at least 1 bit. The optimal fixed remainder is therefore $1/2$ bit, strictly above the adaptive remainder of zero.

3.2 The Gibbs complexity prior

Every prior so far was an arbitrary set of instructively chosen $\{p_j\}$. This section builds the prior the other way around, from a single philosophical principle: *simple explanations are likelier*. How do we define simplicity? With circuit complexity as the measure, made quantitative as a Gibbs ensemble over the hypothesis space. It is an illustration: the later chapters do not depend on the rest of this section.

3.2.1 AIG complexity

A compact circuit description supplies a concrete notion of simplicity. We use an *and-inverter graph* (AIG): a feed-forward circuit with two-input AND nodes and complemented edges, so inversion on any wire is free. Let X_j be the fewest AND nodes in any such graph computing ϕ_j, with constants and input wires available at zero cost. This is the standard AIG-size metric of logic synthesis [15]. It differs from counting ordinary NAND gates, where inversion itself has a cost. Reference values in the AIG convention:

- AND, OR, NAND and NOR all cost 1;
- XOR costs 3;
- constants, projections and negations cost 0.

For $m > 1$ outputs, X_j is the size of one *joint* circuit computing all output bits together, gates shared, which is practical. It is the minimum cost in this AIG convention, generally smaller than the sum of per-output circuits.

The $(n, m) = (4, 1)$ and $(3, 2)$ are the largest tractable setups, at 65,536 maps apiece. The tabulation is feasible because X_j is invariant under relabeling: permuting or negating inputs, negating or (for $m \geq 2$) swapping outputs are all free operations, so the 65,536 maps collapse to 222 and 308 equivalence classes respectively. The $(4, 1)$ class values are the published optimal AIG sizes of [18], cross-checked against our own SAT-based synthesizer on random classes with zero mismatches; the $(3, 2)$ values come from that synthesizer: circuit sizes $X = 0, 1, 2, \ldots$ are encoded as satisfiability instances, and the first satisfiable X is provably minimal, every smaller size having been refuted. Appendix C gives the encoding, the symmetry reduction that makes it tractable, and the provenance of every value.

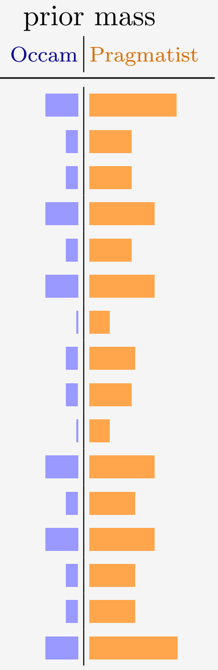
Alongside the interior cost X_j, which takes a SAT solver to see, two descriptors are visible in one pass over the truth table:

- the **footprint** $F_j = n \cdot [\text{Image}] + [\text{Support}]$, where the support counts the input bits the map actually depends on and the image size counts the distinct answers it attains (from 1 for constants up to A). The combination is bijective to the pair, as Support $\leq n$. Compare to base-n digits, quotient Image, remainder Support. The footprint can be viewed as the map's *interface*: does it react to all its inputs, does it use all its values?
- the **bias** B_j: the number of 1s minus the number of 0s among the mQ truth-table bits. This is the symmetry breaker: X_j and F_j are even under negating all outputs, B_j is odd: an energy term coupling to it acts like a magnetic field, favoring 0s or 1s off the bat. In the language of Section 2.5: switching on the odd correlator shells that a pure complexity prior forbids.

Example 30

Take again the $(n, m) = (2, 1)$ space. Each map's data is listed for illustration. We graph two of the Gibbs priors built from these inputs, defined in Subsection 3.2.2. Occam in blue growing left, the Pragmatist in orange growing right. The Pragmatist bars are logarithmic, five decades to the full width, since a linear scale on the same axis as Occam's would show nothing but the two constants.

j	map	X_j	Support	Image	F_j	B_j
0	FALSE	0	0	1	2	-4
1	$\neg(b_1 \vee b_0)$ (NOR)	1	2	2	6	-2
2	$\neg b_1 \wedge b_0$	1	2	2	6	-2
3	$\neg b_1$	0	1	2	5	0
4	$b_1 \wedge \neg b_0$	1	2	2	6	-2
5	$\neg b_0$	0	1	2	5	0
6	$b_1 \oplus b_0$ (XOR)	3	2	2	6	0
7	$\neg(b_1 \wedge b_0)$ (NAND)	1	2	2	6	$+2$
8	$b_1 \wedge b_0$ (AND)	1	2	2	6	-2
9	$b_1 = b_0$ (XNOR)	3	2	2	6	0
10	b_0 (projection)	0	1	2	5	0
11	$b_1 \to b_0$	1	2	2	6	$+2$
12	b_1 (projection)	0	1	2	5	0
13	$b_0 \to b_1$	1	2	2	6	$+2$
14	$b_1 \vee b_0$ (OR)	1	2	2	6	$+2$
15	TRUE	0	0	1	2	$+4$



Occam spreads its belief mostly across the six gate-free maps (0.1106 apiece) and thins gradually with cost; the Pragmatist, punishing footprint and rewarding bits with value 1, piles 0.90 of its mass onto the two constants (0.62 on TRUE against 0.28 on FALSE), the bias breaking the tie. It all but abandons the interior of the table. If the map turns out non-constant, renormalization scales that interior up, and it becomes relevant. Grouped by the full tuple (X, F, B), the sixteen maps fall into six energy classes:

(X, F, B)	members	(X, F, B)	members
$(0, 2, -4)$	1	$(1, 6, -2)$	4
$(0, 2, +4)$	1	$(1, 6, +2)$	4
$(0, 5, 0)$	4	$(3, 6, 0)$	2

Note the table skips $X = 2$ entirely: with two inputs, nothing a two-gate circuit computes is out of reach of one or zero gates.

At the larger sizes the tuple (X, F, B) breakdown runs to hundreds of rows (102 distinct tuples at $(n, m) = (4, 1)$, 196 at $(3, 2)$). It is still interesting to tabulate the complexity X shells alone, with each shell's *aggregate* prior mass under the Occam setting alongside its head count:

EXAMPLE 31

These tables illustrate the compromise between exponentially suppressed weight per map, and combinatorics putting most maps at middling complexity.

X	$(4,1)$ maps	Occam mass
0	10	0.041
1	48	0.072
2	256	0.141
3	940	0.190
4	2,336	0.174
5	6,464	0.177
6	10,616	0.107
7	18,984	0.070
8	17,680	0.024
9	7,882	0.004
10	320	0.000
total	65,536	1

X	$(3,2)$ maps	Occam mass
0	64	0.058
1	432	0.144
2	2,064	0.253
3	4,860	0.219
4	10,832	0.180
5	16,208	0.099
6	17,060	0.038
7	10,672	0.009
8	3,312	0.001
9	32	0.000
total	65,536	1

The head counts are thin at both ends: ten gate-free maps at $(4,1)$ (two constants, eight literals) against only 320 at the ceiling $X = 10$. The aggregate mass shows the thermodynamic compromise this forces on Occam: the count grows faster than e^{-X} decays until $X = 3$ at $(4,1)$ ($X = 2$ at $(3,2)$), so even under a pure simplicity prior Werner has most faith that the map has $X_j \in \{2,3\}$. Shell degeneracy beats shell energy for the simplest maps.

3.2.2 The ensemble

Take the energy linear in the three descriptors and the prior as its Boltzmann weight,

$$E_j = \beta\, X_j + \alpha\, F_j + \mu\, B_j, \qquad p_j = \frac{e^{-E_j}}{\sum_k e^{-E_k}}, \tag{68}$$

with no overall temperature: it's absorbed in the coupling fields. Positive β suppresses complex maps, positive α suppresses large footprints, and μ couples to the signed bias like a magnet. $\mu < 0$ tilts toward 1-heavy tables and thereby breaks the output-negation symmetry. We explore three settings below, chosen out of principle, as well as to exhibit interesting learning behavior.

	β	α	μ
Occam	1	0	0
Occamer	2	0	0
Pragmatist	1.5	1	-0.1

Table 6: The three Gibbs settings of (68), as couplings to complexity, footprint and bias.

Occam suppresses circuit complexity and nothing else. **Occamer** doubles the same field: an even stronger simplicity bias stores more knowledge in the prior, and stores it deeper in the correlation shells, where only the later questions of a run can release it. The **Pragmatist**

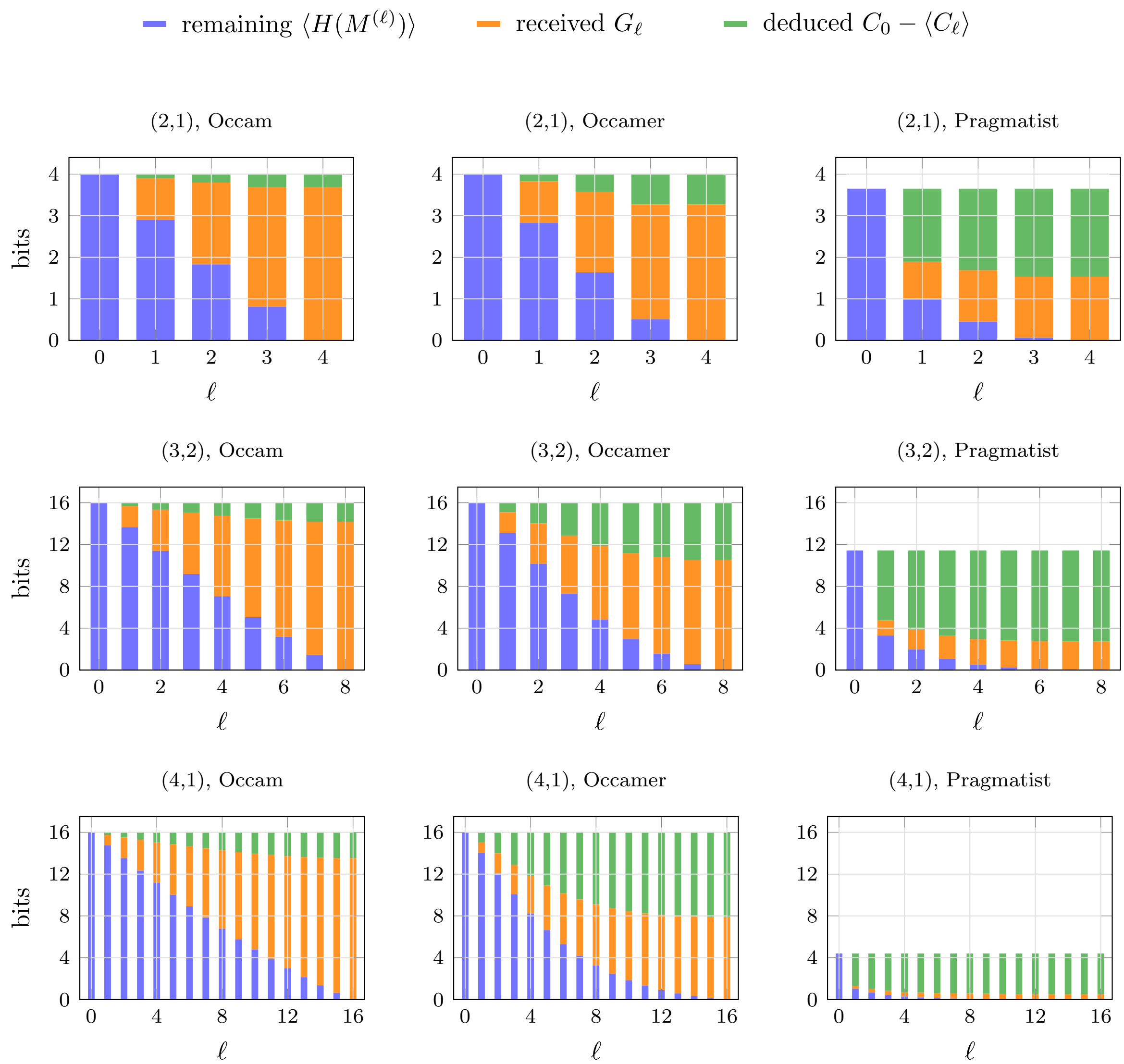
judges by the obvious heuristics instead: a footprint reads the visible interface rather than the interior circuit, a moderate complexity field seconds it, and the tilt $\mu = -0.1$ leans toward 1-heavy tables, breaking the output-negation symmetry that the other two settings keep. This last choice is to illustrate that Werner isn't bound to start with a uniform distribution M before the first question.

3.2.3 The layered entropy books

Let us visualize the uniform question-order average for the three ensembles, at each system size. The following graphs illustrate the flow of bits. The three quantities of (51) partition the starting table entropy at every ℓ: what is still in the table, what was paid for in surprisal, and what was inferred. The third has no standing name, so give it one,

$$\big\langle \text{deduced} \big\rangle_\ell := C_0 - \langle C_\ell \rangle = H\big(M^{(0)}\big) - \big\langle H\big(M^{(\ell)}\big) \big\rangle - G_\ell, \tag{69}$$

the expected withdrawal from the correlation store. The second equality is (31) at both ends together with $\langle \Delta H(p) \rangle = G_\ell$. In expectation, the belief entropy falls by exactly the surprisal received, so whatever else the table lost came out of C. The three therefore sum to $H(M^{(0)})$ at every ℓ, and each panel below is a full-height bar whose composition shifts from all remaining uncertainty to received plus deduced. One row per system size, sharing a vertical scale in bits; one column per prior.



3.2.4 Trajectories

Figures 3–5 show the expected cumulative leverage $\mathcal{L}_\ell$ under the three prior hypotheses, for the uniform question-order average and the strategy grid of Section 3.1. The ℓ-optimal schedule targets the halfway horizon $\ell^* = Q/2$ and is plotted only up to it: past its own horizon the optimizer's continuation is not defined. Inside the optimal set the questions are asked in random order (the graph averages that order). We omit its contingent twin everywhere: wherever we can afford to compute it, it reproduces the fixed optimum's curve to machine precision, and one for which no symmetry argument yet is known. Two consistency checks hold in every panel: all schedules agree at $\ell = 1$, because input relabeling acts transitively on the questions and the Gibbs energies are relabeling-invariant, so single questions are exchangeable; and every full-length curve ends at the same $\mathcal{L}_Q = H(M^{(0)})/H(p)$, since any policy that asks everything receives exactly $H(p)$ on average, and clears the table M. The same invariance acts on whole sets, so the ℓ-optimal objective is a function of a set's relabeling orbit and not of the set: minimizers arrive in orbits.

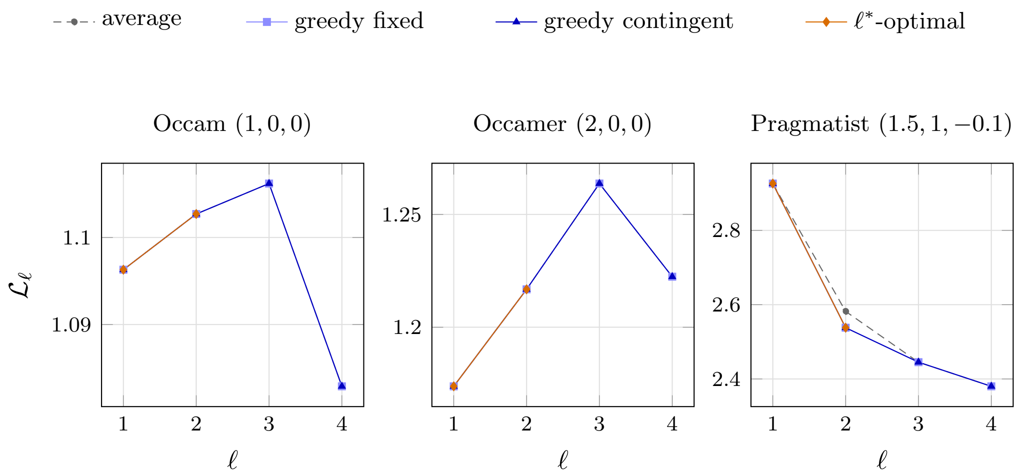


Figure 3: Aggregate leverage under the Gibbs priors at $(2, 1)$, $\ell^* = 2$. In both complexity-only settings every schedule coincides exactly; the Pragmatist's fields finally separate them.

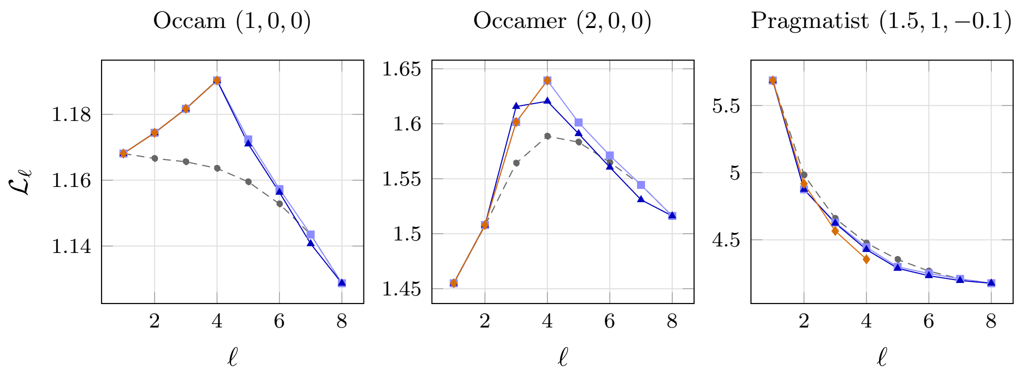


Figure 4: Aggregate leverage under the Gibbs priors at $(3, 2)$, $\ell^* = 4$. Greedy Werner's fixed set is the optimal set under Occam and, up to an input negation, under Occamer. For the Pragmatist, the ℓ^*-optimal has the worst leverage at the horizon, a reminder that we optimize for information received, while accepting that may cost even more surprisal.

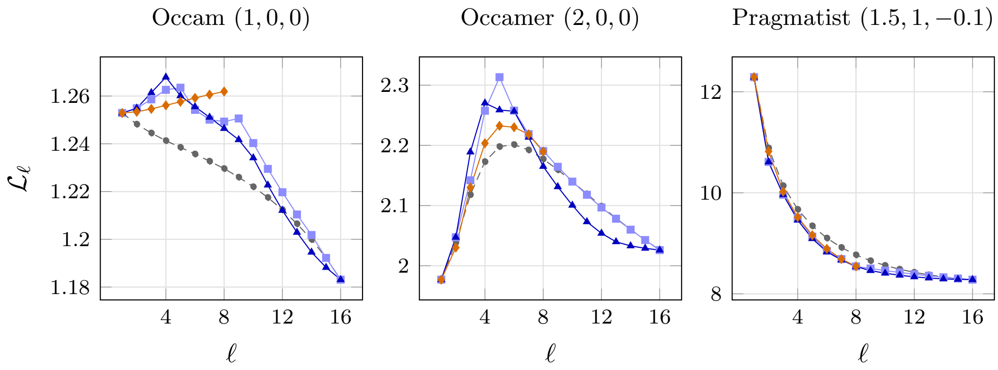


Figure 5: Aggregate leverage under the Gibbs priors at $(4,1)$, $\ell^* = 8$. Occamer's deep correlations C bend even the uniform average curve into an interior hump; the Pragmatist releases all its instinct at the start: whether it's constant.

Three discussions, one per system size.

At $(2,1)$, Occam and Occamer don't have enough structure to tell subsets apart, and all subsets of equal size share a single joint entropy. Occamer shows what doubling the field buys within those constraints. The whole curve lifts (endpoint 1.2224 against Occam's 1.0829, store $C_0 = 0.728$ against 0.306 bits) without creating a single strategic choice to make. The Pragmatist's fields split the pair orbits at last, and the strategy yields a *lower* leverage: greedy's pair $\{00, 11\}$ knows more at $\ell = 2$ than the random order but pays so much more surprisal that it is less leveraged (2.538 against 2.582).

At $(3,2)$ there are exactly two optimal quadruples under *all three* settings: the even-weight inputs $\{000, 011, 101, 110\}$, the maximally spread design, pairwise at Hamming distance two, and its odd-weight image under negating one input bit. Relabeling makes them exactly degenerate. Under the Pragmatist greedy finally falls short: its set $\{000, 011, 101, 111\}$ leaves twice the ignorance of the optimal four (0.405 against 0.198 bits remaining) while receiving 0.099 bits less and it posts the *higher* leverage, 4.445 against 4.355, with the strategyless average higher still at 4.476. The ℓ-optimal schedule maximizes knowledge, not leverage, and a ratio is buoyed by cheaper ignorance.

$(4,1)$ is the richest, with the most questions, the choice becomes impactful. Doubling β triples the store ($C_0 = 8.10$ bits against Occam's 2.48) and bends even the average curve into an interior hump, rising to 2.20 at $\ell = 6$ before decaying. This is the deep-shell signature promised in Subsection 3.2.2, knowledge that only becomes spendable once enough answers are in hand to cash out the higher correlations. The optimal eight questions are a parity class of the input hypercube in every setting, and as at $(3,2)$ both classes minimize exactly; greedy misses the class under both complexity-only settings ($\mathcal{L} = 1.249$ against the optimum's 1.262 at the Occam ℓ^* horizon) but finds it exactly under the Pragmatist whose prior, 0.90 of its mass on the two constants, is a two-sided spike in disguise: $H(p) = 0.53$ bits, leverage running from 12.29 down to 8.28, a great deal of inference about a game that is mostly over after two questions.

4 The thermodynamic limit

The earlier chapters describe learning on a finite question set. This chapter is arguably the real-world payoff. Here we can draw conclusions about general learning, not just boolean maps. This chapter asks what remains when the number of input bits tends to infinity. The hope is to find traces of universality: that character of learning where the microscopic details are coarse-grained out.

Recall definition 23 where $Q = |\mathcal{Q}| = 2^n$, the limit $n \to \infty$ is one of an exponentially growing table of possible questions. Let the measure of time be the fraction of that table already queried,

$$t \;=\; \frac{\ell}{Q} \;\in\; [0,1]. \tag{70}$$

The limit at fixed $t > 0$ is **macroscopic**: it records learning that occupies a nonzero fraction of the run. Learning completed in $O(1)$, in $O(\log Q)$, or more generally in $o(Q)$ questions is compressed into a boundary layer at $t = 0$. That distinction will determine the class of solutions that emerges in the TDL.

Throughout, $u := A^{-1} = 2^{-m}$ abbreviates the reciprocal answer alphabet, and $N = A^Q$ counts the maps $\mathcal{Q} \to \mathcal{A}$, enumerated $\phi_0, \ldots, \phi_{N-1}$ with prior weights $p_j = \Pr(\psi = \phi_j)$. One value of the truth ψ is drawn at the start of a run and supplies every answer in it; posteriors are conditional distributions of that same random map, and expected trajectories average over its prior law and the uniformly random q order.

4.1 A toy example: the spike prior

The most accessible example from micro- to macroscopics, as one calculus exercise, is arguably the spike prior. It places a distinguished weight on one map and spreads the remaining mass uniformly over every other map. The setup is also desirable due to the full support: at every system size, Werner feels the effect of the dimensionality he inhabits.

Take the earmarked boolean map to be the all-zero map, $\phi_0(q) = 0^m$ for every $q \in \mathcal{Q}$, so that its answer to any k questions is the all-zero block 0^{mk}. This is fully general by symmetry: the answer alphabet may be relabeled separately at each question. We then adopt the convention

$$p_0 = p, \qquad p_j = \omega := \frac{1-p}{N-1} \quad (j \neq 0). \tag{71}$$

The plain $p = p_0$ is the mass of the all-zero map alone at start time. We will be interested in the behavior at large (but not necessarily unit) p. The interpretation is that Werner is quite sure of his hypothesis, without ruling anything out.

The example is tractable because it is closed under Bayesian conditioning. A confirming answer produces a sharper spike on a smaller map space; a refuting answer culls (among others) the special map and leaves equal weights on all survivors. The posterior therefore has a two-state flow, in which the uniform family is the stable, absorbing class: once a trajectory enters it, later conditioning keeps it there. An interior spike is a source of one-way probability current into that class, while conditional on avoiding refutation the spike sharpens.

4.1.1 The two-state renormalization flow

What happens when Werner handles the first question? He either gets a zero, or not.

The maps that produce 0^m are a proportion $u = A^{-1}$. One has probability p, and $Nu-1$ have probability ω. If we started in this prior with probability $P_0 = 1$ (full faith), Werner avoids refuting ϕ_0 with probability

$$P_1 = p + (Nu - 1)\,\omega \tag{72}$$

With probability P_1 we can still believe $\psi = \phi_0$. In this case, what exactly is the shape of the posterior? The Bayesian consistency condition holds that unconditional probability cannot be created without measurement. At this point in the ansatz, we must have the same credence in ϕ_0.

At every step, there is a p_0, but as a member of an ever-shrinking cohort. Let r_1 be the height of the peak p_0, after one question. Then we must have $r_1 \cdot P_1 = p$. The rest of the probability is spread again over the $Nu - 1$ other surviving maps. Conversely, with probability $1 - P_1$, we've refuted the zero map, and are in a uniform posterior corresponding to a nonzero answer to the question.

After repeating this process k times, asking the set of questions $\mathcal{S}$, we can infer the probability that ϕ_0 is still in the race. Let P_k be the probability that Werner remains in the confirming scenario: he has only seen zeros. A history remains on the spike branch exactly when $\psi(\mathcal{S}) = 0^{mk}$, and exactly Nu^k maps produce that confirming history, one of which is ϕ_0. The probability of confirmation is then

$$P_k \;:=\; \Pr\left(\psi(\mathcal{S}) = 0^{mk}\right) \;=\; p + (Nu^k - 1)\,\omega. \tag{73}$$

On that event the posterior is another spike on those Nu^k maps,

$$r_k \;=\; \Pr\left(\psi = \phi_0 \mid \psi(\mathcal{S}) = 0^{mk}\right) \;=\; \frac{p}{P_k}, \qquad \omega_k \;=\; \frac{\omega}{P_k}, \tag{74}$$

so r_k is the time-indexed posterior height of the spike evaluated on the confirming branch, with $r_0 = p$ the original weight. It's pegged inversely to P_k for all time, by the same Bayesian consistency:

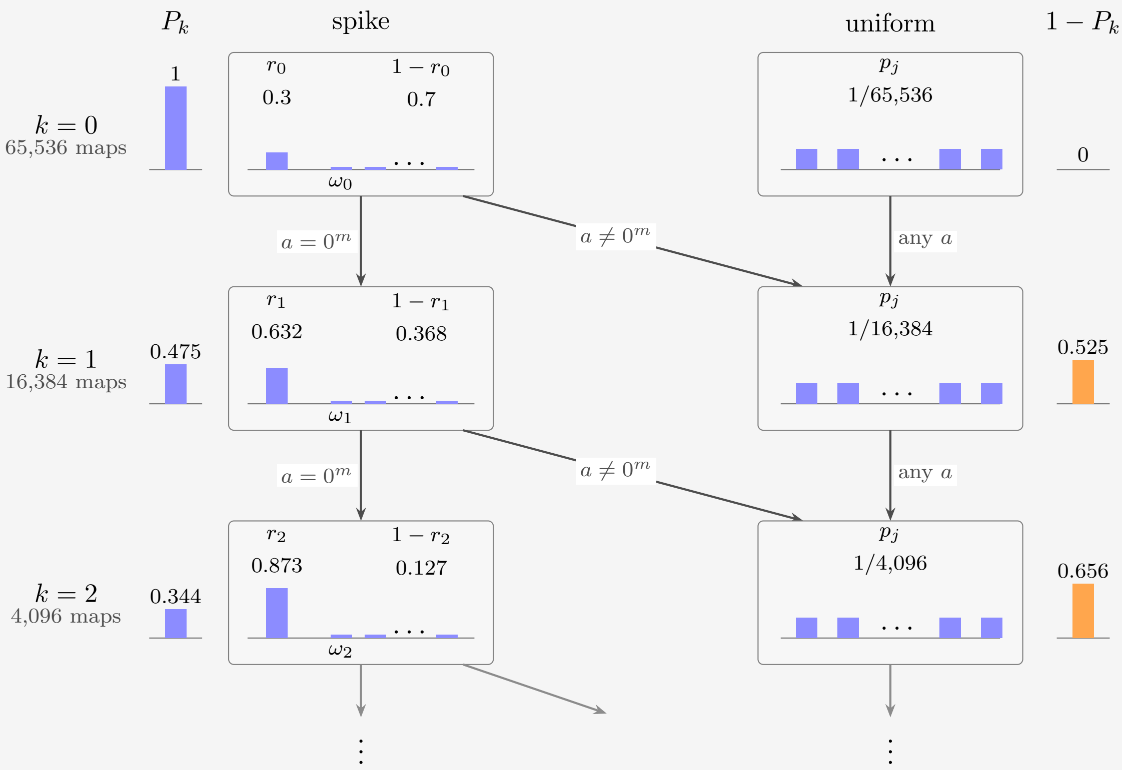
$$P_k\, r_k \;=\; p. \tag{75}$$

If any answer is nonzero then ϕ_0 is eliminated, every compatible survivor had the same mass ω_k, and the posterior is uniform; subsequent conditioning on answers preserves uniformity. With an equivalence across system sizes, this induces a two state flow. Renormalization sharpens the spike. The two macrostates and their directions are

$$\text{spike} \longrightarrow \begin{cases} \text{sharper spike,} & \text{with probability } P_{k+1}/P_k, \\ \text{uniform,} & \text{with probability } 1 - P_{k+1}/P_k, \end{cases} \qquad \Bigg| \; \text{uniform} \longrightarrow \text{uniform.} \tag{76}$$

EXAMPLE 32

The flow at $(n,m)=(3,2)$, so $A=4$ and $N=4^8=65{,}536$, started from $r_0=p_0=0.3$. Each confirmed answer divides the surviving map space by four and renormalizes the spike upward; each refutation drops the belief into the uniform class, which nothing ever leaves.



The two columns of boxes are the macrostates of (76), each arrow is labeled by the answer that drives its transition, and each row's map space is one quarter of the row above. The outer columns track the branch probabilities P_k and $1-P_k$, and the invariant $P_k r_k = p$ is visible between the P_k bars and the $r_k = p_0$ bars. The bars inside the boxes measure the $\{p_j\}$, on the left for the spike ϕ_0 vs others. The spike climbs exactly as fast as its branch loses probability, $0.3 = 1\times 0.3 = 0.475\times 0.632 = 0.344\times 0.873$.

For $p\in(0,1)$, we can combine equations (71) and (73) to get the limiting odds on the surviving branch and, from equation (74), the surviving posterior:

$$P_k = p + (1-p)u^k + O(N^{-1}), \qquad r_k = \frac{p}{p+(1-p)u^k} + O(N^{-1}). \tag{77}$$

Each confirmation supplies approximately m bits of evidence for the spike, in the sense of log odds:

$$\log_2\left(\frac{r_k}{1-r_k}\right) = \log_2\left(\frac{p}{(1-p)\cdot u^k}\right) = \log_2\frac{p}{1-p} + mk. \tag{78}$$

After all Q questions, the only way the spike can survive is by eliminating all alternative maps. This happens with $P_Q = p$ and hence $r_Q = 1$; the histories that confirm through the complete table are precisely those with $\psi=\phi_0$. At the uniform point $p=1/N$ both posterior branches are uniform on their surviving map spaces.

In anticipation of the TDL, we observe that the same P_k has a second reading: it is the probability of the single k-answer block 0^{mk}. For $1 \le k \le Q$ every other answer block is realized by Nu^k background maps and therefore carries the common probability

$$U_k \;:=\; \frac{1-P_k}{A^k-1} \;=\; Nu^k\,\omega, \qquad \text{so} \qquad \Pr(\psi(\mathcal{S})=y) = \begin{cases} P_k, & y = 0^{mk}, \\ U_k, & y \neq 0^{mk}, \end{cases} \tag{79}$$

for any question set $\mathcal{S}$ of size k. The entropy of the above distribution, coarse-grained to blocks of questions, is

$$G_k \;=\; -P_k \log_2 P_k - (A^k-1)\,U_k \log_2 U_k, \tag{80}$$

as there are A^k-1 terms where $y \neq 0^{mk}$. Every choice of k questions obeys the same law, so (80) is also the mean k-question block entropy of Chapter 3. In particular $G_0 = 0$ and $H(M^{(0)}) = Q\,G_1$, with the exact prior entropy

$$G_Q \;=\; h_2(p) + (1-p)\log_2(N-1), \tag{81}$$

extensive in Q as $N = A^Q$ and $P_Q = p$, at fixed $0 < p < 1$. Prior entropy vanishes at $p = 1$.

4.1.2 The fixed-p thermodynamic limit leverage

The thermodynamic limit keeps the prior mass $0 < p < 1$ and asked fraction $t > 0$ fixed, while sending $n \to \infty$. The guiding principle is the exact finite law (64) for $\mathcal{L}$: the aggregate leverage is built from three ingredients,

1. the prior table entropy $H(M^{(0)})$
2. the received entropy G_ℓ
3. the increment $G_{\ell+1} - G_\ell$.

If we can derive each for the example, we are done. The block entropy (80) supplies every G_k through the single counting formula (73). $H(M^{(0)}) = Q\,G_1$ holds for any prior: G_1 is by definition the mean entropy of one column of M. Set $\ell = \lfloor tQ \rfloor$ and take the terms of (64) in turn.

All three derivations run through one exact regrouping of (80). The A^k-1 refuting blocks (79) share the mass $1-P_k$ evenly, so substituting $U_k = (1-P_k)/(A^k-1)$ into (80) and expanding the logarithm regroups the block entropy into the binary entropy of the confirm-or-refute split plus a uniform choice among the refuting blocks:

$$G_k \;=\; h_2(P_k) \;+\; (1-P_k)\log_2\bigl(A^k-1\bigr). \tag{82}$$

At $k=1$, by (73) the spike answer 0^m carries $P_1 = p + (Nu-1)\,\omega \to p + (1-p)u$: the special map's whole mass, plus the fraction u of the background maps that happen to agree with it at any one question. Substituting that limit,

$$\begin{aligned} \frac{H(M^{(0)})}{Q} \;&=\; G_1 \;=\; h_2(P_1) + (1-P_1)\log_2(A-1) \\ &\longrightarrow\; h_2\bigl(p+(1-p)u\bigr) + (1-p)(1-u)\log_2(A-1) \;=:\; \eta_0(p,m). \end{aligned} \tag{83}$$

An η_0-term features in the numerator of every macroscopic leverage, the entropy of the first answer, as in (105). The spike produces this shape; in general η_0 depends on the prior and has no closed form.

The received entropy: Evaluate (82) at $k = \ell = \lfloor tQ \rfloor$. By (77) $P_\ell \to p$, so the split entropy stays bounded while the block-count logarithm grows linearly, $\log_2(A^\ell - 1) = \ell m + o(1)$:

$$\frac{G_{\lfloor tQ \rfloor}}{Q} = \underbrace{\frac{h_2(P_\ell)}{Q}}_{\to\ 0} + \underbrace{(1 - P_\ell)}_{\to\ 1-p} \underbrace{\frac{\log_2\bigl(A^\ell - 1\bigr)}{Q}}_{\to\ t\,m} \longrightarrow t\,(1-p)\,m. \tag{84}$$

This is the two-state flow watched at macroscopic time: for any $t > 0$ the confirm-or-refute lottery has already resolved inside the boundary layer at $t = 0$, where $r_k \to 1$ within $o(Q)$ questions. A surviving spike answers for free; only the refuted class, probability $1 - p$, keeps paying the full m bits per answer.

The increment: Subtract consecutive copies of (82): the bounded $h_2(P_k)$ terms cancel in the limit, and the block-count logarithm grows by $\log_2(A^{\ell+1}-1) - \log_2(A^\ell - 1) \to \log_2 A = m$, paid only by the refuted mass:

$$G_{\ell+1} - G_\ell \longrightarrow (1-p)\,m, \qquad \text{so} \qquad \frac{(Q-\ell)\bigl(G_{\ell+1} - G_\ell\bigr)}{Q} \longrightarrow (1-t)(1-p)\,m. \tag{85}$$

Filling the three limits into (64) term by term,

$$\mathcal{L}_\ell = \frac{\overbrace{H\bigl(M^{(0)}\bigr)}^{\to\ Q\,\eta_0(p,m)} - \overbrace{(Q-\ell)\bigl(G_{\ell+1} - G_\ell\bigr)}^{\to\ Q\,(1-t)(1-p)m}}{\underbrace{G_\ell}_{\to\ Q\,t(1-p)m}},$$

and canceling the common factor Q gives

$$L_p(t) = \frac{\eta_0(p,m) - (1-t)(1-p)m}{t(1-p)m} = 1 + \frac{\eta_0(p,m) - (1-p)m}{t(1-p)m}, \qquad 0 < t \le 1. \tag{86}$$

The $1/t$ term records information released in a vanishing initial fraction of the run, divided by the $O(tQ)$ information received afterwards.

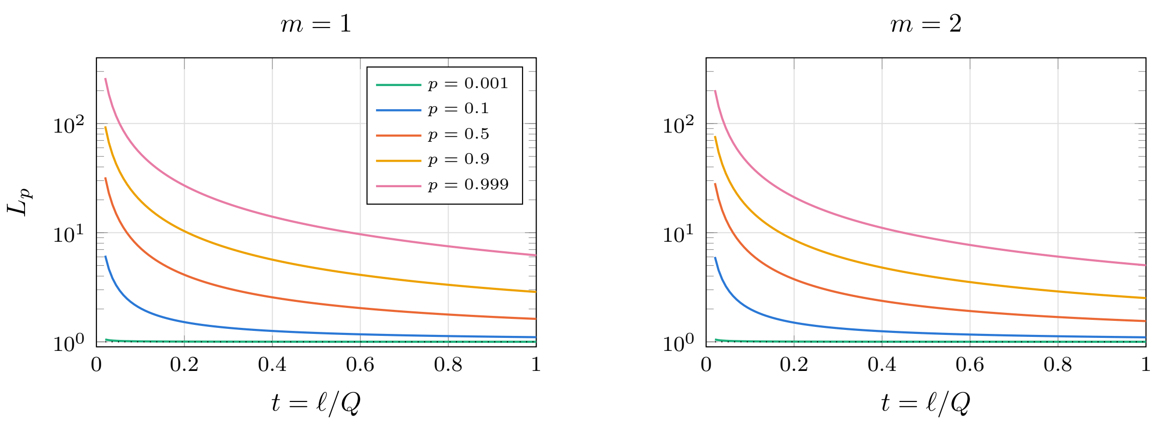
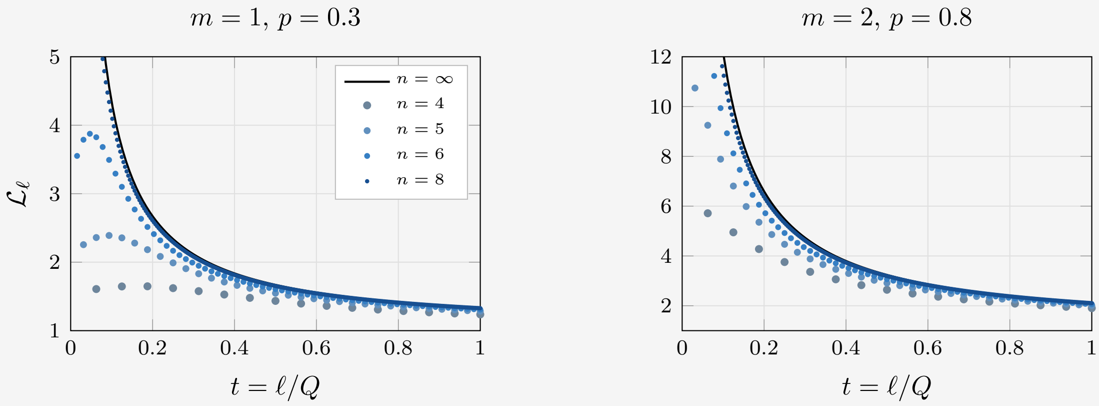


Figure 6: The fixed-p thermodynamic curve (86), on a logarithmic vertical scale. Every weight produces the same $1/t$ shape; the prior sets only the coefficient, which grows from 0.001 at $p = 0.001$ to above 5 at $p = 0.999$.

EXAMPLE 33

The approach to the hyperbolic limit (86). Both panels evaluate the exact finite law (64) on the block entropies (82) at increasing finite n, and set it against the thermodynamic curve in black.



Every finite curve approaches the limit from below, and the gap closes last at the left edge: the hyperbola diverges as $t \to 0$, while a run of Q questions has its first point at $t = 1/Q$ with a finite value there. The boundary layer is a limiting phenomenon.

4.1.3 Exchangeable answers and de Finetti's theorem

The block law (79) has a strong property. Whatever set $\mathcal{S}$ Werner asks, the all-zero pattern $y = 0^{mk}$ carries probability P_k and every other pattern carries U_k. Both depend on $\mathcal{S}$ solely through its size k. Which questions were asked, and in which order, is not relevant. A prior with that property is called **exchangeable**, and a classical theorem constrains such priors to be equivalent to a restrictive construction.

Write the answers in the order they arrive as a sequence of random symbols,

$$\xi_i \;:=\; \psi(q_i) \;\in\; \mathcal{A}, \qquad i = 1, \dots, Q, \tag{87}$$

so that ξ is the random object Werner actually observes (in this case m bits each), one coordinate per question-answer. The sequence is exchangeable when its joint law is unchanged by permuting the indices: for every permutation σ and every pattern y,

$$\Pr\left(\xi_1 = y_1, \dots, \xi_k = y_k\right) \;=\; \Pr\left(\xi_{\sigma(1)} = y_1, \dots, \xi_{\sigma(k)} = y_k\right). \tag{88}$$

Exchangeability is weaker than independence. Independent answers tell each other nothing, whereas exchangeable ones may be strongly dependent; all that is asked is that they be dependent *symmetrically*, so that no question occupies a special position.

De Finetti's theorem [8, 9] states that symmetric dependence, for sequences of unbounded length, is equivalent to a *mixture of coins*. Let ν be a distribution on the answer alphabet $\mathcal{A}$, an **answer law**, writing $\nu(a) = \Pr(a)$ for the weight it puts on the answer a. Let Λ be a distribution over such answer laws, the **directing measure**. Then for an exchangeable sequence there is exactly one Λ with

$$\Pr\left(\xi_1 = y_1, \dots, \xi_k = y_k\right) \;=\; \int \prod_{i=1}^{k} \nu(y_i)\, d\Lambda(\nu), \tag{89}$$

for every k and every pattern. The RHS is executed as a two-stage experiment. Nature first draws one answer law ν from Λ, once and for all, and then answers every question independently from that single ν. Conditional on ν the answers are independent; unconditionally they are correlated. The correlation is mediated by Werner considering each ν possible a priori. The theorem needs the sequence to extend to an infinite one, or equivalently the family of finite laws to be consistent as Q grows; a finite exchangeable law taken in isolation need not be a mixture of independent draws of ν. Sampling without replacement is the standard counterexample. In the infinite set limit, sampling with and without replacement is equivalent.

In general Λ may spread over any collection of answer laws $\nu_1, \nu_2, \ldots$, finite, countable or a continuum, and the richer that collection the more there is to learn about which one was drawn.

Such a ν has appeared once already, at (66), but it is used differently here. There, one draw from ν was copied as the answer to every question, which in the present language is Λ only sampling from such degenerate laws, carrying weight $\nu(a)$ on the law that answers a with certainty: Perfect correlation of answers. Here, after drawing ν from Λ, each answer is independent, the opposite.

The spike prior is the simplest nontrivial such example: Λ is a discrete distribution on two points. Let ν_0 be the spike, and let $\nu_>$ be the uniform law, which allows answers $\neq 0^m$ as well. These distributions on $\mathcal{A}$ have probabilities

$$\nu_0(a) = \delta_{a,0^m}, \qquad \nu_>(a) = u \tag{90}$$

Then

$$\Lambda \;=\; \Big(p - \frac{1-p}{N-1}\Big)\,\delta_{\nu_0} \;+\; (1-p)\,\frac{N}{N-1}\,\delta_{\nu_>}, \tag{91}$$

with δ_ν the point mass at ν, reproduces the spike exactly. Substituting (91) into (89), the confirming pattern 0^{mk} receives full weight from ν_0 and u^k from $\nu_>$, while every other pattern receives nothing from ν_0 and u^k from $\nu_>$:

$$\begin{aligned} \Pr\big(\psi(\mathcal{S}) = 0^{mk}\big) &= \Big(p - \frac{1-p}{N-1}\Big) + (1-p)\,\frac{Nu^k}{N-1} = p + (Nu^k - 1)\,\omega, \\ \Pr\big(\psi(\mathcal{S}) = y\big) &= (1-p)\,\frac{Nu^k}{N-1} = Nu^k\omega \qquad (y \neq 0^{mk}), \end{aligned} \tag{92}$$

which are (73) and (79) exactly. The prior on boolean maps comes back too. A map is answered by ν_0 on all questions only if it is ϕ_0, and by $\nu_>$ with weight $u^Q = 1/N$ whatever its truth table holds. We can find the prior probability of ϕ_0 by reconstructing the probability of only receiving zero answers, through either ν. Starting from (89), substituting (91) at $k = Q$ returns

$$\Pr\big(\xi_1 = 0^m, \ldots, \xi_Q = 0^m\big) \;=\; \Big(p - \frac{1-p}{N-1}\Big)\cdot 1^Q + (1-p)\,\frac{N}{N-1}\cdot u^Q = p, \tag{93}$$

And the reader can convince themselves that the probabilities of every other map (where the first term vanishes) are equal. $p_j = \omega$. Together, these reproduce (71).

Why is the leverage a hyperbola? All the prior knows is which of the two answer laws ν was drawn, and that is settled within $o(Q)$ questions. Afterwards, every remaining answer is an independent draw from a known law and costs the same $\mathbb{E}_\Lambda[H(\nu)] \to (1-p)m$ bits of surprisal, so the profile γ is flat on $(0,1)$ and (105) has no freedom left but a $1/t$ curve. The coefficient can be read off as well. Since

$$\eta_0 - (1-p)m \;=\; H\big(\mathbb{E}_\Lambda\,\nu\big) - \mathbb{E}_\Lambda\big[H(\nu)\big] \;=\; I\big(\nu\,;\,\xi_1\big), \tag{94}$$

the numerator of (86) is the mutual information between the drawn answer law and a single answer: the whole of the spike's leverage is what one question reveals about which world Werner is in, divided by what an answer costs him once he knows.

For the spike, $p_j = f(|j|)$ here in its simplest form: the weight of a map depends on its truth table only through whether the Hamming weight $|j|$ vanishes. This generalizes, and both the exchangeability and its consequence for the limit are recognizable from the prior alone. Any $\{p_j\}$ whose weight depends on the truth table only through the *count* of each answer is exchangeable, and vice versa. Under convergence of those counts in distribution to some Λ as $n \to \infty$, the profile $\gamma(t)$ is flat and the leverage curve is a hyperbola, whether or not any finite prior along the way is literally a mixture of independent draws.

We have explored a number of other characteristics of the spike prior in Appendix A: the branch-by-branch effect of the first question, the extraction ratio, and various further limits, among them the sharp prior $p \to 1^-$ and its noncommutation with $n \to \infty$.

4.2 Thermodynamic-limit calculus

The spike is solvable because all subsets of questions of the same size (k) have the same entropy G_k. Averages are free. The general theory replaces that symmetry by an average over subsets, which in general have distinct entropies. Any symmetry of the prior reduces that average to a sum over orbits of k-subsets, weighted by orbit size, and the spike is the extreme case: invariant under every permutation of the questions, it has one orbit per size and G_k collapses to a single term. Less symmetrical priors divide the number of terms by their order. Nonetheless, this is instrumental in obtaining moderate n results numerically. More comments in Appendix C, for instance.

4.2.1 The finite sequence from which the limit is taken

For each fixed question set $\mathcal{S} \subseteq \mathcal{Q}$ the vector $\psi(\mathcal{S})$ collects the answers to those questions, and its entropy is taken under the prior law of ψ with $\mathcal{S}$ held fixed. The mean block entropy is that of equation (57), now carrying the system size explicitly:

$$G_{n,k} \;:=\; \binom{Q}{k}^{-1} \sum_{|\mathcal{S}|=k} H\big(\psi(\mathcal{S})\big), \qquad 0 \le k \le Q, \tag{95}$$

so that $G_{n,0} = 0$ and $G_{n,Q} = H(\psi) = H(p^{(n)})$, where $p^{(n)}$ is the prior law of the random map at size n, with increments

$$\gamma_{n,k} := G_{n,k+1} - G_{n,k}. \tag{96}$$

By the entropy chain rule, exactly as in the derivation of equation (63), the increment is the mean conditional entropy of one fresh answer,

$$\gamma_{n,k} \;=\; \mathbb{E}_{\mathcal{S},q}\big[H\big(\psi(q) \mid \psi(\mathcal{S})\big)\big], \qquad |\mathcal{S}| = k, \quad q \in \mathcal{Q} \setminus \mathcal{S}. \tag{97}$$

The increments also connect directly to erasure-channel *extrinsic information transfer* (EXIT) functions [12]. Three terms from coding theory are needed first. A **code** of length Q over $\mathcal{A}$ is a subset of $\mathcal{A}^Q$, its members the **codewords**; a truth table is a string of exactly that length, so any set of maps is a code, and the degree classes of Chapter 5 are linear ones. Its rate is $Q^{-1} \log_A$(number of codewords), the information one codeword carries per symbol. An **erasure channel** transmits a codeword one symbol at a time, each symbol independently either delivered unchanged, with probability v, or replaced by a marker recording only that a symbol was lost. An EXIT function then reports how much uncertainty is left in one symbol once every other symbol has passed through such a channel, as a function of the erasure probability $1 - v$; its integral over that probability is the rate, which is the area theorem invoked below. Choose a fresh question q uniformly and reveal each other question independently with probability $v \in [0, 1]$. Think of v as the probability that the question naturally came up, i.e. $v \approx t$, the macroscopic time. Conditional on actually revealing k questions, their set is uniform among the k-sets not containing q. The mean conditional entropy of the fresh answer is therefore

$$\eta_n(v) := \sum_{k=0}^{Q-1} \binom{Q-1}{k} v^k (1-v)^{Q-1-k} \gamma_{n,k}. \tag{98}$$

For binary answers this is the average EXIT function expressed in reveal probability; the usual erasure probability is $1 - v$. Thus $\gamma_{n,k}$ is its fixed-cardinality counterpart. Integrating the binomial weights gives its area law,

$$\int_0^1 \eta_n(v)\, dv = \frac{1}{Q} \sum_{k=0}^{Q-1} \gamma_{n,k} = \frac{H(p^{(n)})}{Q},$$

which is the code rate for a uniform prior on a binary linear code. This is the EXIT and area-theorem connection underlying the Reed–Muller result used in Chapter 5.

We write $\mathbb{E}$ in (97) to emphasize the average over the selected q distinct from $\mathcal{S}$. The additional average here: the conditional entropy averages over answer histories ϕ_j under the prior p_j on the random map. Conditioning reduces entropy, so coupling a random k-set to a random $(k{+}1)$-set yields a **staircase**

$$m \;\ge\; \gamma_{n,0} \;\ge\; \gamma_{n,1} \;\ge\; \cdots \;\ge\; \gamma_{n,Q-1} \;\ge\; 0. \tag{99}$$

This monotonicity is deterministic, the truth and the subset having already been averaged over. Recall, this is a property of the average, not of any constituent terms. Concentration of individual learning trajectories would be a separate result. The two laws (60) and (63) then read

$$\Big\langle \sum_{k=1}^{\ell} s_k \Big\rangle = G_{n,\ell}, \qquad \big\langle H\big(M^{(\ell)}\big) \big\rangle = (Q - \ell)\, \gamma_{n,\ell}, \tag{100}$$

for $0 \le \ell < Q$, with zero remaining entropy at $\ell = Q$, and equation (64) becomes

$$\mathcal{L}_{n,\ell} = \frac{Q\, G_{n,1} - (Q - \ell)\, \gamma_{n,\ell}}{G_{n,\ell}} \qquad (1 \le \ell < Q), \qquad \mathcal{L}_{n,Q} = \frac{Q\, G_{n,1}}{G_{n,Q}}. \tag{101}$$

4.2.2 A precise macroscopic-limit statement

The next step is to find continuous limits of the functions: $\gamma(t)$ from $\gamma_{n,\ell}$ and $g(t)$ from $G_{n,\ell}$.

The leverage has only these two functions in the numerator. In fact, $G_{n,1} = \gamma_{n,0}$ so we might be tempted to express everything in terms of the one function $\gamma(t)$, and find the coarse-grained limit of that. However, that would miss a crucial *delta-like* contribution to the learning.

It is instead useful to keep the initial marginal entropy separate from the macroscopic profile on the right, since that allows an $o(Q)$-question boundary layer at $t = 0$. Let

$$\eta_0 := \lim_{n\to\infty} G_{n,1}, \tag{102}$$

which in general will be allowed to converge to something distinct from $\gamma(t)$. Suppose the staircases have an almost-everywhere limit γ on $(0, 1)$, with $\gamma_{n,\lfloor tQ \rfloor} \to \gamma(t)$ at any particular t where the remaining entropy is evaluated. At such a point $\gamma(t)$ is the expected entropy of one more answer after a random t-fraction has been observed. Four hypotheses suffice:

1. **Fixed answer width.** The integer m is fixed, which supplies the uniform bound $0 \le \gamma_{n,k} \le m$.
2. **Initial-entropy convergence.** The limit η_0 exists.
3. **Profile convergence.** The staircase $t \mapsto \gamma_{n,\lfloor tQ \rfloor}$ converges almost everywhere on $(0,1)$, with direct convergence at any t where the remaining entropy is evaluated. At continuity points of a selected monotone limit this causes no ambiguity.
4. **Positive entropy density.** The limiting area is positive, $\int_0^1 \gamma(x)\,dx > 0$.

The fourth condition is the nondegenerate form of extensivity, in other words, learning continues into the bulk and M doesn't settle within a boundary around $t = 0$. It implies $G_{n,Q}/Q \to \int_0^1 \gamma > 0$.

Extend the finite increments to a staircase on $[0,1]$ by $\gamma_n(t) := \gamma_{n,\lfloor tQ \rfloor}$. Telescoping with equation (96) gives an exact Riemann-sum identity at grid points,

$$\frac{G_{n,\ell}}{Q} = \frac{1}{Q}\sum_{k=0}^{\ell-1} \gamma_{n,k} = \int_0^{\ell/Q} \gamma_n(x)\,dx, \tag{103}$$

ignoring the last undefined endpoint in the limit. Since $0 \le \gamma_n \le m$, dominated convergence gives

$$G_{n,\ell_n}/Q \to g(t) := \int_0^t \gamma; \quad (Q-\ell_n)\gamma_{n,\ell_n}/Q \to (1-t)\gamma(t), \tag{104}$$

using the coordinated construction $\ell_n = \lfloor tQ \rfloor$. Substituting both into (101) proves

$$L(t) \;=\; \frac{\eta_0 - (1-t)\,\gamma(t)}{g(t)}, \qquad g(t) = \int_0^t \gamma(x)\,dx, \qquad 0 < t < 1. \tag{105}$$

Since γ is nonnegative and nonincreasing, a positive integral makes $g(t) > 0$ for every $t > 0$. Let us go over the pieces, for physical intuition:

- $\gamma(t)$: the price of the next answer. It is the expected entropy of one fresh answer after a random t-fraction of the table has already been observed: what Werner still doesn't know about a question he hasn't asked, given everything he has. It starts at most m and decreases monotonically, because each answer can only explain away some of the next one's novelty. γ is the running rate at which surprisal is still available.
- $g(t)$: the total bill so far. Summing the price of each answer as it was bought gives the cumulative surprisal received by macroscopic time t. It is the area under the profile to the left of t, and it sits in the denominator of the leverage because it is exactly what Werner "paid".
- η_0: the size of the job at the outset. It is the marginal entropy of a single answer before anything is learned, so $Q\eta_0$ is the whole table's starting uncertainty $H(M^{(0)})$: the total to be destroyed, by asking or by deducing. It is kept as its own constant rather than read off as $\gamma(0^+)$ because learning finished in $o(Q)$ questions leaves no trace on the profile, and the gap $\eta_0 - \gamma(0^+)$ is precisely that boundary layer.
- Together: $L(t)$ reads as (everything there was to know) minus (what the unasked columns still hold), over what was paid.

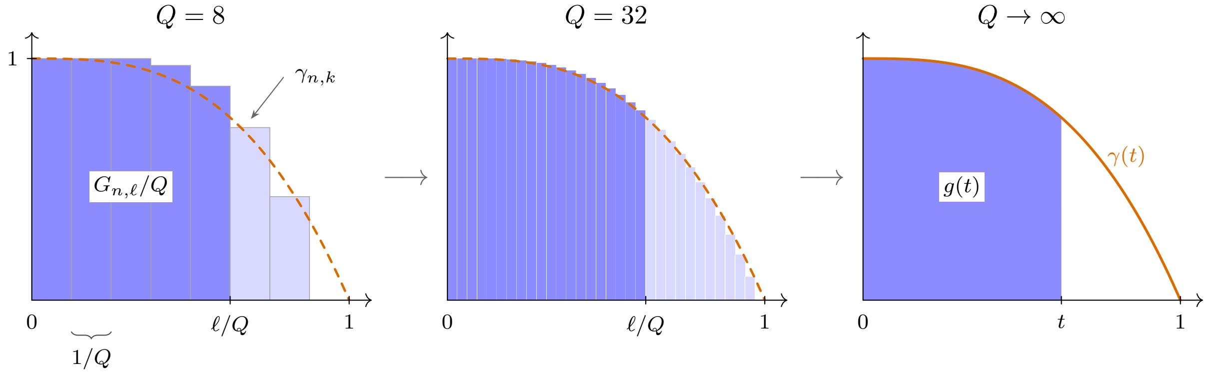


Figure 7: From the finite increments to the profile. Each bar has width $1/Q$ and height $\gamma_{n,k}$, so the shaded bars sum, the telescoping identity (103), to exactly $G_{n,\ell}/Q$, here at $t = \ell/Q = 5/8$. Refining Q shrinks the bars onto the staircase's limit, and dominated convergence turns the shaded sum into the area $g(t) = \int_0^t \gamma$: received information per question becomes the dark area under the profile.

Monotonicity and boundedness guarantee convergent subsequences by a Helly selection argument [7], but they do not guarantee that the full sequence of priors selects a unique profile: a family can alternate between two constructions on even and odd n. At the prior level one therefore needs a coherent construction, or must take profile convergence itself as a hypothesis. In the next chapter, as well as the appendices, we give examples and explain why they converge neatly.

5 Polynomial degree as complexity prior

What could the TDL of an explicit simplicity-based prior look like? Unfortunately, but not unexpectedly, we also must grapple with no-free-lunch. It is not possible to find general circuit complexities as in Subsection 3.2.1 for large n. However, Werner has another avenue at his disposal: write the boolean map as a polynomial in the question inputs, and consider the complexity to be the *degree of that polynomial.* He can construct a prior of decreasing faith with this degree, which turns out to work elegantly. It is *constructive*: computable from the truth table by a fast transform, thus tractable at every n.

There are two guiding insights in this chapter. First, we are interested in demonstrating that the leverage in the TDL can be more than a hyperbola. In order for this to happen, the rate of information $\gamma(t)$ cannot be a flat constant in the bulk. This flatness is conspicuously difficult to escape. Many naive constructions are virtually "finding out what latent set of maps Werner is in". The simplest example was "spike-or-not" of Section 4.1. Even when this is made richer with more variation, it still costs $o(Q)$ bits to learn the set label, and therefore concludes around $t = 0^+$. Once the set is settled (no pun intended), the rate $\gamma(t)$ inside is often constant. Instead, we need a construction that maintains stores of information that don't settle until macroscopic times. The polynomial degree prior has that property.

Second, weighting maps by complexity through a single temperature, with an exponentially vanishing weight with degree like the Gibbs ensemble does, reduces to only effectively consider a thin shell of contributing maps. At low degrees, there are too few maps. At high, there's too little weight.

This may be desirable in statistical physics to find averages, but as a philosophical choice, Werner won't rule anything out in his learning journey. The repair is to *sample* maps from a shared quota of weight per degree, which keeps every shell, and every single map, at positive weight all the way into the thermodynamic limit.

Throughout this chapter the answer size is $m = 1$, so $u = \frac{1}{2}$ and a fresh uninformed answer carries at most one bit. As we saw before, larger A mainly changes the rate of culling, and behaves much like m independent maps being learned simultaneously. The single-bit case is fully instructive, and costs little generality.

5.1 Polynomials over $\mathbb{F}_2$, and the butterfly

All linear algebra in this chapter is over the smallest field there is, the two-element field $\mathbb{F}_2$, whose two operations are

$$\mathbb{F}_2 = \{0, 1\}, \qquad b \oplus b' \;=\; b + b' \bmod 2 \quad (\text{XOR}), \qquad b\,b' \quad (\text{AND}). \tag{106}$$

Three observations:

1. Every element is its own additive inverse, $b \oplus b = 0$, so there is no subtraction and there are no minus signs: a term contributed twice cancels.

2. Every element is idempotent, $b^2 = b$, so no variable is ever squared, and a product of input bits is fixed by *which* bits it contains, never by how often.

3. All matrices, vectors, ranks and inverses below are read mod 2.

These define the rules of this space and the functions that inhabit it.

As a consequence, we only need to consider *monomials* in the input bits $b_{n-1} \cdots b_0$ of a question. Each monomial is labeled by the set of bit positions it contains: $\phi = \prod_i b_i$ is the map answering 1 exactly when every bit is 1, the AND of those bits, and the empty set gives the empty product, the constant map 1. Write, for example, $\mathcal{B} = \{b_1, b_0\}$ for a set of input bits, and the subscript on a coefficient $\rho \in \{0,1\}$ is tantamount to the monomial itself: $\rho_{b_1 b_0}$ represents $\rho_{b_1 b_0} \cdot b_1 \cdot b_0$, the monomial made from $\mathcal{B}$, just as $\rho_\emptyset$ belongs to the constant. There are $Q = 2^n$ bit sets, hence Q monomials. Creating functions in this space is akin to deciding which monomials are present, which $\rho_\mathcal{B}$ are one (remember property 1: no non-unit coefficients). There are $2^Q = N$ ways to switch them on and off: exactly as many as there are maps, and the correspondence is one to one.

One more piece of vocabulary. The **support** of a question is the set of bit positions where it holds a 1,

$$\mathrm{supp}(q) \;=\; \{\, i : b_i(q) = 1 \,\}. \tag{107}$$

A question and a set of bits are the same data in two manifestations, and we use the identification freely: $\mathcal{B}$ names a monomial $\rho_\mathcal{B}$ and equally the question whose 1s sit exactly at $\mathcal{B}$. The condition $\mathrm{supp}(q') \subseteq \mathcal{B}$ then picks out the **subcube below** $\mathcal{B}$: the $2^{|\mathcal{B}|}$ questions q' that are 0 everywhere outside $\mathcal{B}$ and free inside it.

EXAMPLE 34

For instance at $n = 4$, $q = 1010$ has $\mathrm{supp}(q) = \{b_1, b_3\}$. If this is $q \simeq \mathcal{B}$, the subcube below is $q' \in \{1010, 1000, 0010, 0000\}$.

Every map now has a unique vector of single-bit coefficients $\rho_\mathcal{B} \in \{0,1\}$, saying whether monomial $\mathcal{B}$ is present or absent, and the dictionary between the two descriptions is the *algebraic normal form*,

$$\phi(q) \;=\; \bigoplus_{\mathcal{B}} \rho_\mathcal{B} \prod_{i \in \mathcal{B}} b_i(q) \;=\; \bigoplus_{\mathcal{B} \subseteq \mathrm{supp}(q)} \rho_\mathcal{B}. \tag{108}$$

It is worth being explicit about which symbol does what. Above, q is the free variable, the question being asked, and $\mathcal{B}$ a dummy running over all Q monomials. This is *evaluation*, coefficients in, an answer out. The middle expression is the same sum with the dead ($\rho = 0$) terms dropped, since $\prod_{i \in \mathcal{B}} b_i(q)$ equals 1 precisely when $\mathcal{B} \subseteq \mathrm{supp}(q)$. How to go the other way?

$$\rho_\mathcal{B} \;=\; \bigoplus_{\mathrm{supp}(q) \subseteq \mathcal{B}} \phi(q). \tag{109}$$

Conversely, here $\mathcal{B}$ is the free variable, the coefficient being extracted, and q the dummy, running over the subcube below $\mathcal{B}$: this is *inversion*, truth table in, one coefficient out. The dictionary is both identities, the second formula being Möbius inversion. The **degree** $\deg \phi$ is the largest $|\mathcal{B}|$ carrying $\rho_\mathcal{B} = 1$; it is the complexity notion of this chapter.

Both directions are in fact literally the same matrix. Write T for the *truth table* of a map read as a vector, $T[q] = \phi(q)$, with the questions in the descending order of Table 4, and ρ for the coefficient vector with the bit sets in the matching order. Per bit the kernel is upper triangular, and on n bits its tensor power is the subset indicator: with rows and columns descending $11, 10, 01, 00$ (per bit: $1, 0$),

$$Z = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \left(Z^{\otimes n}\right)_{\mathcal{B},q} = \mathbf{1}[\mathrm{supp}(q) \subseteq \mathcal{B}], \qquad Z^{\otimes 2} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{110}$$

where $\mathbf{1}[\cdot]$ evaluates a statement to a bit (1 is True, 0 is False). The matrix is called the *XOR-butterfly*, owing to the mod-2 space. This way, row $\mathcal{B}$ is the indicator of the subcube below $\mathcal{B}$. Under the identification of bit sets with questions the two halves of (108) are the one matrix multiplying the two vectors,

$$\rho \;=\; Z^{\otimes n}\, T, \qquad T \;=\; Z^{\otimes n} \rho, \tag{111}$$

at any n: to read off a map's polynomial, transform its truth table; to tabulate a polynomial, transform its coefficients. The two are the same operation because over $\mathbb{F}_2$ the kernel is its own inverse, as the one-bit case already shows: Z^2 has entries $1, 2, 0, 1$, and $2 = 0$, so $Z^2 = I$ and $(Z^{\otimes n})^2 = (Z^2)^{\otimes n} = I$ by extension.

In practice, computing ρ is an efficient operation up to high n due to its tensor product nature. See an example on the $(n, m) = (2, 1)$ space.

EXAMPLE 35

Take NAND, map $j = 7$ of Table 4, which answers 0 only when both input bits are 1, so that

$$T = \big(\phi(11), \phi(10), \phi(01), \phi(00)\big) = (0, 1, 1, 1).$$

Its four supports are $\mathrm{supp}(11) = \{b_1, b_0\}$, $\mathrm{supp}(10) = \{b_1\}$, $\mathrm{supp}(01) = \{b_0\}$ and $\mathrm{supp}(00) = \emptyset$: the same four bit sets that label the four monomials. For each $\mathcal{B}$, gather the questions whose support fits inside it. As the set grows, so does its subcube:

$$\begin{aligned}
&\mathcal{B} = \emptyset && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{00\},\\
&\mathcal{B} = \{b_0\} && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{01, 00\},\\
&\mathcal{B} = \{b_1\} && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{10, 00\},\\
&\mathcal{B} = \{b_1, b_0\} && \{q : \mathrm{supp}(q) \subseteq \mathcal{B}\} = \{11, 10, 01, 00\}.
\end{aligned}$$

Each coefficient is the XOR of T over its own subcube, so the four lines below have identical shape and only the subcube grows:

$$\begin{aligned}
\rho_\emptyset &= \phi(00) && = 1 && = 1,\\
\rho_{b_0} &= \phi(01) \oplus \phi(00) && = 1 \oplus 1 && = 0,\\
\rho_{b_1} &= \phi(10) \oplus \phi(00) && = 1 \oplus 1 && = 0,\\
\rho_{b_1 b_0} &= \phi(11) \oplus \phi(10) \oplus \phi(01) \oplus \phi(00) && = 0 \oplus 1 \oplus 1 \oplus 1 && = 1.
\end{aligned}$$

Stacking those four lines is (111) at $n = 2$,

$$\begin{pmatrix} \rho_{b_1 b_0} \\ \rho_{b_1} \\ \rho_{b_0} \\ \rho_\emptyset \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \phi(11) \\ \phi(10) \\ \phi(01) \\ \phi(00) \end{pmatrix} = \begin{pmatrix} 0 \oplus 1 \oplus 1 \oplus 1 \\ 1 \oplus 1 \\ 1 \oplus 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

where the two middle rows show the cancellation that makes a monomial absent.

Read the other way, (111) evaluates. Ask $q = 01$: its support is $\{b_0\}$, so row 01 gives $\phi(01) = \rho_\emptyset \oplus \rho_{b_0} = 1 \oplus 0 = 1$, which is what $b_1 b_0 \oplus 1$ returns at $b_1 = 0$, $b_0 = 1$. Hence NAND $= b_1 b_0 \oplus 1$, of degree 2: the top monomial is present, NAND is not affine, and its truth table knows it.

For readers of Section 2.5: Z is the triangular half of the Walsh–Hadamard kernel (40), since in the same $1, 0$ ordering

$$W = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = Z \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix} Z^{\mathsf{T}}. \tag{112}$$

The ledger transform is two Möbius passes with a $(-2)^{|T|}$ rescaling between them, and reducing mod 2 leaves exactly the XOR butterfly.

EXAMPLE 36

The butterfly on every map of the guiding example, truth tables $T = (\phi(11), \phi(10), \phi(01), \phi(00))$ – so that T read as a binary numeral *is* the index j – and coefficients ordered $(\rho_{b_1 b_0}, \rho_{b_1}, \rho_{b_0}, \rho_\emptyset)$.

j	map	T	ρ	polynomial	deg
0	FALSE	$(0,0,0,0)$	$(0,0,0,0)$	0	0
1	NOR	$(0,0,0,1)$	$(1,1,1,1)$	$b_1 b_0 \oplus b_1 \oplus b_0 \oplus 1$	2
2	$\neg b_1 \wedge b_0$	$(0,0,1,0)$	$(1,0,1,0)$	$b_1 b_0 \oplus b_0$	2
3	$\neg b_1$	$(0,0,1,1)$	$(0,1,0,1)$	$b_1 \oplus 1$	1
4	$b_1 \wedge \neg b_0$	$(0,1,0,0)$	$(1,1,0,0)$	$b_1 b_0 \oplus b_1$	2
5	$\neg b_0$	$(0,1,0,1)$	$(0,0,1,1)$	$b_0 \oplus 1$	1
6	XOR	$(0,1,1,0)$	$(0,1,1,0)$	$b_1 \oplus b_0$	1
7	NAND	$(0,1,1,1)$	$(1,0,0,1)$	$b_1 b_0 \oplus 1$	2
8	AND	$(1,0,0,0)$	$(1,0,0,0)$	$b_1 b_0$	2
9	XNOR	$(1,0,0,1)$	$(0,1,1,1)$	$b_1 \oplus b_0 \oplus 1$	1
10	b_0	$(1,0,1,0)$	$(0,0,1,0)$	b_0	1
11	$b_1 \to b_0$	$(1,0,1,1)$	$(1,1,0,1)$	$b_1 b_0 \oplus b_1 \oplus 1$	2
12	b_1	$(1,1,0,0)$	$(0,1,0,0)$	b_1	1
13	$b_0 \to b_1$	$(1,1,0,1)$	$(1,0,1,1)$	$b_1 b_0 \oplus b_0 \oplus 1$	2
14	OR	$(1,1,1,0)$	$(1,1,1,0)$	$b_1 b_0 \oplus b_1 \oplus b_0$	2
15	TRUE	$(1,1,1,1)$	$(0,0,0,1)$	1	0

The guiding example in both bases: every map of Table 4 as a truth table T and as a coefficient vector ρ, with its polynomial and its degree.

Read the table from both ends at once: j and $15-j$ are negations of each other, and their coefficient vectors differ in $\rho_\emptyset$ alone, negation flipping the constant and nothing else. Four maps are fixed points of the butterfly, $\rho = T$: FALSE, XOR, AND and OR. The eight with $\rho_{b_1 b_0} = 0$ are the affine ones, class RM$(1,2)$, and the other eight are the curved ones that the top monomial buys.

5.2 Complexity classes and the sampling rule

Using the dictionary, every map is bijective with a polynomial, and each polynomial has a degree. Grade the hypothesis space by degree. The class of all maps of degree at most d is a linear space of dimension

$$K_d = \sum_{i \le d} \binom{n}{i}. \tag{113}$$

K_d counts one coefficient bit $\rho_\mathcal{B}$ per allowed monomial. It is a linear space because polynomials distribute, and any sum of polynomials, each with degree $\le d$, is again a polynomial of degree $\le d$, and therefore the space is closed.

The classes run from the two constant polynomials $\rho_\emptyset \in \{0,1\}$ $(d=0)$ to the full map space $(d=n)$. In coding theory this class is the Reed-Muller code RM(d,n) [11]. We posit two definitions to avoid confusion:

- the **class** d is the union of all maps of degree at most d. Classes are nested.

- A class's outermost layer, the maps of degree exactly d, is the **shell** d. Shells are disjoint.

The class's **rate** is its dimension per question,

$$R_{n,d} = \frac{K_d}{Q} \in (0,1], \tag{114}$$

and it will turn out to be the fraction of a full run at which the class is "learned out", i.e., Werner has settled whether the map ψ is this class or not. Consecutive rates slice $(0,1]$ into **rate gaps** $(R_{n,d-1}, R_{n,d}]$ of lengths

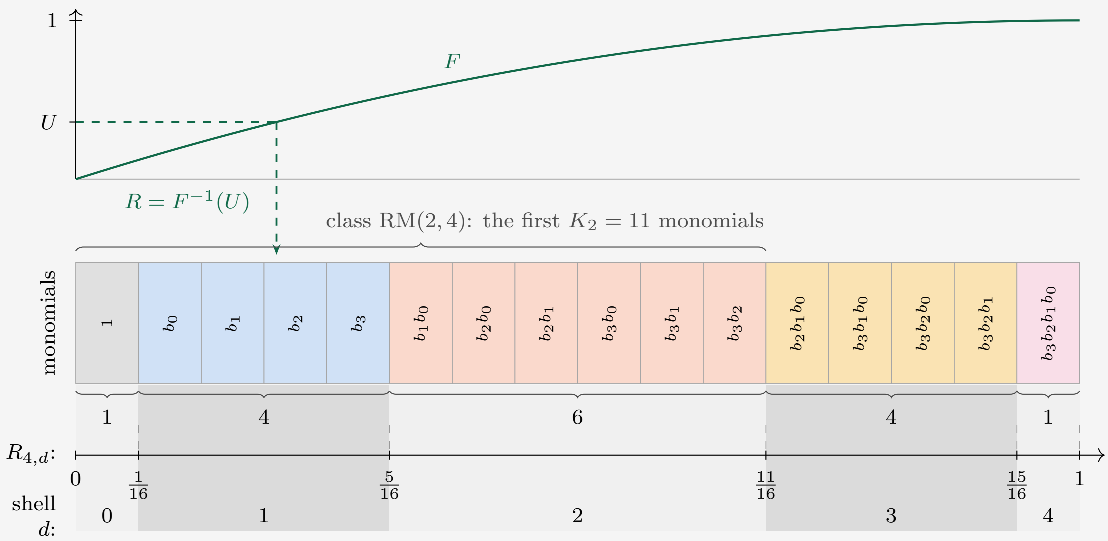
$$R_{n,d} - R_{n,d-1} = \binom{n}{d}/2^n \leq \sqrt{2/\pi n}. \tag{115}$$

This partition refines as n grows. Rate gaps can be seen as the proportional thickness of the shells.

EXAMPLE 37

This graphic explains the stochastic process that samples ψ. Sampling $\sim F$ determines a class (a maximum degree d). Inside that linear space, any polynomial is equally likely. Higher classes have exponentially more maps in them. The graphic illustrates $n = 4$: take one box per monomial, grouped by d into shells. Each box is one coefficient bit, which has 50-50 odds to be included in the true polynomial of ψ. So K_d, the number of monomials under consideration *is* the class $\mathrm{RM}(d,4)$ measured in bits, holding 2^{K_d} maps, and shell d is the $\binom{4}{d}$ boxes the class adds to the previous. The boxes lie ordered along a sampled line. Dividing the position of the edge of a class by $Q = 16$ puts the class boundaries at the rates $R_{4,d} = K_d/16$, with the shells inhabiting the rate gaps.

The shell widths $1, 4, 6, 4, 1$ below the boxes are the populations' logarithmic footprints, while the map *counts* 2^{K_d} explode left to right.



So far, the structure is fixed and none of it depends on F. The statistic enters only through the dart: draw U uniformly on the vertical axis, bounce off the graph of F, and land at $R = F^{-1}(U)$, drawn here for $F(x) = 1 - (1-x)^2$ (the smaller of two uniform draws: the $r = 2$ member of the constant-leverage family (131)), which favors simple (low-degree) maps. The rate-gap the dart hits determines D, the maximum d. For this dart, $D = 1$, and we sample ψ uniformly from polynomials of the form $\rho_\emptyset \oplus \bigoplus_i \rho_{b_i} b_i$.

The exponential explosion in numbers: the share of maps of degree exactly d is $2^{K_d-Q} - 2^{K_{d-1}-Q}$, and

degree	share	$n=2$	$n=4$	$n=6$	$n=8$
n	$\frac{1}{2}$	50%	50%	50%	50%
$n-1$	$\frac{1}{2} - 2^{-(n+1)}$	37.5%	46.9%	49.2%	49.8%
$n-2$	$\approx 2^{-(n+1)}$	12.5%	3.1%	0.78%	0.20%
$n-3$	$\approx 2^{-(n+1+\binom{n}{2})}$	-	$4.6 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$7.3 \cdot 10^{-12}$
$\leq n-4$	rest	-	$3.1 \cdot 10^{-5}$	$2.3 \cdot 10^{-13}$	$1.0 \cdot 10^{-28}$

Table 7: The share of all 2^Q maps carried by each degree. Half sit at full degree n at every size, and the shells below thin out geometrically.

Half of all maps sit at full degree, all but a $2^{-(n+1)}$ sliver in the top two shells, and everything below is doubly exponentially rare: this is the population contour any per-map weight has to compensate. In our case, we allocate a macroscopic amount of weight to each shell by construction. Then divide that weight among the constituent maps. This scales nicely in the TDL, to never effectively rule out any degree, but also keep nonzero weight on each individual map.

1. draw a single random number $R \in [0, 1]$ (the **statistic**) from a fixed law with CDF $F(x) = \Pr(R \leq x)$ and, when it exists, density F';

2. let D be the class whose rate gap contains R;

3. draw the truth uniformly from class D: each *allowed* monomial coefficient ρ by a fair coin.

So R and D only decide which monomials are ruled out as too high a degree. The $R_{n,d}$ form the deterministic grid that R lands in. Only $d \leq D$ are allowed: those with $R_{n,d-1} < R$.

The class label D is *latent*, meaning Werner never observes it. It is one value among $n+1$, worth at most $\log_2(n+1)$ bits. Due to nested classes, a map ϕ_j of degree d can result from any $D \geq d$. To find the probability of that map, we integrate over D.

$$w_d \;=\; F(R_{n,d}) - F(R_{n,d-1}), \qquad p_j \;=\; \sum_{d \geq \deg \phi_j} w_d \, 2^{-K_d}, \tag{116}$$

with $\sum_d w_d = 1$ telescoping. If F increases across every gap (any positive density), then every $w_d > 0$, and two properties follow trivially:

- p_j is *strictly decreasing in the degree*, each step up losing a factor of order $2^{-\binom{n}{d}}$
- $p_j \geq w_n 2^{-Q} > 0$ for every single map: full support, at every n and in the limit.

Every shell, and every map, gets a chance, and still it's a genuine Occam prior. Semantically, F budgets the *shells* while nestedness orders the *maps*: w_d is, up to an exponentially suppressed sliver leaking in from higher classes, the total weight of the degree-d shell, split evenly within it.

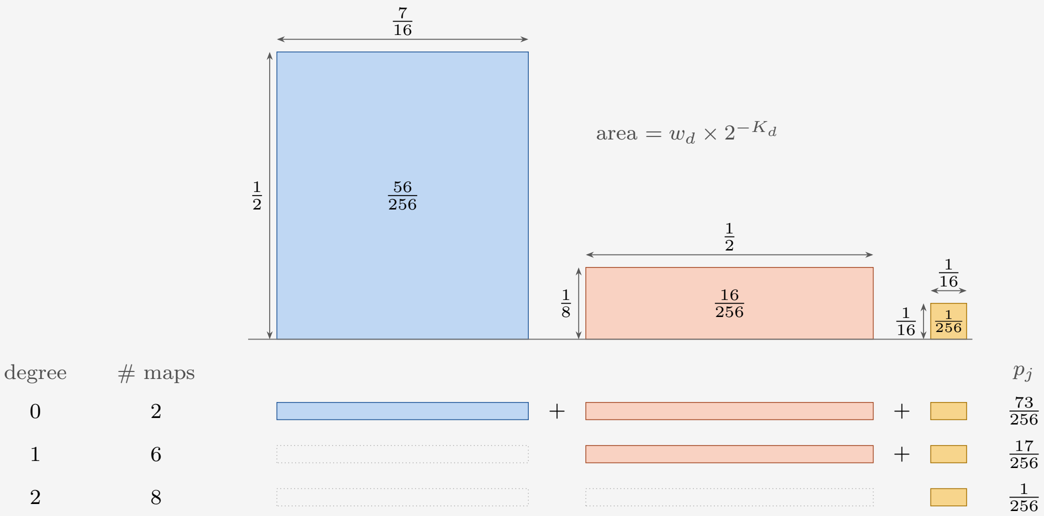
A judicious choice of F is what puts the correct portion of aggregate belief on the complexity that Werner expects from nature. The per-map Occam ordering holds for every F, and F decides only how much aggregate weight each shell carries. The Occam ordering comes from the natural explosion of maps at higher degrees, sharing the shell's marginal weight. The heaviest shell itself is rarely the simplest one: it is whichever rate gap carries the most F-mass. The d maximizing w_d from equation (116) is a contest between the gap width $\binom{n}{d}/2^n$, widest at $d \approx n/2$, and the density of F there. See Example 43 at the end of the chapter for an illustration.

EXAMPLE 38

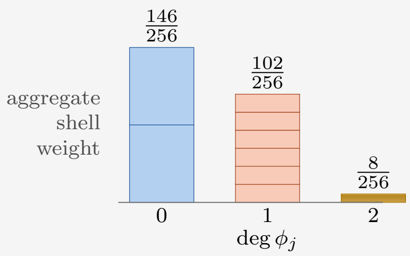
A geometric interpretation of the prior (116) at $(n, m) = (2, 1)$ with the guiding statistic $F(x) = 1 - (1 - x)^2$, which is concave (the $r = 2$ member of (130)). The three classes have

$$(K_0, K_1, K_2) = (1, 3, 4),$$
$$(w_0, w_1, w_2) = (\tfrac{7}{16}, \tfrac{1}{2}, \tfrac{1}{16}),$$

so the rates K_d/Q are $\frac{1}{4}$, $\frac{3}{4}$ and 1. Class d offers each map it contains a rectangle: width w_d, the chance the lottery calls that class, times height 2^{-K_d}, the chance the class then calls that map. A map of degree d lies in every class $D \geq d$, so it collects every block from D rightward: all the blocks a harder map collects, and one more besides.



Each step up in degree curtails the sum from the left, so p_j falls with the degree by construction, regardless of F: the two constants of Table 4 take all three blocks, $\frac{73}{256}$, the six affine maps lose the biggest, $\frac{17}{256}$, and the eight curved maps keep only the last, $\frac{1}{256}$. The blocks are drawn to scale, illustrating the Occam prior. A concave F boosts early widths. The small occupancy of low shells boosts early heights. The check $2 \cdot 73 + 6 \cdot 17 + 8 \cdot 1 = 256$ closes the distribution.



5.3 Three terms for the leverage average

The exact finite law (64), in the TDL chapter's notation (101), needs three ingredients:

1. the prior table entropy $Q\,G_{n,1}$,
2. the received entropy $G_{n,\ell}$,
3. the increment $\gamma_{n,\ell}$.

Each reduces to a statement about F.

5.3.1 The table entropy

A class is a linear code. We'll make the structure explicit. The **generator matrix** $\Omega^{(d)}$ of class d is a $K_d \times Q$ matrix over $\mathbb{F}_2$. Monomials are trivially themselves polynomials, and therefore boolean maps. $\Omega^{(d)}$'s rows are the monomials $\mathcal{B}$ that the class contains, evaluated at every question q.

$$\Omega^{(d)}_{\mathcal{B},q} \;=\; \prod_{i\in\mathcal{B}} b_i(q) \;=\; \mathbf{1}[\,\mathcal{B} \subseteq \operatorname{supp}(q)\,], \qquad |\mathcal{B}| \le d, \tag{117}$$

so that a uniformly random class member is $\psi = \xi^{\mathsf{T}}\Omega^{(d)}$ with ξ uniformly random on $\{0,1\}^{K_d}$. Every column of $\Omega^{(d)}$ is nonzero (its constant-monomial entry is 1), so each single answer is the image of a uniform ξ under a surjective linear map onto $\mathbb{F}_2$: a fair bit, in every class. Any bit that is flipped with 50% chance is fair, which is what happens when the constant bit is added. In other words, the initial M is fully uniform. At every finite n, with no limit taken and no reference to $F(x)$,

$$G_{n,1} = 1, \qquad H\big(M^{(0)}\big) = Q. \tag{118}$$

The prior predicts nothing about any single answer; its entire content is correlational.

EXAMPLE 39

The generator matrix Ω at $n = 2$: the questions across the top in descending order, the monomials down the side in degree order, and the entry where they cross: the value (117) gives. The three boxes are the three classes, each keeping the first K_d rows, which is what makes them nest:

$$\Omega_{\mathcal{B},q} = \begin{array}{c|cccc|l} \mathcal{B} \backslash \mathbf{q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} & \\ \hline 1 & 1 & 1 & 1 & 1 & d=0 \\ b_0 & 1 & 0 & 1 & 0 & \\ b_1 & 1 & 1 & 0 & 0 & d=1 \\ b_1 b_0 & 1 & 0 & 0 & 0 & d=2 \end{array}$$

A class member is $\psi = \xi^\mathsf{T}\Omega^{(d)}$, the element-wise XOR of the rows that the coin-sequence ξ switches on. The K_d rows are independent, so $\xi \mapsto \psi$ is a bijection onto the class: flipping K_d fair coins, one per allowed monomial, lands on each of the 2^{K_d} maps exactly once, which is step 3 of the sampling rule and why no map inside a class is favored over another. Three examples of coin-sequences, yield three maps, at three different class degrees d

$$\begin{aligned} \xi &= (1) && \Rightarrow && \psi = (1,1,1,1) && = \mathrm{TRUE}, \\ \xi &= (1,0,1) && \Rightarrow && \psi = (1,1,1,1) \oplus (1,1,0,0) = (0,0,1,1) && = \neg b_1, \\ \xi &= (1,0,1,1) && \Rightarrow && \psi = (0,0,1,1) \oplus (1,0,0,0) = (1,0,1,1) && = b_1 \to b_0. \end{aligned}$$

Every column carries the constant row's 1, so flipping $\xi_\emptyset$ flips the answer to every question at once: the 2^{K_d} maps pair off into complementary halves and $M_{a,q} = \frac{1}{2}$ at every question, in every class, for every F. The table starts knowing nothing, $H(M^{(0)}) = Q$. Compare to (110), Ω is $Z^{\otimes n}$ transposed, its rows sorted by degree so that every class is a prefix.

5.3.2 The received entropy

Let us approach this answer in steps. For a straightforward intermediate, suppose *Werner were told the class.* His prior would instantly collapse to only those maps. Inside class d the truth is $\psi = \xi^\mathsf{T}\Omega^{(d)}$ with ξ a string of K_d fair coins, so his answers are uniform on the image of the columns of $\Omega^{(d)}$ he has asked and their entropy is that submatrix's rank. Every fresh column (question & answer) independent of its predecessors buys one whole bit; once the rank of the submatrix seen this way reaches K_d, the coins are pinned. ξ is known, and every answer after that is completely determined, and there is no more entropy. Average that behavior over the classes with weight $w_d = \Pr(D = d)$ and the received entropy would be settled.

Two things spoil this reasoning at finite n, and both dissolve in the TDL. Asking K_d questions is not the same as reconstructing a full rank submatrix, because the columns are not in general position: the smallest dependent set has 2^{d+1} members, the minimum distance of the dual code $\mathrm{RM}(n-d-1, n)$ [11], and for the middle classes that falls well short of K_d. For instance, four questions against $K_1 = 5$ at $n = 4$, eight against $K_2 = 37$ at $n = 8$. Four questions whose bit patterns XOR to zero already determine each other for an affine ($d = 1$) map, whatever else is unsettled. So a question can be redundant early and the rank often is not full until late: the switch from one bit per question to none is a smeared step, not a sharp one, though the total entropy encloses area K_d regardless. Only the two ends escape, the constants ($K_0 = 1$) and the full space ($K_n = Q$) being the codes for which every K_d columns really are independent. And Werner is not told the class; he has to read it off the answers, and that costs information.

Again, let $\mathcal{S}$ be the set of questions asked at some point during learning, and D be the class label. Computing $H(\psi(\mathcal{S}), D)$ by the chain rule in both orders prices the second of these,

$$H\big(\psi(\mathcal{S})\big) = \sum_d w_d\, H^{(d)}\big(\psi(\mathcal{S})\big) + I\big(D;\psi(\mathcal{S})\big), \qquad 0 \le I \le \log_2(n+1): \tag{119}$$

mixture entropy is mean class entropy plus what the answers reveal about the label, and the label is cheap: $o(1)$ bits per question. Within class d the answers on $\mathcal{S}$ are $\xi^{\mathsf{T}}\Omega^{(d)}_{\cdot,\mathcal{S}}$, uniform on the image of the column submatrix, so $H^{(d)}(\psi(\mathcal{S})) = \operatorname{rank}(\Omega^{(d)}_{\cdot,\mathcal{S}})$ exactly: the number of $\mathbb{F}_2$-linearly independent columns, the number of bits needed to describe those answers.

Averaging (119) over the $\binom{Q}{\ell}$ sets of size ℓ turns the left side into $G_{n,\ell}$ of (95) and the right side into a mean rank, one per class:

$$\underbrace{G_{n,\ell}}_{\text{Average}} = \underbrace{\sum_d w_d\, G^{(d)}_{n,\ell}}_{\text{told the class}} + \underbrace{I_{n,\ell}}_{\le \log_2(n+1)}, \qquad G^{(d)}_{n,\ell} := \binom{Q}{\ell}^{-1} \sum_{|\mathcal{S}|=\ell} H^{(d)}\big(\psi(\mathcal{S})\big), \tag{120}$$

where $I_{n,\ell}$ is the label term of (119) averaged over the same sets, inheriting its bound. Only $G_{n,\ell}/Q$ ever enters the leverage, and $\log_2(n+1)/Q \to 0$, so the label costs a vanishing fraction of the run: $G_{n,\lfloor tQ\rfloor}/Q$ and $\sum_d w_d\, G^{(d)}_{n,\lfloor tQ\rfloor}/Q$ have the same limit at every t. It is enough to find the class averages.

EXAMPLE 40

Say Werner knows $d = 1$. This means:

$$\psi(q) = \rho_\emptyset \oplus \rho_{b_0} b_0(q) \oplus \rho_{b_1} b_1(q)$$

In the matrix below, see how the sum of all questions vanishes for such affine maps (a linear dependence).

$$\psi(11) \oplus \psi(10) \oplus \psi(01) \oplus \psi(00) = 4\rho_\emptyset \oplus 2\rho_{b_0} \oplus 2\rho_{b_1} = 0$$

Ask any three questions $\mathcal{S}$, the covered submatrix $\Omega^{(1)}_{\cdot,\mathcal{S}}$ is square and of full rank 3.

$$\Omega^{(1)} = \begin{array}{c|cccc} \mathcal{B} \backslash \mathbf{q} & \mathbf{11} & \mathbf{10} & \mathbf{01} & \mathbf{00} \\ \hline 1 & 1 & 1 & 1 & 1 \\ b_0 & 1 & 0 & 1 & 0 \\ b_1 & 1 & 1 & 0 & 0 \end{array}$$

That means the first three questions all have $\gamma_{2,\ell} = 1$, they carry a bit of surprisal. After that, at $t = R_{2,1} = \frac{3}{4}$, through a system of equations we can infer which monomials are present in the polynomial of ψ. The final q cannot have any uncertainty left, and $\gamma_{2,3} = 0$. In total then, the received entropy is capped at $K_1 = 3$.

$$G_{2,\ell} = \min(\ell, 3)$$

Three facts pin the limit of the class average $G^{(d)}_{n,\ell} = \mathbb{E}\,\mathrm{rank}\Omega^{(d)}$. The mean increment $\gamma^{(d)}_{n,\ell}$ = Pr[a fresh column is independent of ℓ random ones] is nonincreasing in ℓ, by rank submodularity (what is dependent on few columns is dependent on more). The increments telescope to the dimension,

$$\sum_{\ell<Q} \gamma^{(d)}_{n,\ell} = K_d, \tag{121}$$

the finite-n area theorem. Inside a class d, there can be at most K_d linearly independent questions, and at most K_d bits to learn. And we must invoke one external result: Reed-Muller codes achieve capacity on the binary erasure channel. Kudekar, Kumar, Mondelli, Pfister, Şaşoğlu and Urbanke [12] showed that revealing any fraction beyond the rate determines a fresh answer with probability tending to one:

$$R_{n,d_n} \to R \in (0,1) \quad \Longrightarrow \quad \lim_{n\to\infty} \gamma^{(d_n)}_{n,\lfloor tQ\rfloor} = 0 \quad \text{for every fixed } t > R. \tag{122}$$

So if the class rate tends to the statistic R, the support of the entropy rate tends to below R. Here $m = 1$, which bounds the increment: $\gamma \in [0, 1]$. Such a nonincreasing sequence with area R_{n,d_n} and no mass to the right of it has no freedom. These characteristics force the shape, so along that same sequence of classes the increments converge pointwise away from $t = R$ to a pure step, written with the Heaviside function θ, which is 1 when its argument is positive and 0 otherwise:

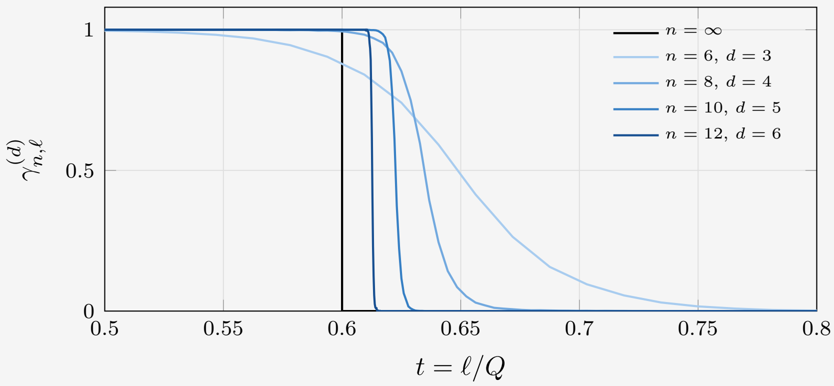
$$\lim_{n\to\infty} \gamma^{(d_n)}_{n,\lfloor tQ\rfloor} \;=\; \theta(R-t), \qquad \frac{G^{(d_n)}_{n,\lfloor tQ\rfloor}}{Q} \;\longrightarrow\; \int_0^t \theta(R-x)\,dx = \min(t, R): \tag{123}$$

each class pays one full bit per question until its tank runs out at its own rate, and nothing after. No pointwise limit is asserted at the step discontinuity $t = R$; its assigned value does not affect the integral.

EXAMPLE 41

Where the step in γ comes from. Conditional on the latent class D, the answers on $\mathcal{S}$ are $\xi^{\mathsf{T}}\Omega^{(d)}_{\cdot,\mathcal{S}}$, so their entropy is the rank of that column submatrix, and $\gamma^{(d)}_{n,\ell}$ is the chance a fresh column is still independent of the ℓ already asked.

To see the limit approach, hold the draw fixed at $R = 0.6$ and let n grow. The sampling rule hands back whichever class's rate gap contains it, so the class index climbs with n, taking $d = 3, 4, 5, 6$ below, while R itself stays put; by (122) the increment tends to the black step at R. Outside the window γ is flat: one to the left, nothing to the right.



Two things converge, at very different speeds. The *shape* goes fast: the drop from $\gamma = 0.9$ down to 0.1 measures six questions at every n here, so as a fraction of the run it closes like $1/Q$, from 0.094 at $n = 6$ to 0.0015 at $n = 12$. The *position* of the transition goes slowly with n: each curve steps at its own rate, which is R rounded up to the grid and so overshoots by at most the gap width $\binom{n}{d}/2^n \le \sqrt{2/\pi n}$. Here $0.056, 0.037, 0.023, 0.013$. Whatever the overshoot, each curve encloses exactly its own rate: the increments telescope to the dimension, $\sum_{\ell<Q}\gamma^{(d)}_{n,\ell} = K_d$, at every n and with no approximation.

So, for a given sampled R, class D is pinned, and it is exactly a sequence of the kind (123) asks for: $R_{n,D}$ is R rounded up to the top of its gap, and by (115) that gap is at most $\sqrt{2/\pi n}$ wide, so $R_{n,D} \to R$ and $G^{(D)}_{n,\lfloor tQ\rfloor}/Q \to \min(t, R)$. All that is left is to integrate over R to get the true prior average.

Since $\min(t, \cdot)$ is 1-Lipschitz the same gap bound controls the expectation, and the class average converges to an average over the statistic,

$$\frac{G_{n,\lfloor tQ\rfloor}}{Q} \;\longrightarrow\; \mathbb{E}_F\big[\min(t, R)\big] \;=\; \int_0^t \big(1 - F(x)\big)\,dx \;=:\; g(t). \tag{124}$$

The last step is achieved through integration by parts. Pointwise

$$\min(t, R) = \int_0^t \theta(R - x)\,dx; \quad \mathbb{E}\,\theta(R - x) = 1 - F(x), \tag{125}$$

by the defining sampling process $R \sim F'$. Received information per question is a mean of the rates per class, weighted by the expected contribution of those classes. The correction from the mutual information term from equation (119) vanishes in the TDL.

5.3.3 The increment

This one now follows for free. The sequence $\ell \mapsto G_{n,\ell}$ is concave (increments nonincreasing), and slopes of concave functions converge wherever the limit is differentiable, so at every continuity point of F

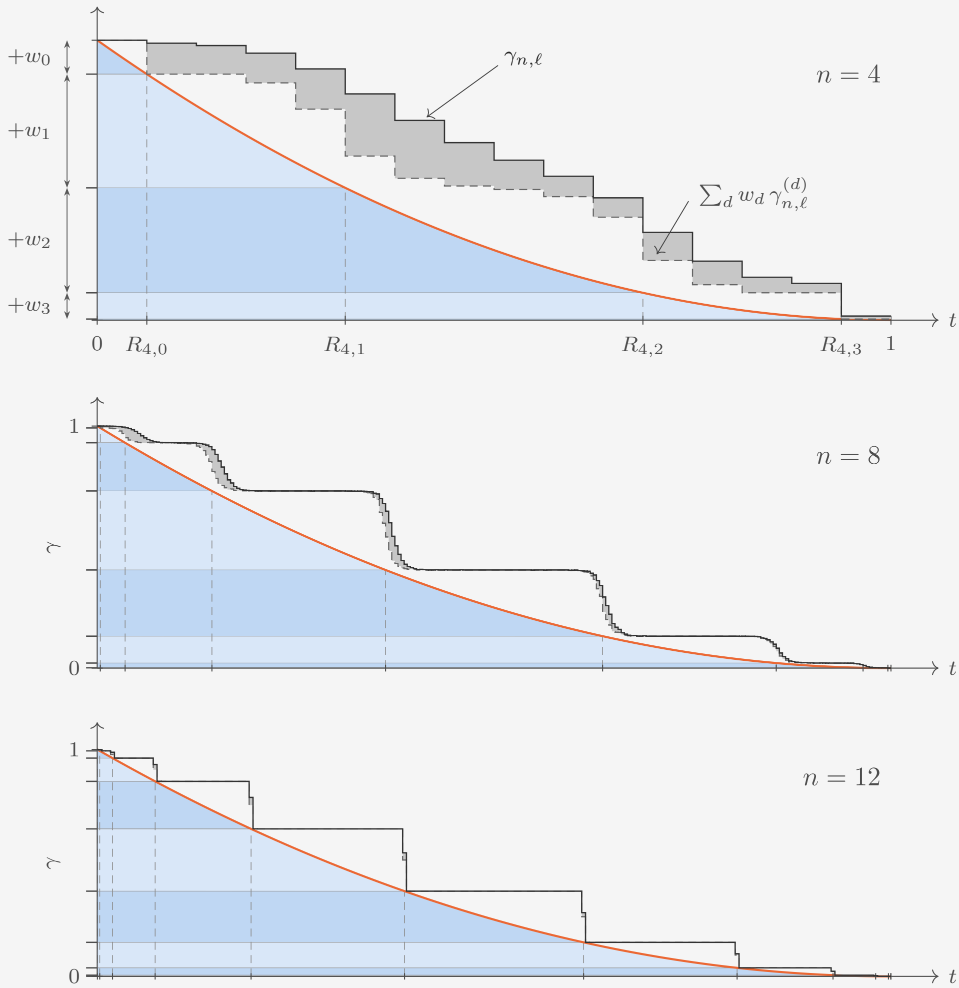
$$\gamma_{n,\lfloor tQ \rfloor} \longrightarrow g'(t) = 1 - F(t) = \Pr(R > t). \tag{126}$$

This caps the construction: the mean conditional entropy of a fresh answer is the probability that the true class is not yet exhausted. The survival function of the statistic is not a postulate; it is the increment term of the finite law.

EXAMPLE 42

This example is meant to illustrate how equation (119) behaves in the TDL. Concisely, the left sum leads to a staircase in $\gamma(t)$, and the mutual information I adds an extra dusting of uncertainty that lies on top.

The shaded region has height $\gamma(t) = 1 - F(t)$: the sampling distribution. The average over D has contributions from all classes $d \le n$ present. Dashed vertical lines are the class rate boundaries $R_{n,d}$. The horizontal lines are the aggregate class weights, largest below, adding lower classes as we rise. This puts the shortest-lived ones on top, so as we move to the right through time, more and more classes "settle", and stop contributing uncertainty. Increasing n cuts the region into more filigree rate gaps horizontally, and more class weights vertically. The two jagged functions on top are two exact increments, the entropy of a single answer, under two states of knowledge. Dashed: what a fresh answer is worth to someone already *told* the class, $\sum_d w_d \gamma^{(d)}_{n,\ell}$. Solid: $\gamma_{n,\ell}$ is what it's worth to Werner, who is not.



By (119) the mixture is more uncertain than its components by exactly what the answers say about the label, so the gray shaded region between the two integrates to $I_{n,\ell}$: the label, and the white between the dashed curve and the region is due to the coarseness of the rate grid. Pretending the steps are smooth is (124). Ignoring the gray is (120).

The relative value of the label vanishes with n as $\log_2(n+1)/Q$. As n grows, the sliver of $I_{n,\ell}$ scales away.

The steps emerge because at a given time a class has either been settled, and contributes no uncertainty, or is still running, and contributes its full bit (scaled by the probability of that class: the step height) at every question, until the next class boundary is reached. The number of questions in which K_d rank (total information) is reached inside a class, statistically, is $o(Q)$, which rounds the step edges. At low n, they bleed together, but at high n, the steps become distinct.

The steps hug $1 - F$ as $n \to \infty$, by construction of the sampling protocol (116).

5.4 The macroscopic curve

Substitute (118), (124) and (126) into (64) and cancel the common factor Q; the numerator tidies from $1 - (1-t)(1-F(t))$ to

$$L_F(t) \;=\; \frac{t + (1-t)\,F(t)}{\displaystyle\int_0^t \big(1 - F(x)\big)\,dx}. \tag{127}$$

This is the master law (105) realized with

$$\eta_0 = 1; \quad \gamma(t) = 1 - F(t); \quad g(t) = \int_0^t (1 - F), \tag{128}$$

Here F must be a monotone (non-constant) CDF, so γ is bounded and decreasing by construction. The sequence of increments, and hence the leverage where its denominator is positive, converges at continuity points of F. Step discontinuities are excluded from this pointwise claim; assigning a value at those does not change the integral. Degenerate zero-entropy laws and singular endpoint behavior require separate treatment. The displayed initial endpoint below assumes $F(0) = 0$ and a finite right derivative there. The numerator has its own reading: destroyed uncertainty density is the asked fraction t plus $F(t)$ per unasked question (column), since an unasked answer is deducible exactly for maps whose class died before t. Asked plus deduced, over received. Zooming in on the endpoints,

$$L_F(0) = 1 + F'(0); \quad L_F(1) = \frac{1}{\mathbb{E}_F[R]}, \tag{129}$$

the curve starts at baseline plus the density of the statistic at zero, and completes at the reciprocal mean class rate.

Conversely, any nonincreasing profile with values in $[0,1]$ is the survival function $\Pr(R > x)$ of exactly one law $F(x)$. This means (127) is invertible: pick any desired increment profile, and (105) at $m = 1$ can achieve it with $F(x) = 1 - \gamma(x)$.

A parallel construction, in which we build priors by postulating that blocks of questions have exchangeable answers, is mathematically complete in the same sense. However, as it is less philosophically aligned with modeling a learner, it is collected in Appendix B.

5.5 Toy statistics: min and max draws

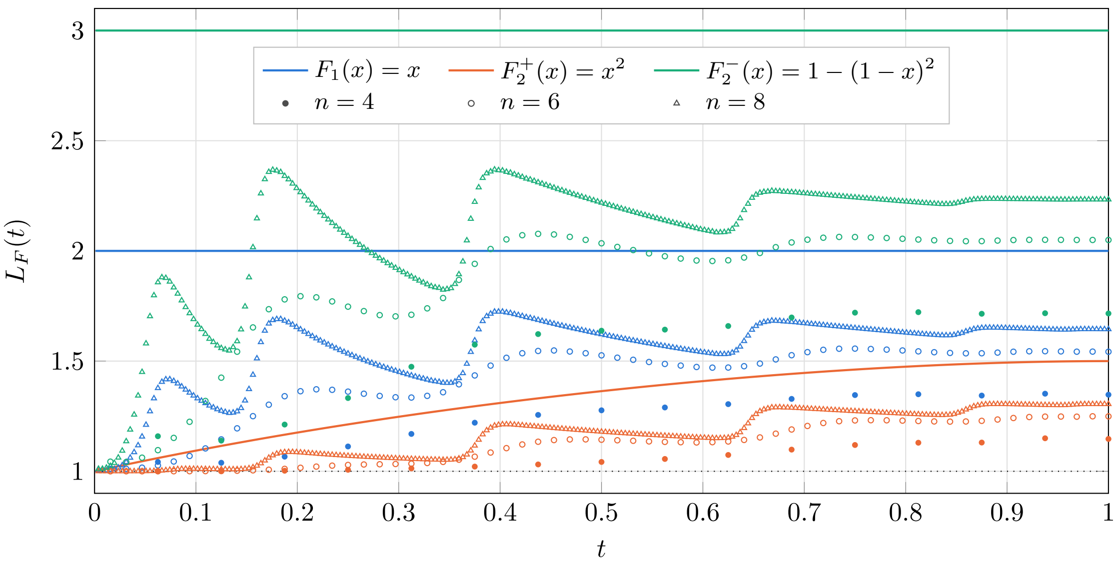


Figure 8: The two families of (131) and (133) at $r = 2$, meeting at $r = 1$; their profiles are in Figure 9. The larger of two draws ($F_2^+(x) = x^2$, orange) delays its deductions and climbs from the baseline to $L_2^+(1) = \frac{3}{2}$; the smaller of two draws ($F_2^-(x) = 1 - (1-x)^2$, green) holds the constant $L_2^- \equiv 3$; between them lies the single uniform draw ($F_1(x) = x$, blue), the one statistic the families share, at $L \equiv 2$. Dots are finite-n leverage: filled $n = 4$ (all 2^{16} question sets enumerated, exact), circles $n = 6$ and triangles $n = 8$ (Monte Carlo over question orders, subset entropies exact via the nested rank profile, latent-label information included). Convergence is slow from below: the label costs $O(\log n)$ of the Q available bits, and the mean rate approaches its limit only as $1/\sqrt{n}$. The left edge is a boundary layer: $L_F(0) = 1 + F'(0)$ presumes $1 \ll \ell \ll Q$, while at $\ell = 1$ the only exhaustible class is the constants and $\mathcal{L}_1 = 1 + (Q-1)\, w_0^2/(2\ln 2) + O(w_0^3)$, the early deductions being spent on identifying the latent label. The finite curves also scallop, most visibly at $n = 8$: each surge is one class being learned out near its rate K_d/Q, a discreteness the limit scrunches away as the rate grid refines.

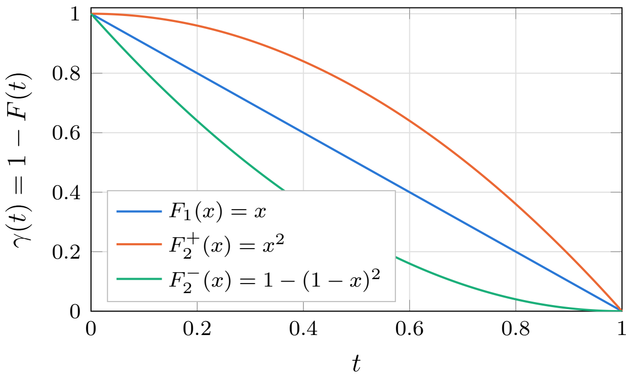


Figure 9: The increment profiles behind Figure 8: the survival function $\gamma(t) = 1 - F(t)$ of each statistic. By (126) it is the mean entropy (expected surprisal) of one fresh answer at macroscopic time t. Convex, straight and concave F give concave, straight and convex profiles respectively; the smaller of two draws spends its predictive power earliest, the larger of two hoards it.

Two one-parameter families are enough to see the law work. Draw r independent uniformly random values on $[0, 1]$ and keep one of them: keeping the smallest suppresses it toward zero, keeping the largest inflates it toward one, r controls how hard. Every curve in Figure 8 comes from this extended family.

The smaller of r draws: All r land above x with probability $(1-x)^r$, so

$$F_r^-(x) = 1-(1-x)^r, \qquad \gamma_r^-(t) = (1-t)^r, \qquad \mathbb{E}[R] = \frac{1}{r+1}. \tag{130}$$

The density $r(1-x)^{r-1}$ is a decreasing function, so budget (weight) is pulled into the low shells. Substituting into (127), the numerator factorizes and the denominator ends up carrying the same bracket,

$$L_r^-(t) \;=\; \frac{t + (1-t)\big(1-(1-t)^r\big)}{\displaystyle\int_0^t (1-x)^r\,dx} \;=\; \frac{1-(1-t)^{r+1}}{\big(1-(1-t)^{r+1}\big)/(r+1)} \;=\; r+1 \tag{131}$$

at every macroscopic time: a whole ladder of constant curves, one rung per r, r deduced bits accompanying each received one. Both endpoint identities below (127) return that same number, $1 + F'(0^+) = 1 + r$ at the start and $1/\mathbb{E}_F[R] = r + 1$ at the finish, and with a constant curve they have no choice but to agree.

The larger of r draws: All r land below x with probability x^r, so

$$F_r^+(x) = x^r, \qquad \gamma_r^+(t) = 1 - t^r, \qquad \mathbb{E}[R] = \frac{r}{r+1}, \tag{132}$$

an increasing density, budget on the high shells. Nothing telescopes now and the leverage varies with t,

$$L_r^+(t) \;=\; \frac{t + (1-t)\,t^r}{t - t^{r+1}/(r+1)} \;=\; \frac{(r+1)\big(t + t^r(1-t)\big)}{(r+1)\,t - t^{r+1}}, \tag{133}$$

rising monotonically across the run. At the start the density vanishes and $L_r^+(0) = 1 + F'(0) = 1$ for every $r > 1$: no leverage at all, each early answer releases no direct deductions. At the finish $L_r^+(1) = 1/\mathbb{E}_F[R] = (r+1)/r$, so the deductions are late-loaded, but even the last of them buys only $1/r$ extra. The two finishes are complements, $1/L_r^-(1) + 1/L_r^+(1) = 1$, because the mean of the smaller and the mean of the larger of the same r draws sum to one.

Pushing r below 1 gives up the min-max reading and flips the tilt of F_r^+ back to concave, but with an infinite density at the origin ($F = \sqrt{x}$ at $r = \frac{1}{2}$), and $L_r^+(0)$ diverges: front-loaded deduction, the shape of a boundary layer, no longer a constant.

Where do the two choices meet? At $r = 1$ the min is the max, and both families collapse onto the single uniform draw $F(x) = x$, whose numerator $t(2-t)$ cancels its denominator $t(2-t)/2$ to leave $L \equiv 2$ exactly. That is the hinge of Figure 8: the blue line is the shared $r = 1$ case, and the curves either side of it are the $r = 2$ members of the two families, $L_2^+(t)$ climbing from 1 to $\frac{3}{2}$ and $L_2^-(t) \equiv 3$. The uniform draw is also the flattest Occam on offer, each shell receiving exactly the share of coefficient space its degree inhabits, $w_d = \binom{n}{d}/2^n$. An observed flat leverage does not mean we had this microscopic prescription: Appendix B produces the same constant 2 by a completely different mechanism.

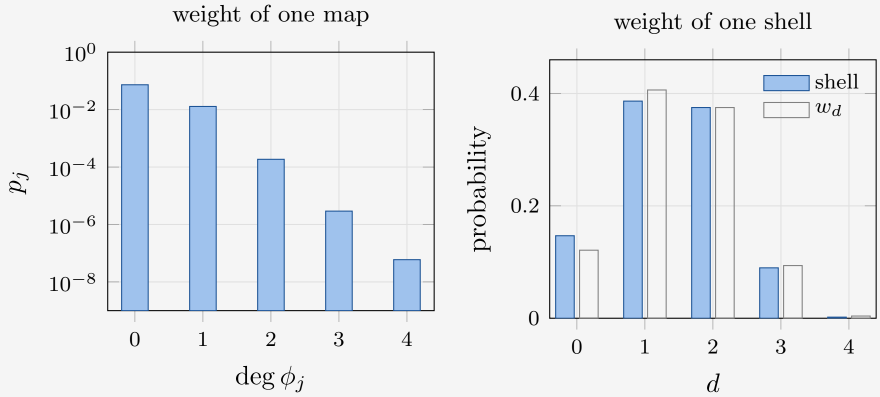
The concave direction may look like the natural one, a simplicity prior ought to favor simple worlds, but F never decides *whether* Werner is an Occamist. The classes are nested, so the weight p_j of a single map falls with its degree for every F whatsoever, and even the convex F_r^+, which loads the high shells, still ranks each complicated (high-degree, highly composite) map as less likely than each simple one. What F moves is the aggregate: which shell carries the belief, and with it the leverage, from r deduced bits per received one under F_r^- down to at most $1/r$ under F_r^+.

EXAMPLE 43

How is belief split over the aggregate degree shells? We will illustrate for $n = 4$, $m = 1$, with the statistic: $F(x) = 1 - (1-x)^2$, the smaller of two uniform draws. The five classes have dimensions

$$K_d \in \{1, 5, 11, 15, 16\},$$
$$w_d \in \left\{\tfrac{31}{256}, \tfrac{104}{256}, \tfrac{96}{256}, \tfrac{24}{256}, \tfrac{1}{256}\right\},$$

using (116) with rates $R_{4,d} = K_d/16$. We can visualize what prior weight maps of each polynomial degree receive.



On the left, per map: the weight drops by a factor 5.7, then 69, 64, 49, matching the factor $2^{\binom{4}{d}}$ that the next class costs in dimension, tilted by the budget ratio w_{d-1}/w_d. Six decades across five degrees, which is why the axis is logarithmic. Occam is priced per map, and it is highly undemocratic. On the right, one shell: the $2^{K_d} - 2^{K_{d-1}}$ maps of each degree carry the belief in blue on a linear axis, peaking at the affine ($d = 1$) maps.

d	maps in shell	shell weight	budget w_d
0	2	14.7%	12.1%
1	30	38.6%	40.6%
2	2,016	37.5%	37.5%
3	30,720	9.0%	9.4%
4	32,768	0.2%	0.4%
total	65,536	100%	100%

In general, population can beat simplicity in the aggregate, while the per-map ordering never wavers. The white bars are the budgets w_d, and the slight difference between the two is the $2^{-\binom{4}{d}}$ leakage into a degree from higher classes.

6 Discussion and Next Steps

6.1 Stochastic truth maps

To all questions in this essay, nature answers deterministically: ψ is a map, and an answer that disagrees with a candidate kills it outright. Replace the truth by a conditional law $\Psi(a \mid q)$ and the update rule (28) becomes genuine Bayes conditioning, reweighting each candidate by its likelihood instead of eliminating it. The hypothesis for the single truth space becomes a simplex rather than the finite set of N maps. The chain rule still gives (56) and the received-entropy law for the observed answer string y. The remaining-entropy law (63), however, needs a revised prediction target: an already asked question need not have zero entropy for a fresh response a'. If M predicts fresh responses at all questions, the asked columns must still be included; if the target consists only of unasked questions, that restriction must be stated. Another identification that breaks is $G_Q = H(p)$. Answers no longer pin down the truth fully, so a complete run does not end with the belief settled. The profile inherits that: no class is ever finished, so γ stays positive forever, as faith in individual answers grows asymptotically. The question is what becomes of the master law (105) when the remaining predictive entropy need not reach zero.

The denominator needs reconsidering as well. Under elimination the surprisal of an answer is also the relative entropy between prior and posterior, so one number both prices the answer and measures how much it shifted the Bayesian belief. A soft likelihood (not 0 or 1) separates them, and this is exactly the distinction Baldi and Itti draw between actual informational content and surprise [4]: a datapoint can carry a large $-\log_2 M_{a,q}$ and barely disturb the belief, or be unremarkable and shift the posterior by a landslide. Leverage then has a choice of denominator, and the two choices are different theories. Pricing by surprisal keeps the accounting of Chapter 2 and the telescoping identity (56) with it; pricing by belief movement measures what the learner absorbed rather than what it paid, and gives up that identity. It would be interesting to research how to reconcile these methods.

6.2 A prior that is wrong

Every average in this work assumes Werner's prior is that from which nature samples. This assumption is woven into all results. It is what makes the expected surprisal of a question equal to the entropy of its own column. Let nature draw from $\{p'_j\}$ while Werner believes $\{p_j\}$, and his expected surprisal at q becomes a cross-entropy, $H(\pi_q)+D(\pi_q \| M_{\cdot,q})$ [2]: costing him the true uncertainty plus the divergence of his belief from it. The numerator of (38) is still measured in his own units, so the ratio falls twice over, once for the inflated bill and once for deductions drawn on correlations that are not there. Some quantities analogous to the KL-divergence could be derived. The interesting assumption is a prior with full support, so under elimination Werner can't rule out the truth outright, which would leave $M_{a,q} = 0$ at some later question: an infinite surprisal and an undefined update. Two questions in particular: does aggregate leverage $\mathcal{L}_\ell$ ever fall below one? If so, is the deficit a divergence, or a sum of per-question KL terms? For the denominator alone the answer is already known: the cumulative excess surprisal of a wrong model is exactly the relative entropy between the two joint laws, which is the redundancy of universal prediction [5]. What that literature does not price is the numerator.

6.3 Designing a curve, shell by shell

The truncation remark of Chapter 3 cuts both ways. Read forward it says the first ℓ rounds cannot see correlators spanning more than $\ell+1$ questions; read backward it says the shell of order $\ell+1$ is the first place that has an influence on learning at step ℓ. Suppose then that we want a particular sequence of leverages. Equation (64) inverts,

$$G_{\ell+1} \;=\; G_\ell \;+\; \frac{Q\,G_1 - \mathcal{L}_\ell\, G_\ell}{Q-\ell}, \tag{134}$$

so the demanded curve fixes the whole block-entropy sequence recursively from G_1. Each $G_{\ell+1}$ in that recursion, with the previously fixed correlators, forms a set of conditions on shell $\ell+1$. One could then try to build the prior a shell at a time: shell 1 to fix G_1, shell 2 to fix G_2, and onward, leaving the earlier rounds untouched. One constraint is submodularity on G, which forces it to be concave with increments in $[0,m]$. What makes this nontrivial is that not every set of correlators $\chi_{\mathcal{S}}$ is admissible. The shells cannot be chosen freely: they must be the transform of an actual distribution. When all correlators are specified, the inverse Walsh transform recovers the candidate prior, $p = N^{-1}W^{\otimes mQ}\chi$; with $\chi_\emptyset = 1$, admissibility is checked by testing every recovered $p_j \geq 0$. The challenge in a shell-by-shell construction is completing the as-yet unspecified higher correlators while preserving these global inequalities. Constraints on the already chosen shells alone need not establish that such a completion exists. The geometry of partial correlation constraints is discussed in [13]. The same language is worth carrying into the limit. To second order the block entropy reads

$$G_k \;=\; km \;-\; \frac{1}{2\ln 2}\sum_{\mathcal{V}\neq\emptyset} \Pr\left[\mathcal{V}\subseteq\mathcal{S}\right]\chi_{\mathcal{V}}^2 \;+\; O(\chi^3), \tag{135}$$

where the remainder begins with cubic products of correlators in general. The inclusion probability of a correlator touching r questions tends to t^r at macroscopic time. Each shell enters the profile suppressed by its own power of t. This codifies the idea of storing knowledge in deep correlations that do not surface until late. For this approach to work, along the way, we would develop TDLs of the correlators themselves, which would be an interesting exercise in its own right.

6.4 Measuring leverage in a real learner

The protocol of Chapter 2 is an experiment, not only a calculation, and a language model is a convenient Werner: its answer matrix is directly observable, the token probabilities in response to a prompt being the column $M_{\cdot,q}$ itself. Fix a finite universe of questions with a latent rule, ask them one at a time, record the surprisal of each answer and the drop in total predictive entropy over the questions not yet asked, and both halves of (38) are measured rather than modeled. The same protocol runs on a network during training, reading the predictive entropy off a held-out question set between updates. Tracking a learner's predictive entropy is established ground, and its sub-extensive part is already the standard measure of how much there was to learn [6]. That work models learning as mutual information between past and future, in a statistical physics paradigm. It reproduces the de Finetti result for $o(Q)$ settling of the label. In our language, that would emerge as hyperbolic leverage. Let us now apply these ideas to LLMs. Perhaps they store their correlations deeper. The obstacle is scale: $t = \ell/Q$ needs a question universe small enough to exhaust, or the whole experiment lives in the boundary layer at $t = 0^+$ and reports nothing about macroscopic learning. Nonetheless, we think the TDL is the correct regime to understand neural networks, as the scale should be large enough to coarse-grain away the microscopic details of boolean maps.

Appendices

Appendix A continues the spike-prior example of Section 4.1. Appendix B collects the block-prior construction and its Bernstein profile language, a block-based route to the same family of macroscopic curves. Appendix C records how the circuit complexities of Subsection 3.2.1 were obtained, the symmetry reduction and the satisfiability encoding that make a minimum over all circuits computable at all. Appendix D collects every symbol used in this work on one page, in the order the argument introduces them. Appendix E lists the scripts that produce the numbers and the curves.

A More characteristics of the spike prior

This appendix continues the toy example of Section 4.1: the anatomy of a single question, the fraction of the dependency stock it extracts, and the sharp-prior limit with its order-of-limits structure.

A.1 One question, branch by branch

For a single question abbreviate $P := P_1$ and $U := U_1$, so that $P + (A-1)U = 1$: the zero answer has probability P, while each particular nonzero answer has probability U. Here *correct* means that the observed answer agrees with the spike map at this question; it does not assert that the entire spike map is the truth. The surprisals are

$$s_{\mathrm{correct}} = -\log_2 P, \qquad s_{\mathrm{wrong}} = -\log_2 U, \qquad \langle s\rangle = P\, s_{\mathrm{correct}} + (A-1)U\, s_{\mathrm{wrong}} = G_1, \tag{136}$$

a particular wrong answer having probability U, not $1-P$. Before asking, every one of the Q columns carries the same answer distribution $(P, U, \ldots, U)$ and hence entropy G_1, so $H_{\mathrm{before}} = H(M^{(0)}) = Q\, G_1$. Conditional on confirmation the next unasked answer has probabilities P_2/P for zero and U_2/P for each nonzero symbol, with entropy

$$H_{\mathrm{c}} := -\frac{P_2}{P}\log_2\frac{P_2}{P} - (A-1)\frac{U_2}{P}\log_2\frac{U_2}{P}, \qquad H_{\mathrm{after,correct}} = (Q-1)H_{\mathrm{c}}, \tag{137}$$

the asked column being fixed while all $Q-1$ unasked columns share that conditional marginal. Refutation instead leaves equal posterior weights on all compatible maps, so every unasked answer is uniform on A symbols and $H_{\mathrm{after,wrong}} = (Q-1)m$. Weighting the two branches,

$$\big\langle H\big(M^{(1)}\big)\big\rangle = P\, H_{\mathrm{after,correct}} + (A-1)U\, H_{\mathrm{after,wrong}} = (Q-1)(G_2 - G_1), \tag{138}$$

the first equality ordinary branch averaging, the second the $\ell = 1$ case of (63).

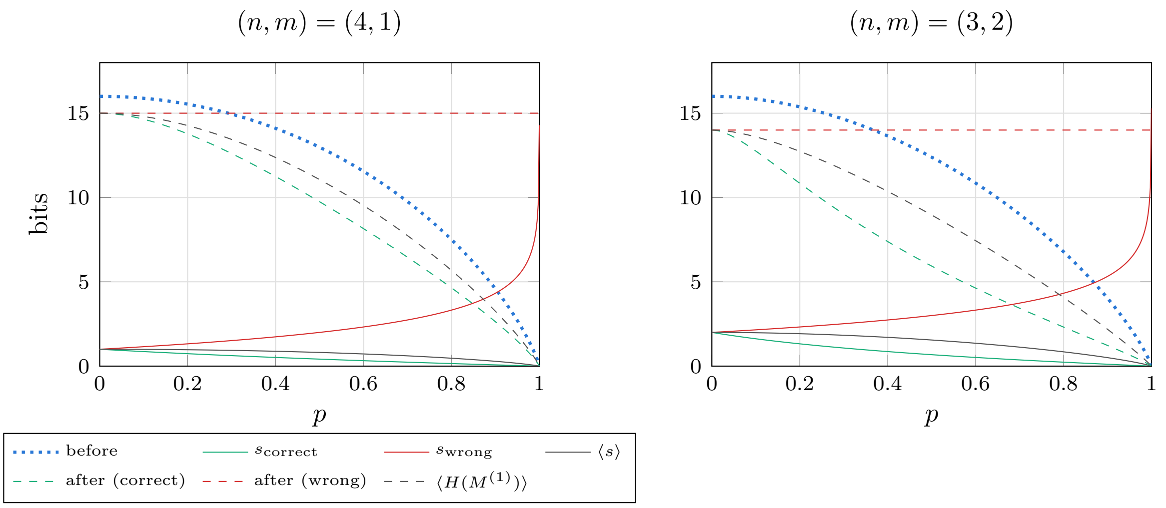


Figure 10: One-question anatomy of the spike prior against the special map's mass p: the branch surprisals (136) (solid), the answer-matrix entropy before the question (dotted) and on each branch after it (137) (dashed). Both panels carry $Qm = 16$ bits of table entropy at the uniform point.

For $p < 1$ the branch leverages divide the vertical distance between the pre-question curve and each branch curve by that branch's own surprisal,

$$L_{\mathrm{correct}} = \frac{QG_1 - (Q-1)H_{\mathrm{c}}}{-\log_2 P}, \qquad L_{\mathrm{wrong}} = \frac{QG_1 - (Q-1)m}{-\log_2 U}, \tag{139}$$

whereas the aggregate leverage is the ratio of the two expected totals,

$$\mathcal{L}_1 = \frac{P\big[QG_1 - (Q-1)H_c\big] + (A-1)U\big[QG_1 - (Q-1)m\big]}{G_1} = \frac{QG_1 - (Q-1)(G_2 - G_1)}{G_1}. \tag{140}$$

The branch formulas use realized surprisals; the expected formula divides the expected entropy drop by the expected surprisal G_1 and is therefore *not* the probability-weighted average of the two branch ratios.

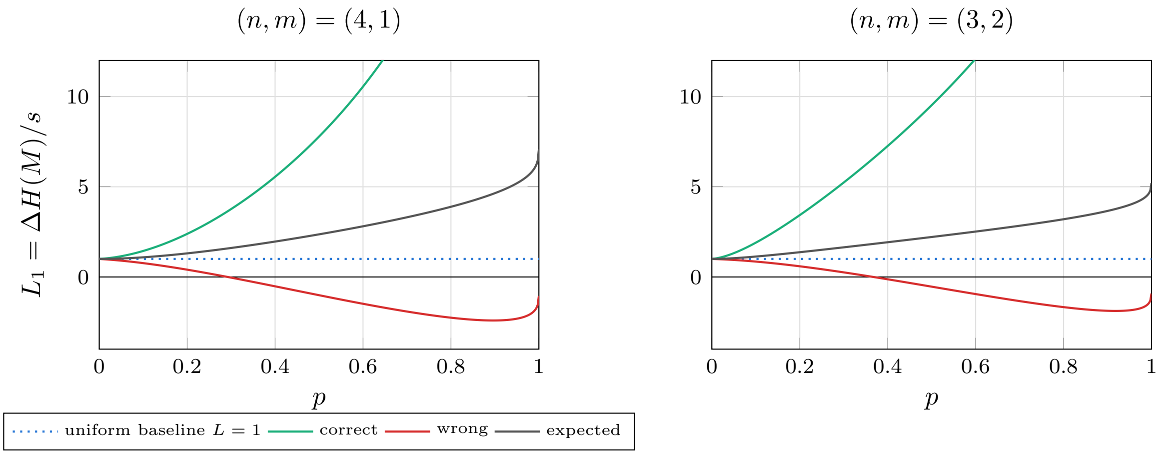


Figure 11: Branchwise (139) and expected (140) leverage of the first question. A refuting answer carries so much surprisal that its own ratio falls below the uniform baseline and eventually below zero, yet the expectation stays well above 1 because the confirming branch is both likely and cheap.

A.2 The extraction ratio

The total correlation $C := H(M^{(0)}) - G_Q = QG_1 - G_Q$ is the information stored in dependencies among answers rather than visible in their separate marginals. Only the part of the expected entropy drop beyond the received entropy G_1 comes out of that store, so when $C > 0$ the share of it freed by a single question is

$$R_1 := \frac{\langle \Delta H(M) \rangle - G_1}{C} = \frac{(Q-1)(2G_1 - G_2)}{QG_1 - G_Q}. \tag{141}$$

The two ratios differ only in their denominator: R_1 measures against the whole dependency stock, leverage against the received surprisal, and

$$R_1 = \frac{G_1}{C}(\mathcal{L}_1 - 1), \qquad \mathcal{L}_1 = 1 + \frac{C}{G_1} R_1. \tag{142}$$

Thus R_1 is a stock-extraction fraction while $\mathcal{L}_1 - 1$ is released dependency information per received bit. At $p = 1$ both the numerator of R_1 and C vanish; expanding (80) with $\varepsilon = 1 - p \to 0^+$ gives the exact finite-(n,m) ratio limit

$$\lim_{p \to 1^-} R_1(p) = \frac{(Q-1)N(1-u)^2}{QN(1-u) - (N-1)}, \qquad \frac{(Q-1)(1-u)^2}{Q(1-u) - 1} \tag{143}$$

the second expression being the additional large-N approximation. At $(n,m) = (2,1)$ the exact value is $12/17 \simeq 0.706$ against the approximation's 0.75. Both share the thermodynamic limit

$$\lim_{n \to \infty} \lim_{p \to 1^-} R_1(p) = 1 - u = 1 - 2^{-m}. \tag{144}$$

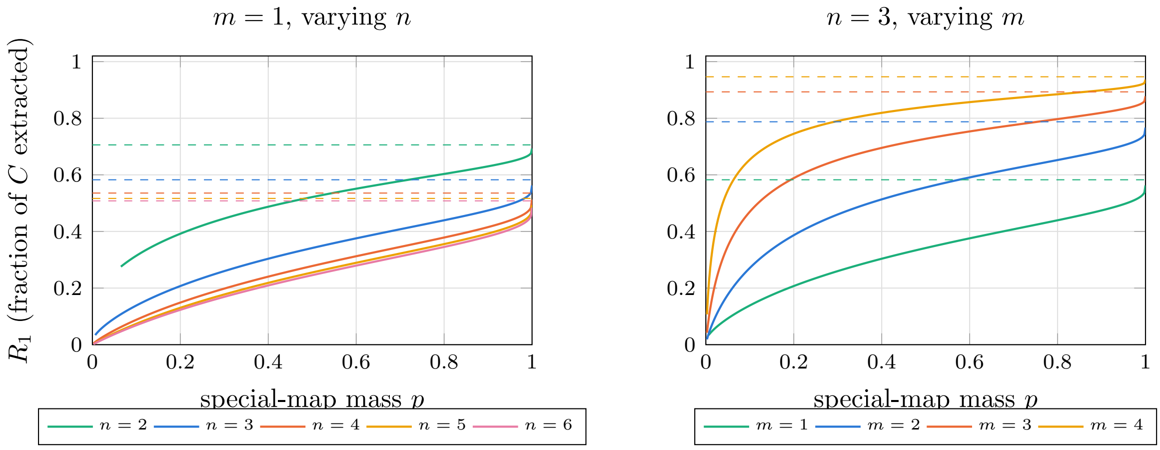


Figure 12: The exact one-question extraction ratio (141) of the spike prior, plotted only where it is defined ($1/N < p < 1$). Dashed lines are the exact finite-N limits (143) as $p \to 1^-$. Growing n drives them down to $1 - u = 1/2$; growing m lifts them toward one.

For the spike-plus-uniform family (144) is a theorem. Extending it to a broader class of uninformative hedges would require a precise definition and a separate extremal argument: full support alone is not sufficient.

A.3 The sharp-prior limit and the order of limits

The sharp-prior calculation takes $\varepsilon = 1 - p \to 0^+$ with n and m fixed. It is a limit toward the delta prior, not evaluation at $p = 1$, where leverage is $0/0$. With the entropy coefficient $\iota_k := \frac{N}{N-1}(1 - u^k)$ we have $P_k = 1 - \varepsilon\iota_k$, the other $u^{-k} - 1$ outcomes share mass $\varepsilon\iota_k$, and

$$G_k = h_2(\varepsilon\iota_k) + \varepsilon\iota_k \log_2(u^{-k} - 1) \;\sim\; \varepsilon\, \iota_k \log_2 \frac{1}{\varepsilon} \qquad (\varepsilon \downarrow 0) \tag{145}$$

at fixed finite Q and k. Substituting into (64) cancels the common factor $\varepsilon \log_2(1/\varepsilon)$ and leaves, for $1 \le \ell \le Q$,

$$\lim_{p \to 1^-} \mathcal{L}_\ell = (1 - u)\, \frac{Q - (Q - \ell)u^\ell}{1 - u^\ell} = \underbrace{Q(1 - u)}_{\text{extensive plateau}} + \underbrace{(1 - u)\frac{\ell\, u^\ell}{1 - u^\ell}}_{O(1)\ \text{boundary layer}} . \tag{146}$$

The second term decays geometrically in ℓ and does not depend on Q at all. In particular $\lim_{p\to 1^-} \mathcal{L}_1 = Q - (Q - 1)2^{-m}$, so

$$\lim_{n\to\infty} \lim_{p\to 1^-} \frac{\mathcal{L}_1}{Q} \;=\; 1 - 2^{-m}. \tag{147}$$

The coefficient $1 - 2^{-m}$ is the fraction of the hypothesis count eliminated by one answer. Its equality with the normalized leverage follows from the entropy asymptotics and should not be read as literally settling that fraction of every individual column. The limiting operation is necessary: at $p = 1$ the learner already assigns probability one to the truth, so received and destroyed information both vanish and leverage is $0/0$. The finite value is the ratio at which the two vanish as $p \to 1^-$. Taking $\ell = \lfloor tQ \rfloor$ in (146) gives the same constant at every $t > 0$, with the uniform bound

$$\sup_{1 \le \ell \le Q} \left| \frac{1}{Q} \lim_{p\to 1^-} \mathcal{L}_\ell - (1 - u) \right| = O(Q^{-1}), \tag{148}$$

so the microscopic boundary term survives only as an $O(1)$ correction to the unnormalized leverage.

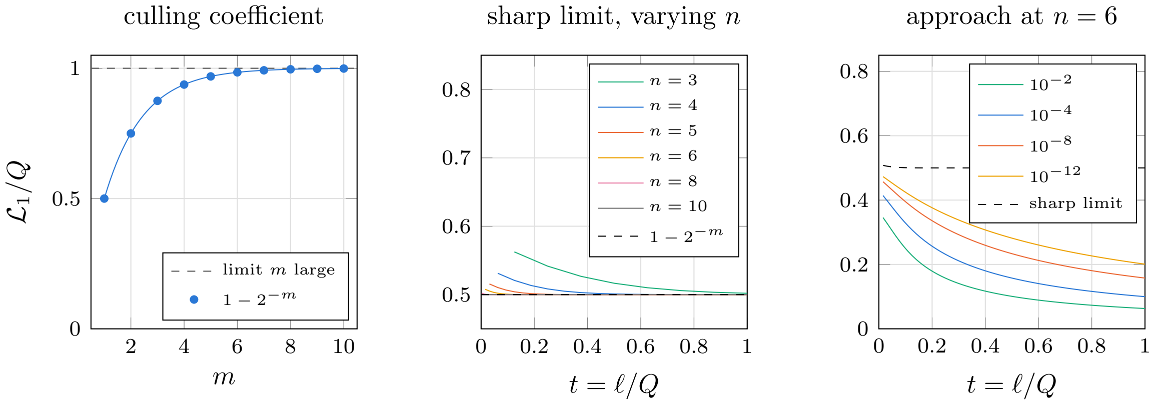


Figure 13: Left: the culling coefficient (147) against the answer width. Middle: the sharp-prior curve (146) normalized by Q, whose $O(1)$ boundary layer is squeezed into the origin as n grows, leaving the plateau at $1-u$. Right: the approach at fixed $n = 6$, $m = 1$, labeled by $1-p$. Tightening the prior lifts the curve only logarithmically, so the plateau is approached pointwise in t but never uniformly.

As $p \to 1^-$ the fixed-p coefficient of (86) diverges, $\big[\eta_0(p,m) - (1-p)m\big]/\big[(1-p)m\big] \sim \frac{1-2^{-m}}{m} \log_2 \frac{1}{1-p}$, which is the noncommutation in explicit form. Writing $L_{p,n}(t) := \mathcal{L}_{\lfloor tQ \rfloor}$ and comparing under the same normalization,

$$\lim_{n\to\infty} \lim_{p\to 1^-} \frac{L_{p,n}(t)}{Q} = 1 - 2^{-m}, \qquad \lim_{p\to 1^-} \lim_{n\to\infty} \frac{L_{p,n}(t)}{Q} = 0. \tag{149}$$

Before dividing by Q the first ordering is extensive while the second diverges only logarithmically in $1/(1-p)$. The two orders are the endpoints of a family of joint scalings. Choose $p_n \to 1$ with prescribed exponential sharpness

$$\frac{\log_2 \big(1/(1-p_n)\big)}{Q} \longrightarrow \lambda \in (0,\infty), \tag{150}$$

so that $\lambda = 0$ recovers the scaling behind the second limit and the formal endpoint $\lambda = \infty$ the first. Putting $\varepsilon_n = 1 - p_n$, the leading entropies for $0 < t < 1$ and $\ell = \lfloor tQ \rfloor$ are $G_1 \sim (1-u)\varepsilon_n \lambda Q$, $G_\ell \sim \varepsilon_n Q(\lambda + mt)$ and $G_{\ell+1} - G_\ell \sim \varepsilon_n m$, and substitution into (64) gives the crossover

$$\frac{L_{p_n,n}(t)}{Q} \longrightarrow (1-u)\,\frac{\lambda}{\lambda + mt}. \tag{151}$$

At $t = 1$ the increment $G_{\ell+1} - G_\ell$ is undefined and the same value follows instead from the completion formula. The two familiar orders are thus endpoints of a larger family.

B Block priors and the Bernstein profile language

Subsection 4.1.3 read the spike prior as a mixture over two answer laws. Nothing there forced the number to be two, and the whole exchangeable family is worth seeing before it is left behind. Let Λ be any distribution over answer laws ν on $\mathcal{A}$. Reading (89) at $k = Q$, the prior it induces on maps is

$$p_j \;=\; \int \prod_{q\in\mathcal{Q}} \nu\big(\phi_j(q)\big)\, d\Lambda(\nu), \tag{152}$$

a map drawn by fixing one answer law and then answering every question independently from it, and by de Finetti's theorem [8, 9] every projectively consistent exchangeable prior is of this form and no other.

The construction is generous. Support is full wherever Λ's is, with no sliver to argue about. Consistency across sizes holds by definition, since Λ carries no reference to n, so the sequence of priors converges without the separate hypothesis that Section 4.2 otherwise has to assume. And every block law depends on $\mathcal{S}$ through its size alone, so G_k is the entropy of A^k block probabilities, each a moment of Λ, and the limit closes in form:

$$\gamma(t) \;\equiv\; \mathbb{E}_\Lambda\big[H(\nu)\big], \qquad \eta_0 \;=\; H\big(\mathbb{E}_\Lambda\,\nu\big), \qquad L(t) \;=\; 1 + \frac{I(\nu\,;\,\xi_1)}{\mathbb{E}_\Lambda\big[H(\nu)\big]\;t}, \tag{153}$$

the numerator once more a mutual information, as at (94). Take $m = 1$ and Λ uniform on the coin biases, for instance. The prior is then $p_j = \big((Q+1)\binom{Q}{|j|}\big)^{-1}$, with $|j|$ the number of ones in the truth table, positive on every one of the N maps, and $\eta_0 = 1$ against $\mathbb{E}_\Lambda[H(\nu)] = \int_0^1 h_2 = 1/(2\ln 2)$ gives

$$L(t) \;=\; 1 + \frac{2\ln 2 - 1}{t} \;=\; 1 + \frac{0.3863}{t} \tag{154}$$

exactly: a full-support prior whose thermodynamic limit is closed form, with no external theorem needed to take it.

And there the family stops. Every member has a flat interior profile and therefore a hyperbola, so a curve that changes throughout $0 < t < 1$ needs a prior that retains structure tied to particular groups or addresses of questions, and a fixed partition into local blocks is the simplest solvable construction. The escape is narrower than it looks. Failing to be a mixture of independent draws is not enough: a prior supported on the truth tables with an even number of ones is exchangeable and is no such mixture at any $Q \geq 4$, since giving zero weight to a single one forces the mixing law onto the two constant answers and thereby kills every other count as well, yet its frequencies still converge and its profile still goes flat. What does escape is a family whose count law never settles, alternating between constructions on even and odd n; that sequence has no limiting frequency law, and so no thermodynamic limit to speak of.

B.1 Independent blocks of questions

Partition $\mathcal{Q}$ into independent blocks drawn from a fixed finite set of types, writing $\mathcal{P}_n$ for the partition and indexing types by κ. At finite n let $K_{\kappa,n}$ count the type-κ blocks and set $w_{\kappa,n} := r_\kappa K_{\kappa,n}/Q$, the fraction of all questions belonging to them, assuming $w_{\kappa,n} \to w_\kappa$ with $\sum_\kappa w_\kappa = 1$. A type κ carries a size r_κ, a fixed joint distribution on its r_κ answers shared by every block of that type up to relabeling, and a mean subset-entropy profile defined as follows. Let $\mathcal{V} \subseteq \mathcal{Q}$ be one particular block of type κ, fix an ordering of it once and for all, and give every subset $\mathcal{J} \subseteq \mathcal{V}$ the inherited order, so that $\psi(\mathcal{J}) \in \mathcal{A}^{|\mathcal{J}|}$ is its random answer vector. For $j \in \{0, \ldots, r_\kappa\}$ define

$$H_\kappa(j) := \binom{r_\kappa}{j}^{-1} \sum_{\substack{\mathcal{J} \subseteq \mathcal{V} \\ |\mathcal{J}|=j}} H\big(\psi(\mathcal{J})\big). \tag{155}$$

Here j is the *number of questions selected from the block*, not a question label, and $H(\psi(\mathcal{J}))$ measures the learner's uncertainty in their *joint answers*, not uncertainty about which questions were selected: the set $\mathcal{J}$ is known. No symmetry is required for this averaged definition, although every zoo member below has the stronger property of **entropy homogeneity**, meaning that $H(\psi(\mathcal{J}))$ itself depends only on $|\mathcal{J}|$. Full permutation invariance is sufficient for that but not necessary: MDS code blocks [11] are entropy homogeneous even though their coordinate distribution is not invariant under every permutation.

The profile records how much answer uncertainty remains visible after looking at j locations inside a block, and its purpose is that successive differences have an operational meaning. Averaging the chain rule over a uniform i-subset $\mathcal{J}$ and then a fresh question drawn uniformly from $\mathcal{V} \setminus \mathcal{J}$ gives the within-block increments

$$\zeta_\kappa(i) := H_\kappa(i+1) - H_\kappa(i) = \mathbb{E}_{\mathcal{J},q}\big[H\big(\psi(q) \mid \psi(\mathcal{J})\big)\big], \qquad 0 \le i \le r_\kappa - 1, \tag{156}$$

the mean uncertainty in one fresh answer after i of its block partners have been answered. It's γ restricted to a block type. This is why the block-entropy profile is useful: it converts a joint law on an entire block into exactly the conditional entropies the learning curve needs. The endpoints make the meaning concrete, $H_\kappa(0) = 0$ while $H_\kappa(r_\kappa) = H(\psi(\mathcal{V}))$ is the entropy of the whole block taken jointly. Two independent fair binary answers have profile $(0, 1, 2)$ and increments $(1, 1)$; the same fair bit copied into both questions has profile $(0, 1, 1)$ and increments $(1, 0)$, since after either answer is known the other carries no remaining uncertainty.

For a fresh question in a block of size r_κ, the number of its $r_\kappa - 1$ partners already present in a uniformly random ℓ-set is hypergeometric, so exactly at finite Q, for $0 \le \ell < Q$,

$$\gamma_{n,\ell} = \sum_\kappa w_{\kappa,n} \sum_{i=0}^{r_\kappa - 1} \frac{\binom{r_\kappa - 1}{i}\binom{Q - r_\kappa}{\ell - i}}{\binom{Q-1}{\ell}}\, \zeta_\kappa(i). \tag{157}$$

Holding every r_κ fixed, taking $\ell = \lfloor tQ \rfloor$ and assuming $w_{\kappa,n} \to w_\kappa$, sampling without replacement approaches independent sampling,

$$\frac{\binom{r-1}{i}\binom{Q-r}{\ell-i}}{\binom{Q-1}{\ell}} \longrightarrow \underbrace{\binom{r-1}{i} t^i (1-t)^{r-1-i}}_{\beta_{r-1,i}(t)}, \tag{158}$$

which simultaneously defines the Bernstein basis polynomial $\beta_{r-1,i}$ and, probabilistically, is the limiting probability that exactly i of the fresh question's $r - 1$ partners have been asked by macroscopic time t. The limiting profile is therefore

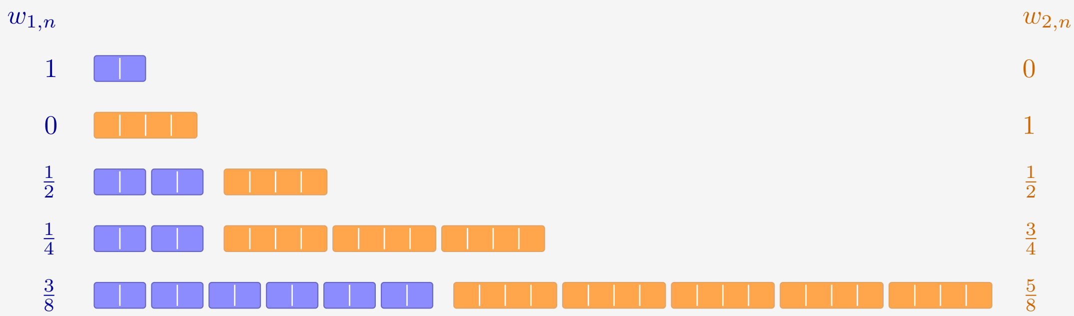
$$\gamma(t) = \sum_\kappa w_\kappa \sum_{i=0}^{r_\kappa - 1} \beta_{r_\kappa - 1, i}(t)\, \zeta_\kappa(i). \tag{159}$$

At the start of the run no partner has been asked, so only the $i = 0$ term survives and $\eta_0 = \gamma(0) = \sum_\kappa w_\kappa \zeta_\kappa(0) = \sum_\kappa w_\kappa H_\kappa(1)$: the same block-entropy profile supplies both the initial marginal entropy and the entire macroscopic conditional-entropy curve.

To construct a sequence with prescribed fractions, choose integer block counts with $r_\kappa K_{\kappa,n}/Q \to w_\kappa$ and $\sum_\kappa r_\kappa K_{\kappa,n} \le Q$, assigning only $o(Q)$ leftover sites, whose treatment does not affect the limit; at finite n any leftovers can be recorded as an additional block type, so (157) still includes every question and remains exact. If block sizes grow with n the proof changes, a useful sufficient condition for the hypergeometric-to-binomial approximation being $\left(\max_{\mathcal{V}\in\mathcal{P}_n} |\mathcal{V}|\right)^2/Q \to 0$. For countably many types, pointwise frequency convergence must be replaced by an ℓ^1 or tightness condition.

EXAMPLE 44

Two types tile the growing table: pairs ($r_1 = 2$, blue) and quads ($r_2 = 4$, orange), aiming at $w_1 = 3/8$ and $w_2 = 5/8$. Integer counts force detours. At $n = 1$ only a pair fits and at $n = 2$ only a quad, while at $Q = 8$ and $Q = 16$ the best leftover-free tilings still miss the target shares by $1/8$.



Each cell is one question and each rounded group is one block, so the counts $K_{\kappa,n}$ can be read off: $(K_1, K_2) = (1,0),\ (0,1),\ (2,1),\ (2,3),\ (6,5)$. The pair share $w_{1,n} = 1,\ 0,\ \frac{1}{2},\ \frac{1}{4},\ \frac{3}{8}$ closes in on its target by halving steps, and at $n = 5$ the tiling is exact: $6 \times 2 + 5 \times 4 = 32 = Q$.

B.2 Block laws and their map priors

The profile machinery never displayed the prior itself. It is constructive: each block law prices the restriction of a map to its block, and independence across blocks multiplies the prices. A map ϕ_j restricts to the answer tuple $\phi_j(\mathcal{V})$ on block $\mathcal{V}$, and

$$p_j \;=\; \prod_{\mathcal{V}\in\mathcal{P}_n} \Pr\big(\psi(\mathcal{V}) = \phi_j(\mathcal{V})\big). \tag{160}$$

The classical block laws, with the factor each contributes to (160) and the curve it produces:

block law	a size-r block is sampled by	factor $\Pr(\psi(\mathcal{V}) = y)$	leverage
clique	one value $a_{\mathcal{V}} \sim \nu_{\mathcal{V}}$ copied to every question	$\nu_{\mathcal{V}}(a)$ if $y = (a, \ldots, a)$, else 0	$L \equiv r$, exact at every n
parity (σ)	uniform on the coset $\bigoplus_q y_q = \sigma$	$2^{-(r-1)}$ on the coset, else 0	rises $1 \to r/(r-1)$, late
tilted parity	i.i.d. Bernoulli(θ) bits conditioned on the coset	$\theta^{\lvert y \rvert}(1-\theta)^{r-\lvert y \rvert}/\mathcal{Z}_{r,\sigma}$ on the coset, else 0	same shape, bent by $\varepsilon = 1 - 2\theta$
$[r,k]_A$ MDS	uniform on an MDS code ($A \geq r$)	A^{-k} on codewords, else 0	$L(1) = r/k$; interior peak when mixed with fresh sites
lapse (δ)	any law above, replaced by the uniform block with probability δ	$(1-\delta)P_{\mathcal{V}}(y) + \delta A^{-r}$	full support; $\gamma(1) > 0$

Table 8: The classical block laws: how a block of size r is sampled, the factor it contributes to (160), and the leverage that follows.

The supports multiply as well: cliques carry the $A^{Q/r}$ maps constant on every block, parity carries the $2^{(r-1)Q/r}$ maps landing in every coset, a code carries $A^{kQ/r}$ maps, and the lapse restores $p_j > 0$ everywhere, playing the role the top class played in Chapter 5.

EXAMPLE 45

Cliques at $(n,m) = (2,1)$: partition the four questions by their high bit, $\mathcal{V}_1 = \{00, 01\}$ and $\mathcal{V}_2 = \{10, 11\}$, with $\nu_{\mathcal{V}_1} = (0.8, 0.2)$ and $\nu_{\mathcal{V}_2} = (0.55, 0.45)$ on the answers $(0,1)$. A supported map must be constant on both blocks, so it can depend only on b_1: of the sixteen maps of Table 4, only FALSE ($j = 0$), $\neg b_1$ ($j = 3$), b_1 ($j = 12$) and TRUE ($j = 15$) survive, with product weights $p_0 = 0.8 \times 0.55 = 0.44$, $p_3 = 0.11$, $p_{12} = 0.36$, $p_{15} = 0.09$, and $p_j = 0$ for the other twelve. One answer from either block settles that block's remaining question for free: $L \equiv 2$.

From any row of the table the route to leverage is the one already walked in Chapter 5. The block law fixes the subset entropies (155), their differences give the increments ζ (156), and the hypergeometric average (157) is exact at every finite n; its binomial limit is the Bernstein profile (159). No capacity theorem is needed: blocks are bounded, so elementary hypergeometric-to-binomial convergence replaces it. The finite law (101) then gives $\mathcal{L}_\ell$ exactly at every n, and the master law (105) gives the curve, with $\eta_0 = \sum_\kappa w_\kappa \zeta_\kappa(0)$.

B.3 Bernstein polynomials as a profile language

For a single block type, suppressing the type label, equation (159) reads in the Bernstein basis [10]

$$\gamma(t) = \zeta(0)(1-t)^2 + 2\zeta(1)t(1-t) + \zeta(2)t^2 \tag{161}$$

at $r = 3$, and $\zeta(0)(1-t)^3 + 3\zeta(1)t(1-t)^2 + 3\zeta(2)t^2(1-t) + \zeta(3)t^3$ at $r = 4$. Some useful coefficient vectors, with h_{fresh} the entropy of one independent fresh answer and h_{clique} the entropy of the value copied through a clique:

block	$(\zeta(0), \ldots, \zeta(r-1))$	profile
independent fresh answers	$(h_{\text{fresh}}, \ldots, h_{\text{fresh}})$	h_{fresh}
clique	$(h_{\text{clique}}, 0, \ldots, 0)$	$h_{\text{clique}}(1-t)^{r-1}$
parity	$(m, \ldots, m, 0)$	$m(1-t^{r-1})$
(r, k) MDS	k entries m, then zeros	$m \Pr[\mathrm{Bin}(r-1, t) \leq k-1]$

Table 9: Coefficient vectors ζ for the block laws above, and the profile each one generates.

Entropy submodularity gives $m \geq \zeta_\kappa(0) \geq \cdots \geq \zeta_\kappa(r_\kappa - 1) \geq 0$, and the derivative identity for Bernstein polynomials makes the macroscopic monotonicity explicit,

$$\gamma'(t) = \sum_\kappa w_\kappa (r_\kappa - 1) \sum_{i=0}^{r_\kappa - 2} \big(\zeta_\kappa(i+1) - \zeta_\kappa(i)\big) \beta_{r_\kappa - 2, i}(t) \;\leq\; 0, \tag{162}$$

so γ is nonincreasing as the general entropy calculus requires. Leverage itself need not be monotone.

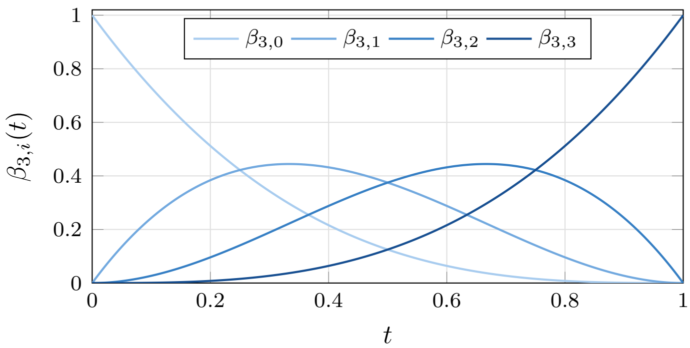


Figure 14: The degree-three Bernstein basis: bump functions scanning left to right, the i-th peaked at $i/3$. A block profile is a positive combination of such bumps weighted by the microscopic conditional entropies $\zeta(i)$, so the shape of L becomes a question about *where the coefficients drop.*

Equation (159) describes every prior in the stated class of independent blocks drawn from a fixed finite list of bounded types, and mixing types mixes their profiles linearly. There is also a useful approximation result when the answer alphabet is allowed to grow: mixtures of common-length MDS blocks realize any nonincreasing Bernstein coefficient sequence, and Bernstein polynomials approximate every continuous nonincreasing profile. A Reed–Solomon block of length r needs $A \geq r$, however, so sending $r \to \infty$ in that approximation also requires $m \to \infty$, or a different family of realizable entropy vectors. It is not a completeness theorem at fixed answer width.

B.4 Leverage curves at finite n

Because (157) is exact, the finite-n curves need no simulation: the hypergeometric increments feed the finite law (101) directly, and the limits below are the master law (105) applied to the Bernstein profiles of the table above. Three representative members, at $Q = 16$ and $Q = 64$ against their limits:

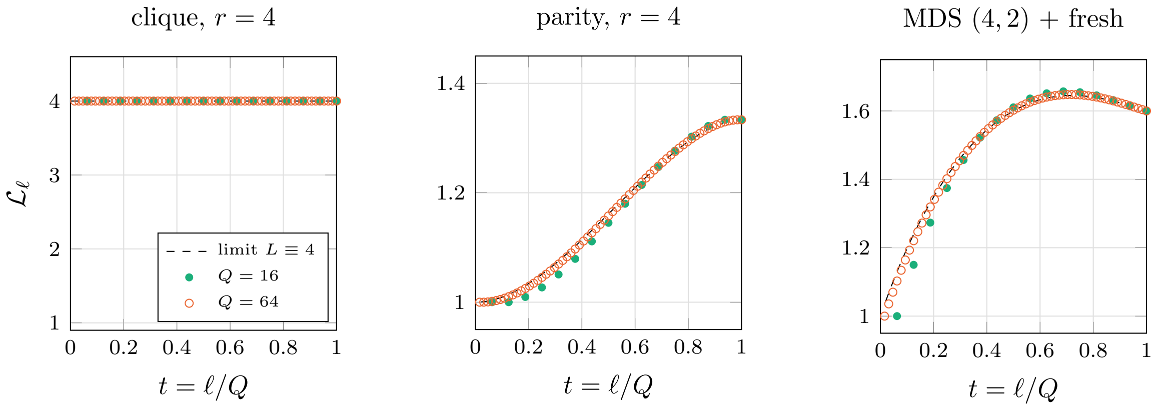


Figure 15: Exact finite-n leverage (dots, $Q = 16$ and 64) against the macroscopic limits (dashed), from (157) and (101) with no simulation. Left: cliques of four with a fair shared bit; the dots sit exactly on the constant $L \equiv 4$ at every finite n. Middle: parity blocks of four; deductions concentrate late and the finite curves approach the rise to $r/(r-1) = \frac{4}{3}$ from below. Right: a 3:1 mixture of $[4, 2]$ MDS code blocks ($m = 2$) with fresh questions; the code releases its deductions around the threshold and the fresh sites keep receiving afterwards, an interior maximum of 1.645 at $t \simeq 0.72$ against the completion value 1.6.

C Computing the circuit complexities

We read the footprint and the bias of Subsection 3.2.1 off a truth table in one pass. The complexity X_j we cannot: it is a minimum over all circuits, and we have to compute it. This appendix records how we obtained the 65,536 values in each of the $(4,1)$ and $(3,2)$ tables. We describe the pipeline in `gibbs_complexity.md`, and implement it in `aig_synth.py`, `groups.py`, `run_synth_3to2.py` and the `build_table` drivers.

Reduction by symmetry: Relabeling wires is free in an and-inverter circuit: permuting inputs is rewiring, negating an input or an output is an edge attribute, and for $m = 2$ swapping the two outputs renames which wire is which. So X_j is constant on the orbits of the relabeling groups

$$|G_{4,1}| = 4! \cdot 2^4 \cdot 2 = 768, \qquad |G_{3,2}| = 3! \cdot 2^3 \cdot 2^2 \cdot 2 = 384, \tag{163}$$

the first being the standard NPN group of logic synthesis [14]. In `groups.py` we apply every group element to every truth table and keep the numerically smallest image as the class representative, which collapses 65,536 maps to 222 classes at $(4,1)$ and 308 at $(3,2)$. We then run exact synthesis a few hundred times instead of 65,536.

Exact synthesis as satisfiability: For a fixed gate budget r, in `aig_synth.py` we build a formula that is satisfiable exactly when some r-gate and-inverter circuit computes the target. Three families of variables carry it. First $t_{s,\rho}$, the bit that signal s holds on truth-table row ρ, with signal 0 the constant and signals $1,\ldots,n$ the inputs, all pinned by unit clauses. Then a one-hot selector $\sigma_{g,a,b,\iota_a,\iota_b}$ saying that gate g draws its two fanins from *earlier* signals $a \le b$, inverted according to ι; restricting to earlier signals is what makes the circuit acyclic by construction, so we need no ordering constraints. Last a one-hot selector $o_{k,s,\iota}$ naming which signal, at which polarity, carries output k. The AND semantics costs three clauses per gate, candidate and row, and the output constraint pins the chosen signal to the target column on every row. We add two structural clauses that tighten the search without excluding any minimum circuit: every gate must feed a later gate or an output, and no two gates may carry identical truth tables. Then

$$X_j \;=\; \min\{\, r \ge 0 \;:\; \text{the } r\text{-gate formula is satisfiable} \,\}, \tag{164}$$

which we search by trying $r = 0, 1, 2, \ldots$ upward. We get a proven minimum rather than a best effort: the solver refuted every smaller budget, and did not merely fail to find one. We run CaDiCaL [17] as the back end, through PySAT [16]. Run as a script, the synthesizer reproduces the textbook two-input values, AND, OR and NAND at 1 and XOR at 3.

Provenance: At $(4,1)$ we adopt the class values from the published reproducibility dataset [18] of the 222 NPN classes, and in `crosscheck.py` we re-derive ten of them at random with our own synthesizer, finding no mismatch. At $(3,2)$ there is no reference table, so we synthesized all 308 classes from scratch with `run_synth_3to2.py` over a process pool. The resulting ranges are $X \le 10$ at $(4,1)$ and $X \le 9$ at $(3,2)$, and we tabulate the head counts in Subsection 3.2.1.

Why we adopt the $(4,1)$ values rather than recompute them: All 222 entries of [18] are proven optimal, the two hardest classes included: `0x1669` and `0x166b`, at 10 gates each, which is the ceiling of the range. Our own synthesizer agrees wherever it can reach, but it cannot reach that far. For `0x1669` it refutes $r = 5$ in under a second and $r = 7$ in half a minute, and does not settle $r = 9$ within fifteen: the instance grows steeply with the budget, since the selector family for gate g already carries $O\big((n+g)^2\big)$ candidates and its at-most-one encoding is quadratic in that. Hence our division of labour. We take the $(4,1)$ column from the reference dataset and sample it against our own synthesizer, while the $(3,2)$ classes, being one input bit smaller and so a much smaller formula, we synthesize here from scratch.

D Symbols

n	number of bits in a question
m	number of bits in an answer
q, a	one question and one answer
$Q = 2^n$	how many distinct questions exist, the width of M
$A = 2^m$	how many answers any one question admits
$N = A^Q$	how many candidate maps exist, the size of the hypothesis space
ψ	the truth: the single map nature uses to answer, unknown to Werner
ϕ_j	candidate map j, indexed by its truth table read as a numeral
p_j	Werner's prior belief that $\psi = \phi_j$
$M_{a,q}$	the answer matrix: his belief that question q is answered a
$H(M)$	table entropy, the column entropies of M added
$H(p)$	entropy of the belief itself, over the N maps
C	correlation store $H(M) - H(p)$, knowledge no single column sees
$\chi_{\mathcal{S}}$	correlator: the mean parity of the answers on a set of sites
$\mathcal{S}$	the set of questions asked so far
ℓ	how many have been asked, $\lvert\mathcal{S}\rvert$
y	the pattern of answers received on $\mathcal{S}$
s_k	surprisal of round k, minus the log of the probability that answer had
L_ℓ	leverage: table entropy destroyed per bit of surprisal received
$\mathcal{L}_\ell$	ratio of expected table-entropy reduction to expected surprisal
$\eta_n(v)$	fresh-answer entropy at independent reveal probability v
G_k	mean entropy of the answers to k questions, averaged over which k
$\gamma_{n,k}$	increment $G_{n,k+1} - G_{n,k}$, what one fresh answer is worth
$t = \ell/Q$	macroscopic time, the fraction of the table already asked
$\gamma(t)$	the profile: the increment's limit, entropy of a fresh answer at t
$g(t)$	$\int_0^t \gamma$, the surprisal received per question by time t
η_0	marginal entropy of one answer before anything has been asked
ν	an answer law: a discrete distribution on the alphabet $\mathcal{A}$
Λ	the directing measure: a distribution over answer laws ν
F, R	the statistic: a law on $[0, 1]$ and the value drawn from it
D	the complexity class that draw selects
w_d	prior weight of class d, the F-measure of its rate gap
K_d	dimension of class d, the number of monomials of degree $\leq d$
$R_{n,d}$	rate of class d, its dimension per question K_d/Q
$\rho_{\mathcal{B}}$	coefficient of the monomial $\mathcal{B}$ in a map's polynomial
Ω	generator matrix: a class's monomials evaluated at every question
ξ	the coin string that picks a map uniformly inside a class

Table 10: Every symbol used in this work, in the order the argument introduces it.

E Scripts

Numbers, curves, and combinatoric sets in this work are produced by the scripts below. They are hosted with the source, at `https://github.com/DMChernowitz/boolean-maps-learning`.

script	what it does
`tikz_data.py`	Tabulates the spike-prior curves of Appendix A, the one-question anatomy, the branch leverages, the extraction ratio and the sharp-prior flow, and the Bernstein basis of Appendix B.
`spike_finite_data.py`	Exact finite-n leverage of the spike prior from its closed block entropies.
`rm_gamma_data.py`	Mean rank increment $\gamma_{n,\ell}^{(d)}$ of one Reed-Muller class along a random question order.
`rm_finite_data.py`	Exact finite-n leverage of the polynomial-degree prior, for the three statistics of Figure 8.
`block_finite_data.py`	Exact finite-n leverage of the clique, parity and MDS block priors, through the hypergeometric increment (157).
`rm_label_data.py`	Splits the received entropy into the class average and the label term $I_{n,\ell}$ of (120).
`rm_carpet_fig.py`	Draws the layered increments that illustrate (119) in the limit, from those tables.
`gibbs_curves.py`	Aggregate leverage trajectories under the three Gibbs settings, and the descriptor tables they are read from.
`gibbs_stack_data.py`	The three-layer split of the table entropy at every ℓ, for the panels of Subsection 3.2.3.
`aig_synth.py`	Exact minimum-size and-inverter synthesis as satisfiability, one formula per gate budget.
`groups.py`	Canonical representatives of the relabeling classes, of order 768 at $(4,1)$ and 384 at $(3,2)$.
`run_synth_3to2.py`	Runs the synthesizer over all 308 classes of the $(3,2)$ space on a process pool.
`crosscheck.py`	Re-derives random classes of the published $(4,1)$ dataset with our own synthesizer.
`build_table_4to1.py`	Assembles the 65,536-row $(4,1)$ table of complexity, footprint and bias.
`build_table_3to2.py`	The same at $(3,2)$, where the complexity is one joint circuit for both output bits.
`build_table_3to1.py`	The same at $(3,1)$, which the checks below use as a small test space.
`expected_trajectory.py`	Checks the two exact laws, and the expected run of the guiding example, by enumerating every question set.
`walsh_update.py`	Checks the correlator ledger and the two-term update rule (46) at $m=1$.
`correlator_shells.py`	The same at general m, on the $(3,2)$ space, against (50).
`spike_flow.py`	Checks the two-state renormalization flow (76) and the invariant $P_k r_k = p$.

Table 11: The scripts behind this work, grouped by role: the tables read at compile time, the figures generated and pasted in, the circuit-complexity pipeline of Appendix C, and the checks.

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